Deep Holes of Projective Reed–Solomon Codes Beyond Redundancy Four: Recursive Carriers and Exact Classifications Through Redundancy Ten

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Abstract

We classify projective Reed–Solomon split-free syndrome directions, in the stated field ranges, from the first previously open case, redundancy five, through redundancy ten, and obtain deep-hole classifications wherever the covering-radius gate is available. A syndrome is split-free precisely when its two-row Hankel kernel contains no completely split squarefree form. At redundancy five, this kernel is a pencil of cubics. At higher redundancy, coherent polar contraction retains a removed root as a marker, allowing a lower split witness to lift without repetition.

The split members of the R5 pencil are counted exactly. On the trivial-gcd separable stratum,

$$\#Y_f(\mathbb{F}_q) = 6N_f + 3d_2 + d_3,$$

where N_f counts the completely split squarefree members, d_2 those with a rational double root, d_3 the perfect cubes, and Y_f is the off-diagonal fiber square. The proof counts rational roots fiber by fiber and uses no discriminant, so it holds in every characteristic. The identity class has splitting density $1/6$, giving the Chebotarev main term $(q+1)/6$; the identity bounds N_f two-sidedly around that main term at Weil scale, lowers the geometric threshold for a split witness to $q \geq 20$, and in characteristic two leaves no split-free S_3 pencil once $q \geq 16$. The finite bridge is therefore $q \in \{7, 8, 9, 11, 13, 17, 19\}$.

For every $r \geq 6$, the reduced recursively contained locus is exactly the union of the catalecticant rank-two scheme and one maximal adjacent-zero Lucas carrier. Dense squarefree-marker contractions select one terminal component, while Pascal nesting merges all modular descendants into that carrier. This component theorem is unconditional. The finite-field escape statement is separate: if the explicit pointed lower packages exist at every intermediate redundancy, then

$$q \geq 6r - 15 + \lfloor 2\sqrt{6r-17} \rfloor$$

forces every split-free syndrome into this carrier. Under the same hypothesis, when $\operatorname{char} \mathbb{F}_q > r-1$, the Lucas carrier is empty; the radius theorem then leaves exactly the tangent and conjugate-secant deep-hole families, with $q(q+1)^2/2$ projective directions.

The required packages are discharged at the fixed levels. Redundancies five and six are classified for every $q \geq 7$; redundancy seven has a complete split-free classification for every $q \geq 7$, which is a deep-hole classification for $q \geq 11$; at $q = 8$, exact distance extraction leaves precisely ten deep directions in two orbits. Redundancies eight, nine, and ten have exact deep-hole classifications for $q \geq 43, 53, 59$, respectively. Separately, on the even diagonal $r = q - 1$, the known dimension-two radius is r; in the present syndrome coordinates the distinguished tangent direction e_{r-2} has that exact distance, giving an explicit certified deep direction at arbitrary redundancy (for example R15 over $\mathbb{F}_{16}$). Certificates close the bounded R5–R7 residues, the full degree-nine Lucas carrier at $q = 16, 32$, and its invariant block at $q = 64$. A final-pair Artin–Schreier argument then proves that at redundancy ten the Hankel kernel of every point on the full degree-nine Lucas carrier contains a split squarefree form over $\mathbb{F}_{2^m}$, $m \geq 4$.

Index Terms

catalecticants, covering radius, deep holes, Hankel matrices, maximum-distance-separable extensions, normal rational curves, projective Reed–Solomon codes.

I. Introduction

This paper gives a recursive obstruction theorem for projective Reed–Solomon deep holes at arbitrary redundancy and exact classifications through redundancy ten. The common mechanism is a coherent polar flag, a marked contraction that propagates splitting information without forgetting the factor needed for a squarefree lift. The contained part of the recursion is exactly the persistent catalecticant scheme together with one maximal Lucas carrier. At a fixed depth, the transverse part escapes by a uniform point count once the explicit one-step lower packages at every intermediate stage have been established.

Let $\mathrm{PRS}(k)$ denote the projective Reed–Solomon code of length $q+1$ obtained by evaluating homogeneous binary forms of degree $k-1$ on $\mathbb{P}^1(\mathbb{F}_q)$. If the redundancy is

$$r = q + 1 - k,$$

a parity-check matrix has as columns the $\mathbb{F}_q$-points of the degree-$(r-1)$ normal rational curve

$$\mathcal{R}_{r-1} = \{(1, t, \ldots, t^{r-1}) : t \in \mathbb{F}_q\} \cup \{(0, \ldots, 0, 1)\} \subset \mathbb{P}^{r-1}.$$

Call a projective syndrome direction *split-free* if it is not contained in the span of $r-2$ distinct columns. When $\rho(\mathrm{PRS}(q+1-r))=r-1$, the split-free directions are precisely the projective syndrome directions of deep-hole cosets. The logical separation used throughout the paper is

$$\boxed{\text{split-free classification} \ + \ \rho(\mathrm{PRS}(q+1-r))=r-1 \quad \Longrightarrow \quad \text{deep-hole classification.}}$$

The first term is proved geometrically; the covering-radius equality is a separate coding-theoretic input and is never inferred from a finite split-free census.

Two logical layers. The arbitrary-redundancy result separates an unconditional component theorem from a conditional finite-field escape theorem. Propositions VI.1–VI.3 identify the reduced recursively contained locus without a field-size bound or a lower-package hypothesis. The theorem below then assumes the pointed lower packages and its displayed bound on q to show that every split-free point lies in that locus. The fixed-level theorems discharge those extra inputs only through redundancy ten.

Theorem I.1 (Recursive obstruction and fixed-level consequences)**.** Let $r\geq 6$, and let

$$q\geq 6r-15+\left\lfloor 2\sqrt{6r-17}\right\rfloor .$$

Every split-free redundancy-r syndrome belongs to

$$\mathcal{P}_r\cup\mathbb{P}\left\langle e_j:\binom{r-2}{j}=\binom{r-2}{j-1}=0 \pmod{\operatorname{char}\mathbb{F}_q}\right\rangle. \tag{1}$$

provided the explicit one-step lower packages of Definition V.4, with the stated deletion bounds, exist at every intermediate redundancy. Under the same package hypothesis, if $\operatorname{char}\mathbb{F}_q>r-1$, the second term is empty and the deep-hole syndromes are exactly the persistent tangent and conjugate-secant families, of total cardinality $q(q+1)^2/2$.

At fixed levels, the classifications are exact at redundancies five and six for every $q\geq 7$, at redundancy seven in the split-free sense for every $q\geq 7$, and at redundancies eight, nine, and ten for $q\geq 43, 53, 59$, respectively. The redundancy-seven result promotes to deep holes whenever the radius is six, in particular for $q\geq 11$.

Theorems I.2, A.10, A.16, B.11, B.22, and Corollary C.18 give the fixed levels. The redundancy-ten conclusion uses Theorem C.13: the first fresh higher Lucas carrier is entirely shallow, including the two-dimensional quotient missed by the projective-additive section.

The correspondence between deep holes, one-symbol MDS extensions, and uncovered points of a normal rational curve was developed in [1, Prop. 1] and [2, Lem. II.4 and Thms. I.4–I.6]; it belongs to the regime in which the covering radius is r, and every radius proved here is $r-1$, where Dür's equivalence instead forbids the extension. See Remark IV.15. Zhang, Wan, and Kaipa classified projective Reed–Solomon deep-hole directions through redundancy four and identified redundancy five as the next case [2]. Theorem I.2 closes that case; Theorem I.3 gives the complete redundancy-six classification and the radius-delimited redundancy-seven result. Their lower tangent and conjugate-secant families remain inputs to the terminal redundancy-five analysis. Once that terminal classification is known, the finite-depth theorem propagates it through marked contractions; the prior lower-redundancy theory thus supplies the base data for the new recursive mechanism.

Thus redundancy five is the first previously open level in the sequence, and Theorem I.2 classifies it over every prime power $q\geq 7$.

Theorem I.2 (Complete redundancy-five classification)**.** Let $q\geq 7$ be a prime power. The covering radius of $\mathrm{PRS}(q-4)$ is 4. Its projective deep-hole syndromes are exactly the following disjoint families:

(i) the tangent family T, of size $q(q+1)$;
(ii) the conjugate-secant family S, of size $q(q^2-1)/2$;
(iii) if $\operatorname{char}\mathbb{F}_q\neq 3$, the rational osculating-pair family O^+ when $q\equiv 2 \pmod 3$, of size $q(q+1)/2$, or the conjugate osculating-pair family O^- when $q\equiv 1 \pmod 3$, of size $q(q-1)/2$;
(iv) in characteristic three, one $\mathrm{PGL}_2(q)$-fixed nucleus point and one wild Artin–Schreier orbit W of size $(q^2-1)/2$;
(v) the sporadic $\mathrm{PGL}_2(q)$-orbits in Table IV, occurring exactly for $q\in\{7,8,9,11,13,17,19\}$.

There are no other deep-hole syndromes. The three corresponding nonsporadic totals are

$$\frac{q^2(q+3)}{2},\qquad \frac{q(q+1)(q+2)}{2},\qquad \frac{q^3+3q^2+q+1}{2},$$

according as $q\equiv 1 \pmod 3$ with characteristic not three, $q\equiv 2 \pmod 3$, or $\operatorname{char}\mathbb{F}_q=3$.

At $q=8$ the classification gives 1116 deep-hole directions, splitting as 360 nonsporadic and 756 sporadic ones.

The Hankel dictionary turns redundancy five into a pencil of cubics. Cyclic descent gives the arithmetic families in Theorem I.2; the S_3 case is controlled by the off-diagonal fiber square and a genus-one point bound. At higher redundancy, coherent polar contraction lowers the degree while retaining the removed root as a forbidden marker. The recursive carrier theorem identifies the polar lines that remain trapped; the fixed-level arguments evaluate the transverse packages and Lucas-carrier arithmetic. Section II gives the short proof route.

Theorem I.3 (Redundancy-six deep holes and redundancy-seven split-free syndromes)**.** The following statements hold.

(a) For every prime power $q \geq 7$, the projective deep-hole directions of $\mathrm{PRS}(q-5)$ are exactly the persistent tangent and conjugate-secant families, the five finite-field exception rows of Theorem A.10, and, when $q = 2^m$ with odd $m \geq 5$, one characteristic-two nucleus orbit.

(b) For every prime power $q \geq 7$, Theorem A.16 gives the complete split-free redundancy-seven classification. For $q \geq 13$ only the persistent families and, when $q = 2^m$ with odd m, one central nucleus point remain. The result is a deep-hole classification for $q \geq 11$. At $q = 8$ the known radius is seven; exact extraction leaves the nine-direction diagonal tangent orbit and the fixed central nucleus, so the classification is complete there as well. The radii at $q = 7, 9$ remain separate.

Table II separates the split-free result, radius gate, deep-hole conclusion, and remaining field gap at every level. The R6 and R7 conclusions are all-field structural classifications: their polar arguments exhaust every field outside a bounded range, and deterministic certificates close the remaining prime powers.

II. Reading map

Here is the argument in one pass. A syndrome f determines a linear system W_f of binary forms through its two-row Hankel matrix. The Hankel criterion says that a completely split squarefree form in W_f is a witness that f lies in the span of the required number of distinct normal-rational-curve points; we call such a syndrome *shallow*. Thus the geometric problem is to find such a form. Only after the separate covering-radius gate does its absence mean that f is a deep-hole syndrome.

At redundancy five, W_f is a pencil of cubics. Its off-diagonal fiber square records ordered pairs of distinct roots lying in one pencil member. The exact count in Proposition IV.6 turns rational points on that genus-one curve into split cubics and makes R5 the terminal model for every later level.

At higher redundancy, choose a root λ as a marker and contract the syndrome. A split witness in the lower system lifts after multiplication by λ, and it stays squarefree precisely when it avoids that marker. As λ varies, the contracted syndromes trace a polar line. There are then two branches. We use *transverse* as shorthand for *noncontained*: such a line meets the lower bad locus in only finitely many marker parameters. After discarding those parameters and finitely many points on the lower splitting cover, a rational-point bound supplies a witness. If the line is contained, the component calculation places the original syndrome in the persistent catalecticant locus or a characteristic-dependent Lucas carrier. The first is the rank-two catalecticant locus carrying the tangent, conjugate-secant, and shallow rational-secant strata. The second is a linear coordinate space forced by adjacent binomial coefficients that vanish in the field characteristic; this is why a new obstruction can appear in small characteristic. The classification of the contained branch is unconditional. Proving that every point outside it escapes additionally requires the pointed lower packages and field-size bounds stated in Theorem I.1. Figure 1 displays this single recursion.

$$\begin{array}{rcl} \text{coding} & : & [f] \longmapsto W_f \supset \text{split squarefree } g \longmapsto [f] \text{ is shallow}, \\ \text{recursion} & : & [f] \xrightarrow{\text{contract at } \lambda} \text{lower splitting problem}, \\ & & \begin{array}{rcl} \text{contained} & \longmapsto & \text{persistent or Lucas carrier}, \\ \text{transverse} & \longmapsto & \text{lower witness } g,\ \lambda \nmid g \longmapsto \lambda g \in W_f. \end{array} \end{array}$$

Fig. 1. The conceptual route through the paper. The first line changes from coding language to a splitting problem. Marked contraction then separates the contained branch, classified by the carrier theorem, from the transverse branch, where finite avoidance and point counting produce the lower witness g.

The proof has one recursive spine and six evaluated levels. Table I shows what changes from one level to the next; its deletion budget is the number of points excluded from the terminal cover. Table II separately records which geometric classifications promote to code deep-hole classifications.

TABLE I

RESULTS AND MECHANISMS. THE CLOSING-GATE COLUMN RECORDS THE FIRST FIELD RANGE IN WHICH THE STATED GEOMETRIC ARGUMENT CLOSES; BOUNDED CERTIFICATES HANDLE ONLY THE LISTED LOWER FIELDS.

Level	New obstruction	Main geometric mechanism	Closing gate
R5	cubic splitting	cubic pencil and off-diagonal $(2,2)$ fiber square; exact fiber-square counting identity	branch correction at most 12, or 6 in characteristic two; geometric threshold 20 and seven-field certificate
R6	one retained marker	one contraction to the same R5 curve	deletion budget 19; geometric threshold 29 and finite bridge
R7	two retained markers	two coherent contractions to the same R5 curve	deletion budget 25; geometric threshold 37 and finite bridge
R8	three retained markers	the same integral $(2,2)$ terminal cover	deletion budget 30; $q \geq 43$
R9	four markers and a characteristic-seven carrier	terminal cover off the carrier; residual-quadratic genus-one slice on it	deletion 36, $q \geq 53$; in characteristic seven, $q = 343$ is geometric and the below-range $q = 49$ row is certified
R10	five markers and the first higher Lucas carrier	odd terminal cover; binary ordered-Hessian genus-one slice; final-pair Artin–Schreier trace test on the carrier	odd: deletion 42, $q \geq 59$; binary: selector bound 40, deletion 23, $q \geq 64$; stated carrier certificates

TABLE II

CLASSIFICATION STATUS, WITH $p = \operatorname{char} \mathbb{F}_q$. A SPLIT-FREE EQUALITY BECOMES A CODE DEEP-HOLE EQUALITY ONLY AFTER THE RADIUS GATE.

Red.	Field range	Split-free geometry	Deep holes and radius	Gap
5	$q \geq 7$	Complete	Complete; $\rho = 4$	None
6	$q \geq 7$	Complete	Complete; $\rho = 5$	None
7	$q \geq 7$	Complete	Complete for $q \geq 11$ and at $q = 8$; there $\rho = 7$ and exactly ten directions in two orbits are deep	Radius at $q = 7, 9$
8	$q \geq 43$	Persistent-only	Complete; $\rho = 7$	Below range
9	$q \geq 53$	Persistent-only	Complete; $\rho = 8$	Below range
10	$q \geq 59$	Persistent-only	Complete; $\rho = 9$	Below range
$r \geq 6$	$q \geq Q_r$	Contained in persistent plus maximal Lucas; persistent-only if $p > r-1$, conditional on the intermediate lower packages	Complete under that hypothesis if $p > r-1$	Stagewise packages; Lucas arithmetic in small characteristic

Here $Q_r = 6r - 15 + \lfloor 2\sqrt{6r-17} \rfloor$. The R7 finite record is exposed only as a four-field exception delta beyond the uniform T/T^6 spine: the full fourteen-field ledger remains in the supplement. At the even diagonal $r = q-1$, equivalently the dimension-two code, [3, Thm. 17(1)] gives $\rho = r$, and [4] gives the corresponding explicit deep family. In the syndrome coordinates used here its distinguished direction is e_{r-2}: a split degree-$(r-1)$ locator would require a $(q-2)$-subset of $\mathbb{F}_q$ of sum zero, whose two-element complement would consist of distinct elements with zero sum, impossible in characteristic two. This supplies an explicit deep direction at every such redundancy (for example R15 over $\mathbb{F}_{16}$); it is an imported diagonal family, not a complete higher-redundancy classification. At $q = 8$/R7 the finite extraction is complete: among the 46 frozen nonpersistent semilinear representatives, only the fixed central nucleus has distance seven, and among the four persistent representatives only the nine-direction diagonal tangent orbit does. Thus there are exactly ten projective deep-hole directions.

Relation to previous work. The tangent and conjugate-secant families and the coding/normal-rational curve dictionary come from [1, Prop. 1] and [2, Lem. II.4 and Thms. I.4–I.6]; the covering-radius input in the high-rate range combines the Seroussi–Roth extension theorem [5, Thm. 1] with Dür's completeness–covering-radius equivalence [6], as stated by Kaipa [1, Sec. IV and Prop. 4(1)]. The broader arc/MDS and code-automorphism background is surveyed or established in [7], [8]. Earlier explicit-family results for projective and generalized projective Reed–Solomon codes include [9], [10], [4]; the recent extension framework of [3] further clarifies the deep-hole/MDS-extension interface. These works construct families, treat punctured or lower-redundancy regimes, or analyze the extension criterion; they do not supply the complete redundancy-five cubic-pencil exhaustion proved here. Classical apolarity, catalecticants, and binary-form rank strata are treated in [11], [12]; our use of divided powers keeps the small-characteristic contraction formulas explicit.

A second body of work meets this paper at redundancy five rather than at its coding statement. A redundancy-five witness system is a pencil of binary cubics, that is a line of $\mathrm{PG}(3, q)$ relative to the twisted cubic, and split-freeness asks that the pencil contain no completely split squarefree member, equivalently that the line lie in no plane meeting the curve in three distinct rational points. In that form the question belongs to the twisted-cubic line-orbit literature and has been answered there. Blokhuis, Pellikaan, and Szőnyi partition the lines into classes and settle, for $q \geq 23$ and in every characteristic, which classes lie in such a plane, by way of the double point scheme of the associated degree-three rational function [13, Thm. 8.1, Prop. 8.4 and Rem. 7.12]; the same table is confirmed by the classification of degree-three permutation rational functions [14]. Exact incidence counts, rather than the vanishing test, are supplied for the non-generic classes by Davydov, Marcugini, and Pambianco [15], [16], [17] and by Günay and Lavrauw [18], and for the generic class by Kaipa, Patanker, and Pradhan [19] and Kaipa and Pradhan [20], whose Proposition 4.5 and Theorem 5.1 express the count through an associated binary quartic and elliptic curve; the generic class in characteristic three is [21] and in characteristic two is Ceria and Pavese [22]. Section IV states the correspondence class by class in Remark IV.14. Blokhuis–Pellikaan–Szőnyi formulate the codimension-four generalized Reed–Solomon problem in syndrome and coset-leader language and compute its extended coset-leader weight enumerator. What redundancy five adds here is the passage from a syndrome to its pencil: the divided-power Hankel dictionary, the resulting restriction on which lines occur, the classification of split-free *syndromes* with their orbit sizes, stabilizers, and Frobenius fusion, and the promotion of split-free to deep through the radius gate.

Three companion papers provide comparisons with relative conic completion, inverse rigidity, and conic-ideal factorization memory [23], [24], [25]; none is an input here. The MDS–CSS companion [26] proves general rigidity and conversion theorems for MDS–CSS codes; this paper draws no consequence from them, since the one-column extensions that would have supplied their input do not exist at these radii (Remark IV.15).

From the viewpoint of binary forms, the question is which linear systems contain a totally split squarefree member. Factorization statistics in linear families [27] and normal-rational-curve nuclei [28, Lem. 1 and Thm. 1] are inputs. For splitting families, we use Wang's Theorem 3.6 [29]: on the étale locus, the factorization type of a specialized polynomial is the cycle type of its Frobenius conjugacy class. The coding-theory references above identify deep syndromes with uncovered points of a normal rational curve and settle the lower-redundancy families; they do not classify the redundancy-five cubic pencils. The orbit-theoretic and factorization references supply normal forms and distribution tools, but not the coherent marked contraction used here. The nucleus results locate characteristic-dependent kernels, while the present work computes which of their coherent lifts meet the PRS splitting incidence. Accordingly, the mathematical spine consists of the redundancy-five terminal classification, finite-depth coherent polar escape, the reduced recursive carrier theorem, and the arithmetic evaluation of that carrier through redundancy ten. Wang–Wu–Hu's projective-subline criterion is imported at its exact endpoint; the adjacent-zero carrier, its coherent nesting, and the final-pair split-squarefree classification are the paper-owned steps.

III. The syndrome–Hankel dictionary

Let E be a two-dimensional vector space. We use divided powers for syndromes and ordinary symmetric powers for root forms. This avoids invalid binomial rescalings in small characteristic. For $f = (a_0 : \cdots : a_{r-1}) \in \mathbb{P}(\Gamma^{r-1}E)$, write $e_0, \ldots, e_{r-1}$ for the divided-power coordinate basis and $\nu(t) = (1, t, \ldots, t^{r-1})$, with $\nu(\infty) = e_{r-1}$. If x, y is a basis of E with dual basis X, Y, our perfect divided-power/symmetric-power pairing is

$$\left\langle x^{[n-i]}y^{[i]}, X^{n-j}Y^j \right\rangle = \delta_{ij}, \qquad \left\langle \sum_{i=0}^{n} a_i x^{[n-i]}y^{[i]}, \sum_{i=0}^{n} b_i X^{n-i}Y^i \right\rangle = \sum_{i=0}^{n} a_i b_i.$$

Thus the pairing is coefficient extraction, with no binomial factors in any characteristic. Define

$$H_f = \begin{pmatrix} a_0 & a_1 & \cdots & a_{r-2} \\ a_1 & a_2 & \cdots & a_{r-1} \end{pmatrix}, \qquad W_f = \ker H_f \subset \operatorname{Sym}^{r-2} E^\vee.$$

TABLE III
NOTATION USED FROM THE HANKEL DICTIONARY ONWARD.

Symbol	Meaning
q	field size; all numerical thresholds are thresholds on prime powers
k	code dimension in $\mathrm{PRS}(k)$
$r = q + 1 - k$	redundancy
$n = r - 1$	divided-power degree of a syndrome representative
E	a two-dimensional vector space over the current base field
$f \in \Gamma^n E$	divided-power representative of a projective syndrome
$W_f \subset \operatorname{Sym}^{n-1} E^\vee$	Hankel/apolar witness system annihilated by f
g, δ, κ	genus, deletion degree, and point-bound correction
d	degree of the unavailable marker scheme

Read $f = (a_0, \ldots, a_{r-1})$ as a finite sequence. A form $g \in W_f$ is the characteristic polynomial of a recurrence satisfied by that sequence, while a squarefree split g has a basis of geometric-sequence solutions. Thus split-free means that the syndrome sequence satisfies no order-$(r-2)$ recurrence with squarefree completely split characteristic polynomial.

Lemma III.1 (Hankel criterion)**.** Let $\lambda_1, \ldots, \lambda_{r-2}$ be distinct points of $\mathbb{P}(E^\vee)(\mathbb{F}_q)$ and put $g = \prod_i \lambda_i$. Then

$$f \in \operatorname{span}\bigl(\nu(\lambda_1), \ldots, \nu(\lambda_{r-2})\bigr) \iff g \in W_f.$$

The same statement with repeated factors and Hasse derivatives gives the corresponding osculating spans. Both statements commute with $\mathrm{P\Gamma L}_2(q)$.

Proof. On an affine chart, the geometric sequences $\nu(\lambda_i)$ solve the recurrence with characteristic polynomial g; their independence is the Vandermonde determinant. A root at infinity gives the reversed-chart recurrence. For repeated roots, the confluent Vandermonde basis is formed by Hasse derivatives, so the same argument works in every characteristic. Equivariance follows because substitution transports both the normal rational curve and the recurrence. □

Corollary III.2 (Split-free criterion)**.** Assume $\rho(\mathrm{PRS}(q+1-r)) = r-1$. A projective syndrome f is deep if and only if W_f contains no completely split squarefree form of degree $r-2$ over $\mathbb{F}_q$.

We use *split-free* without a covering-radius hypothesis and call a direction *shallow* when W_f contains a completely split squarefree form.

The first locus that persists under contraction is already visible here. If every member of W_f has a common quadratic factor, then that factor is either a rational square, an irreducible quadratic, or a product of two distinct rational factors. These are respectively the tangent deep family, the conjugate-secant deep family, and a shallow rational secant. The common factor cannot grow with redundancy. If H_f has rank one, then $\dim W_f = r-2$, which is incompatible with a common quadratic factor because the space of degree-$(r-2)$ forms divisible by that factor has dimension $r-3$. Otherwise $\dim W_f = r-3$, and

$$r - 3 \le r - 1 - \deg \gcd(W_f),$$

so its degree is at most two.

A. Coding consequences and recognition

The dictionary has two immediate coding consequences. First, a split-free direction is a deep hole whenever the stated covering-radius gate applies, so the classification theorems give exact projective deep-hole counts and not only normal forms. It is not a one-column MDS extension: an extension needs a point outside the span of every $r-1$ columns, while split-freeness gives only $r-2$, and at radius $r-1$ no point of the larger kind exists. Dür's completeness–covering-radius equivalence makes that a biconditional, so each radius gate proved here simultaneously rules the corresponding extensions out; see Remark IV.15. Second, they give recognition procedures. From a syndrome one forms H_f, computes W_f, and tests the listed geometric families: a common quadratic factor detects the persistent tangent and conjugate-secant families; the relevant contraction kernel detects the modular locus; and the finite exceptional fields are decided by the canonical orbit representatives printed in Table IV and recorded in the supplement.

Recognition at the fixed redundancies proved here begins with row-reduction of H_f and the listed family-membership tests. Coherent polar induction replaces a direct search through the projective members of W_f.

IV. Redundancy five: exceptional cubic covers

Here $f = (a_0 : \cdots : a_4)$ and W_f is the pencil annihilated by

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \end{pmatrix}.$$

These are divided-power coordinates; the coefficients of a cubic in W_f are ordinary symmetric-power coordinates. Factoring the formal quartic $\sum a_i T^i U^{4-i}$ would therefore be noninvariant in characteristics two and three.

a) The criterion in finite-geometric form: At $r = 5$ the witness system $W_f \subset \mathrm{Sym}^3 E^\vee$ is a pencil of binary cubics, that is a line of $\mathrm{PG}(3,q)$ relative to the twisted cubic; and a completely split squarefree cubic is a point of the $\mathrm{PGL}_2(q)$-orbit $\mathcal{O}_3 = \mathrm{PGL}_2(q) \cdot XY(X-Y)$ of size $(q^3-q)/6$, equivalently a plane meeting the twisted cubic in three distinct rational points [13, Prop. 4.1 and Rem. 4.2], [20, Lem. 4.1]. Corollary III.2 therefore reads

$$f \text{ split-free} \iff W_f \cap \mathcal{O}_3 = \varnothing. \tag{2}$$

In this form the redundancy-five question is a point-orbit distribution question for line orbits of $\mathrm{PG}(3,q)$, and as such it has been answered in the finite-geometry literature; the relation is set out in Remark IV.14. What that literature does not supply, and what this section adds, is the passage from a redundancy-five syndrome to its pencil: which lines arise as some W_f, and which *syndromes* are thereby split-free.

Proposition IV.1 (Radius gate)**.** For every prime power $q \ge 7$,

$$\rho(\mathrm{PRS}(q-4)) = 4.$$

Proof. The coding/arc criterion identifies radius four with completeness of the quartic normal rational curve. The Seroussi–Roth extension theorem [5, Thm. 1], applied to the dual five-dimensional GRS code, gives nonextendability when

$$5 \le \left\lfloor \frac{q}{2} \right\rfloor + 2,$$

apart from its even-field dimension-three exception. The inequality holds for $q \ge 7$, and the exception does not occur. Dür's completeness–covering-radius equivalence [6], as stated by Kaipa [1, Sec. IV and Prop. 4(1)], now gives radius four. Thus Corollary III.2 applies. The finite certificate independently checks the same conclusion only in its declared finite domain; it is not the all-field radius proof. ☐

A. Gcd strata

Proposition IV.2 (Quadratic gcd)**.** If $\deg \gcd(W_f) = 2$, then

$$W_f = Q\langle T, U\rangle.$$

If Q is split squarefree, f is shallow. If Q is irreducible, f belongs to the conjugate-secant family; if $Q = \lambda^2$, it belongs to the tangent family. Conversely, every point in those families has the displayed gcd. Their deep-family sizes are

$$|\mathrm{T}| = q(q+1), \qquad |\mathrm{S}| = \frac{q(q^2-1)}{2}.$$

Proof. The equality follows by dimension. A split Q and a linear cofactor avoiding its two roots give a split squarefree cubic. An irreducible Q survives in every member, while a square Q forces a repeated root in every member. Lemma III.1, including its confluent and conjugate-root forms, identifies the corresponding syndromes with the secant and tangent spans. The sizes are the established tangent/conjugate-secant counts from the lower-redundancy classification [1, Prop. 1] and [2, Thms. I.4–I.6]. ☐

Proposition IV.3 (Linear gcd)**.** Suppose $\deg \gcd(W_f) = 1$. Then f is shallow for $q \ge 8$. At $q = 7$ the same conclusion is part of Certificate R5.

Proof. Write $W_f = \lambda_0 V$ with V a base-point-free quadratic pencil. If its degree-two map $\psi : \mathbb{P}^1 \to \mathbb{P}^1$ is separable, its rational deck involution ι gives the graph

$$\Gamma_\iota(\mathbb{F}_q) = \{(t, \iota(t)) : t \in \mathbb{P}^1(\mathbb{F}_q)\},$$

which has exactly $q+1$ rational points. A safe union bound deletes at most two fixed graph points, at most four graph points over the branch fibers, and at most the two graph points $(\lambda_0, \iota(\lambda_0))$ and $(\iota(\lambda_0), \lambda_0)$. Thus fewer than $q+1$ graph points are deleted when $q \ge 8$. A surviving point gives a split quadratic with two distinct roots avoiding λ_0, and hence a split squarefree cubic in W_f. In characteristic two an inseparable quadratic pencil is equivalent to $\langle T^2, U^2\rangle$; the two overlapping Hankel equations then force f onto the normal rational curve, contrary to the rank-two pencil hypothesis. ☐

B. The trivial-gcd cubic cover

Choose a basis c_1, c_2 of a trivial-gcd pencil and put

$$X_f = \{([u:v], t) : uc_1(t) + vc_2(t) = 0\} \subset \mathbb{P}^1 \times \mathbb{P}^1.$$

Proposition IV.4 (Cubic incidence and off-diagonal fiber square)**.** The curve X_f is geometrically integral of bidegree $(1,3)$ and arithmetic genus zero. Projection to the root coordinate is birational, while projection to the pencil coordinate gives a degree-three morphism

$$\phi_f : \mathbb{P}^1 \longrightarrow \mathbb{P}^1.$$

If ϕ_f is separable, division of

$$c_1(t_1)c_2(t_2) - c_2(t_1)c_1(t_2)$$

by the diagonal bracket defines a residual curve $Y_f \subset \mathbb{P}^1 \times \mathbb{P}^1$ of bidegree $(2,2)$ and arithmetic genus one. Its geometric components are the geometric-monodromy orbits on ordered pairs of distinct roots.

Proof. A component of X_f of bidegree $(0,e)$ would be a common factor of c_1, c_2; hence none exists. The bidegree genus formula gives genus zero. A root determines the unique pencil member containing it, so the root projection is birational. For a separable map the diagonal factor in the fiber-square determinant has multiplicity one. Removing its bidegree $(1,1)$ from the fiber square leaves bidegree $(2,2)$, whose arithmetic genus is one. The standard fiber-square/monodromy correspondence identifies its components with orbits on distinct ordered sheets. ☐

Lemma IV.5 (Cyclic stratum)**.** Suppose ϕ_f is geometrically cyclic.

(i) In characteristic different from three there are two ramification points. If they are rational, the corresponding osculating-plane intersection is deep exactly when $q \equiv 2 \pmod 3$; if they are conjugate, it is deep exactly when $q \equiv 1 \pmod 3$.

(ii) In characteristic three all osculating planes meet in $n = (0:0:1:0:0)$, whose pencil is $\langle T^3, U^3\rangle$; this fixed point is deep.
(iii) The remaining wild characteristic-three pencils have the form $\langle U^3, T^3 + aTU^2\rangle$. They are deep exactly when $-a$ is nonsquare, forming one orbit of size $(q^2-1)/2$.

Proof. Tame characteristic. Assume that the characteristic is not three. Riemann–Hurwitz gives two totally ramified points t_1, t_2. The confluent Hankel criterion places f in $\operatorname{Osc}(t_1) \cap \operatorname{Osc}(t_2)$; after transporting the pair to $\{0, \infty\}$, the pencil is $\langle T^3, U^3\rangle$ and a geometric deck generator is $t \mapsto \zeta t$.

For a rational ramification pair, Frobenius fixes the points and the deck map descends exactly when $\zeta^q = \zeta$, namely $q \equiv 1 \pmod 3$. Thus its complement is deep exactly for $q \equiv 2 \pmod 3$. The setwise stabilizer of the ordered normal form is the split dihedral group of order $2(q-1)$, so the orbit size is

$$\frac{q(q^2-1)}{2(q-1)} = \frac{q(q+1)}{2}.$$

For a conjugate pair Frobenius interchanges the fixed points and inverts the multiplier; descent is instead $q \equiv 2 \pmod 3$. The nonsplit dihedral stabilizer has order $2(q+1)$ and the orbit size is $q(q-1)/2$.

The fixed wild point. Suppose the characteristic is three. The second Hasse derivative of

$$\nu(t) = (1, t, t^2, t^3, t^4)$$

is $(0, 0, 1, 3t, 6t^2) = e_2$, including in the reversed infinity chart. Thus every osculating plane contains $n = [e_2]$. Substitution shows that n is $\mathrm{PGL}_2(q)$-fixed and $W_n = \langle T^3, U^3\rangle$, so it is deep.

The remaining wild orbit. A remaining separable cyclic cover has one wild totally ramified point. Sending it to infinity and comparing coefficients under a nontrivial translation gives

$$W_f = \langle U^3, T^3 + \alpha T^2 U + \beta T U^2\rangle, \qquad \alpha = 0, \quad \kappa^2 = -\beta.$$

It is therefore represented by

$$f_a = e_2 - a e_4, \qquad W_{f_a} = \langle U^3, T^3 + aTU^2\rangle.$$

The affine members are $z^3 + az + s$. Such a member has three rational roots exactly when the additive map $z \mapsto z^3 + az$ has a nonzero rational kernel, equivalently when $-a$ is a square. Hence the nonsquare parameters form the deep orbit. Its stabilizer is $t \mapsto \pm t + \beta$, of order $2q$, and orbit–stabilizer gives $(q^2-1)/2$. ☐

The count of split members of the pencil is not merely positive or zero: it is determined exactly by the fiber square, and the following identity is what drives both the threshold and the shape of the exceptions.

Proposition IV.6 (Exact split-witness count)**.** Let $\deg \gcd(W_f) = 0$ and let ϕ_f be separable; the characteristic is arbitrary. Write

- N_f for the number of completely split squarefree members of W_f,
- d_2 for the number of members with a rational double root and a distinct rational simple root,
- d_3 for the number of members that are perfect cubes of a rational linear form.

Then

$$\#Y_f(\mathbb{F}_q) = 6N_f + 3d_2 + d_3. \tag{3}$$

In particular f is split-free if and only if $\#Y_f(\mathbb{F}_q) = 3d_2 + d_3$.

Proof. Each $x \in \mathbb{P}^1(\mathbb{F}_q)$ lies on exactly one member g_x of the pencil, because the pencil is base-point-free; write $g_x = \ell_x h_x$ with ℓ_x the linear form at x. The fiber of Y_f over x under the first projection is the set of rational roots of h_x, so $\#Y_f(\mathbb{F}_q) = \sum_x \#\{\text{rational roots of } h_x\}$, and it suffices to total that sum member by member. A member with three distinct rational roots is met at each of them, and there h_x has the two remaining roots, contributing $3 \cdot 2 = 6$. A member with a rational double root r and distinct rational simple root s is met at both: over r the residual is $\ell_r \ell_s$ with two distinct rational roots, over s it is ℓ_r^2 with one, contributing $2 + 1 = 3$. A perfect cube is met only at its root, where the residual is a square, contributing 1. A member with one rational root and an irreducible quadratic cofactor contributes 0 there, and a member with no rational root is met at no rational x. Summing gives (3). ☐

The proof exhibits the fiber square as a double cover, and that is what identifies it with the genus-one curve of the incidence literature.

Proposition IV.7 (The fiber square is the incidence curve)**.** Keep the hypotheses of Proposition IV.6, assume $\operatorname{char} \mathbb{F}_q \neq 2$, and let $D_f(x) = \operatorname{disc}(h_x)/4$. The restriction is needed here and not in Proposition IV.6: in characteristic two the quotient by four is undefined, and a separable quadratic cover is Artin–Schreier rather than the Kummer cover displayed below. The first projection realizes Y_f as the double cover

$$w^2 = D_f(x)$$

of $\mathbb{P}^1$, branched where D_f vanishes. If moreover W_f misses the twisted cubic and $\operatorname{char}\mathbb{F}_q \neq 2, 3$, then D_f is the discriminant quartic that Kaipa and Pradhan attach to the line, so the normalization of Y_f is their elliptic curve E_{W_f} and

$$\#Y_f(\mathbb{F}_q) = \#E_{W_f}(\mathbb{F}_q).$$

Proof. The fiber of Y_f over x under the first projection is the root set of the residual quadratic h_x, and its two roots are interchanged by the choice of a square root of $\operatorname{disc}(h_x)$; so the projection is the stated double cover, and it is branched exactly where that discriminant vanishes. Kaipa and Pradhan form the discriminant of the same residual quadratic and define E_L as the nonsingular model of $w^2 = D_L$ [20, §4–§5], taking as we do one quarter of the discriminant of the same residual quadratic. They therefore differ by the square of the nonzero rational function $\lambda \in \mathbb{F}_q(x)^\times$ that relates the two normalizations of the residual quadratic,

$$D_f = \lambda^2 D_L \quad \text{in } \mathbb{F}_q(x)^\times,$$

with equality when the representative of g_x is normalized as they normalize it. Equality of square classes in the function field, rather than pointwise at rational x, is what rules out a quadratic twist and leaves the cover unchanged. For the point counts, note that on that stratum D_f has nonzero discriminant, so the double cover displayed above is already smooth and is its own normalization, and Y_f is smooth there as well; the two smooth projective models are then isomorphic and their counts agree. Away from that stratum the identification is birational and the counts may differ at a rational singularity, which is why the exact count of Proposition IV.6 is stated for $\#Y_f(\mathbb{F}_q)$ rather than for the count on a normalization. ☐

Lemma IV.8 (Branch budget)**.** Under the hypotheses of Proposition IV.6,

$$d_2 + 2d_3 \leq 4,$$

so a split-free f has $\#Y_f(\mathbb{F}_q) \leq 12$, with equality only when $d_2 = 4$ and $d_3 = 0$.

Proof. Riemann–Hurwitz gives different degree four for the separable degree-three map ϕ_f. A member with a double root has one ramification point of index two; it spends different degree one when that ramification is tame and at least two when it is wild, which happens exactly in characteristic two. We avoid the term *simple ramification*, which is usually taken to include the tame exponent. A member that is a perfect cube is totally ramified; it spends two when tame and at least three when wild, which happens exactly in characteristic three. Distinct members are distinct branch values, so $d_2 + 2d_3 \leq 4$ in every characteristic, and characteristic two gives the sharper $d_2 + d_3 \leq 2$. Substituting into $\#Y_f(\mathbb{F}_q) = 3d_2 + d_3$ and maximizing over the integer points of that region gives 12, attained only at $(d_2, d_3) = (4, 0)$. ☐

Corollary IV.9 (Split members of an S_3 pencil)**.** If ϕ_f has geometric monodromy S_3, then

$$\frac{q+1-2\sqrt{q}-B}{6} \;\leq\; N_f \;\leq\; \frac{q+1+2\sqrt{q}}{6},$$

where $B = 12$, and $B = 6$ when $\operatorname{char}\mathbb{F}_q = 2$.

Proof. Proposition IV.6 gives $6N_f = \#Y_f(\mathbb{F}_q) - 3d_2 - d_3$, and Lemma IV.8 bounds $3d_2 + d_3$ above by 12 in general and by 6 in characteristic two, where $d_2 + d_3 \leq 2$; it is bounded below by 0 in every characteristic. The S_3 hypothesis makes Y_f geometrically integral of arithmetic genus one, so Aubry–Perret confines $\#Y_f(\mathbb{F}_q)$ to $q + 1 \pm 2\sqrt{q}$ [30, p. 468]. ☐

Corollary IV.10 (Binary S_3 shallowness above $q = 8$)**.** If $\operatorname{char}\mathbb{F}_q = 2$ and ϕ_f has geometric monodromy S_3, then f is shallow for every $q \geq 16$.

Proof. A split-free f has $\#Y_f(\mathbb{F}_q) = 3d_2 + d_3 \leq 6$ by Lemma IV.8, while Aubry–Perret gives $\#Y_f(\mathbb{F}_q) \geq q + 1 - 2\sqrt{q}$. For $q = 16$ the latter is 9, and it increases with q, so no binary field with $q \geq 16$ admits a split-free S_3 pencil. ☐

The identity class has S_3 splitting density $1/6$, so $(q+1)/6$ is the expected count. A completely split fiber is one whose Frobenius class is the identity, and that class has density $1/|S_3| = 1/6$ on the unramified locus, by the factorization-type correspondence of [29, Thm. 3.6] together with the Chebotarev count and density of [29, Thm. 3.7 and Cor. 3.8], whose geometric Galois group is S_3 under the present hypothesis; so $(q+1)/6$ is a Chebotarev main term rather than a count of fibers. Corollary IV.9 makes it quantitative: split members occur with that expected density up to a Weil-scale error and a bounded ramification correction. Split-freeness is the extreme small-field case in which the lower bound has turned nonpositive, which is why the exceptional fields are small ones. Lemma IV.11 records only the sign of this statement. We claim the density only for the completely split class; the other factorization types are not counted here.

Lemma IV.11 (S_3 stratum)**.** If ϕ_f has geometric monodromy S_3 and $q \geq 20$, then the pencil contains a completely split squarefree cubic. For odd q the same conclusion holds for every $q \geq 20$ without the monodromy hypothesis whenever Y_f is geometrically integral.

Proof. By Proposition IV.4, components of Y_f correspond to monodromy orbits on distinct ordered roots. The S_3 action is two-transitive, so Y_f is geometrically integral and has arithmetic genus one. The Aubry–Perret singular-curve bound, in the form

$$\left|\#C(\mathbb{F}_q) - (q+1)\right| \le 2\pi_C\sqrt{q}$$

for a reduced geometrically integral projective curve of arithmetic genus π_C, gives

$$\#Y_f(\mathbb{F}_q) \ge q + 1 - 2\sqrt{q}$$

[30, p. 468]. Suppose f were split-free. Then $\#Y_f(\mathbb{F}_q) \le 12$ by Lemma IV.8, so $q + 1 - 2\sqrt{q} \le 12$, that is $(\sqrt{q} - 1)^2 \le 12$ and $q \le (1 + 2\sqrt{3})^2 < 20$. ☐

Corollary IV.12 (Forced fiber-square invariants)**.** Let q be odd with $17 \le q \le 19$ and let f be split-free with Y_f geometrically integral. Then

$$\#Y_f(\mathbb{F}_q) = 12, \qquad d_2 = 4, \qquad d_3 = 0.$$

Proof. Lemma IV.8 restricts $\#Y_f(\mathbb{F}_q) = 3d_2 + d_3$ to the values $0, 1, 2, 3, 4, 6, 7, 9, 12$. Aubry–Perret forces $\#Y_f(\mathbb{F}_q) \ge q + 1 - 2\sqrt{q}$, which exceeds 9 for $q \ge 17$; the only admissible value left is 12, and by Lemma IV.8 that value is attained only at $(d_2, d_3) = (4, 0)$. ☐

Proposition IV.6 is a counting identity valid in every characteristic, and the quantitative splitting law of Corollary IV.9 follows from it directly. In the tame range it is equivalent to an incidence count in the twisted-cubic line literature. Precisely: when W_f misses the twisted cubic, so that $d_3 = 0$, and $\operatorname{char}\mathbb{F}_q \ne 2, 3$, the quantity d_2 is the invariant η_L attached to the binary quartic of the line and $\#Y_f(\mathbb{F}_q)$ is the point count of the associated elliptic curve, so that (3) reads

$$N_f = \frac{\#E_L(\mathbb{F}_q) - 3\eta_L}{6},$$

which is the incidence count of [20]. That count should be taken in the form derived in their proof, namely $|S \cap \mathcal{O}_3| = (\nu_L - \eta_L)/3$ together with $\nu_L = (\#E_L(\mathbb{F}_q) - \eta_L)/2$ [20, Prop. 4.5(3) and Thm. 5.1]; the denominator printed in the displayed statement of their Theorem 1.3(3) is 3, which is a typographical slip, since 6 is what their proof gives and what makes their five incidence counts sum to $q+1$. Proposition IV.6 itself drops the restriction $\operatorname{char}\mathbb{F}_q \ne 2, 3$ altogether, brings in lines meeting the curve through the d_3 term, and needs no invariant theory of quartics: its proof counts rational roots in fibers and uses neither a discriminant nor a division by two. What does stay inside $\operatorname{char}\mathbb{F}_q \ne 2, 3$ is the comparison itself, through Proposition IV.7, since the elliptic model is built from a discriminant. An exhaustive check of (3) and of Lemma IV.8 over every point of $\mathrm{PG}(4, q)$ for the prime powers $4 \le q \le 32$, in every characteristic, is recorded in the supplement.

Proposition IV.13 (Exhaustion and finite bridge)**.** The preceding cases exhaust every redundancy-five pencil for $q \ge 20$, and in characteristic two for $q \ge 16$ by Corollary IV.10. The finite residue is therefore

$$q \in \{7, 8, 9, 11, 13, 17, 19\},$$

which Certificate R5 closes. The certificate additionally covers $q = 16$, where it finds no sporadic orbit; that field is no longer logically certificate-dependent, and the computation is retained as an independent regression check. Sporadic S_3-stratum orbits occur exactly at $q = 7, 8, 9, 11, 13, 17, 19$.

Proof. The gcd degree is zero, one, or two. Propositions IV.2 and IV.3 settle the positive-gcd cases. In the trivial-gcd case the cubic map is inseparable or separable. The inseparable case occurs only in characteristic three and the overlap equations give the all-cube nucleus pencil. A separable transitive cubic has geometric monodromy C_3 or S_3, settled by Lemmas IV.5 and IV.11. The stated finite residue, canonical representatives, stabilizers, histograms, and Frobenius links are exactly the domain of Certificate R5. ☐

Remark IV.14 (Relation to the twisted-cubic line literature). Two conventions for "the line of $\mathrm{PG}(3, q)$ attached to a pencil of binary cubics" are in use, and they are exchanged by a polarity, so the class names below must be read in the right one. Günay and Lavrauw, and Blokhuis, Pellikaan, and Szőnyi, send a cubic to a *plane*: the twisted cubic is a curve of points, a cubic is the plane meeting it in that cubic's divisor, and a pencil of cubics becomes a pencil of planes, whose axis is the line [18, §2], [13, Prop. 6.5]. Kaipa and Pradhan send a cubic to a *point*, so that the twisted cubic consists of the perfect cubes and the pencil *is* the line [20, §1]. The bridge is the symplectic polarity σ of $\mathrm{PG}(3, q)$, which carries each point of the twisted cubic to its osculating plane and carries the chords of the twisted cubic to the axes of the osculating developable [18, §2].

For us the translation is concrete. The osculating plane at the point λ^3 of the twisted cubic is $\lambda \operatorname{Sym}^2 E^\vee$, so

$$\operatorname{Osc}(\lambda_1) \cap \operatorname{Osc}(\lambda_2) = \lambda_1\lambda_2\langle T, U\rangle$$

is exactly a quadratic-gcd pencil, which is therefore an axis in the point convention; by [18, Lem. 2] its polar is the chord joining λ_1^3 and λ_2^3, which is what the plane convention calls it. Symmetrically $\langle T^3, U^3\rangle$ joins two points of the twisted cubic and so is a chord in the point convention and an axis in the plane convention. The class names in this remark follow the plane convention of the source being cited; the split members of a pencil are of course the same either way.

Under (2) every stratum above is a class of lines of $\mathrm{PG}(3,q)$ in the sense of Blokhuis, Pellikaan, and Szőnyi, who partition the lines into classes $O_1,\dots,O_8$ and determine for $q \geq 23$, in every characteristic, which classes lie in a plane meeting the twisted cubic in three rational points [13, Thm. 8.1 and Prop. 8.4]. The correspondence is exact. Our quadratic-gcd trichotomy of Proposition IV.2 is their degree-one case: real chords O_1 are shallow, tangents O_2 give the tangent family, imaginary chords O_3 give the conjugate-secant family. The linear-gcd unisecants of Proposition IV.3 are their O_4 and O_5. The two cyclic cases of Lemma IV.5(i) are their real axes O_1' and imaginary axes O_3', with the same congruences mod three; the characteristic-three nucleus of Lemma IV.5(ii) is their axis $O_7(3)$; the wild pencils of Lemma IV.5(iii) are their $O_{8.1}(3)^{\pm}$ and $O_{8.2}(3)$. Lemma IV.11 is their O_6, and their Remark 7.12 reaches the threshold $q \geq 23$ by the same route: the double point scheme of the degree-three map is a genus-one curve, at most twelve points must be excluded, and the Hasse–Weil bound then produces a fiber with three distinct rational points. Two differences remain. Aubry–Perret in place of Hasse–Weil drops their simplicity hypothesis, which is what lets one estimate carry the singular and wild cases uniformly; and the exact count of Proposition IV.6 replaces the deletion bookkeeping by an identity, which lowers the threshold to $q \geq 20$ and, through Corollary IV.12, pins the fiber square of every remaining exception at $q = 17, 19$. Independent confirmation of the same table comes from the classification of degree-three permutation rational functions [14], as recorded in [13, Rem. 8.6].

Exact counts, rather than the vanishing test, are available too. For the ten non-generic line classes Davydov, Marcugini, and Pambianco give the exact number of planes of each orbit through each line for $q \geq 5$ [16, Thm. 3.3], building on their line-orbit classification [17], [15]; Günay and Lavrauw give the point-orbit and plane-orbit distributions of the same classes for odd q prime to three [18]. For the generic class and $\mathrm{char}\,\mathbb{F}_q \neq 2,3$, Kaipa and Pradhan attach to a line L a binary quartic φ_L and an elliptic curve E_L, and count $|W_f \cap \mathcal{O}_3|$ in terms of $\#E_L(\mathbb{F}_q)$ and the invariant $\eta_L \in \{0,1,2,4\}$ recording the rational factorization type of φ_L. That count is the specialization of Proposition IV.6 discussed after Corollary IV.12, and Proposition IV.7 identifies their curve with our fiber square. By (2) a generic line is split-free exactly when $\#E_L(\mathbb{F}_q) = 3\eta_L \leq 12$, which the Hasse bound $\#E_L(\mathbb{F}_q) \geq q + 1 - 2\sqrt{q}$ forbids for $q \geq 20$ and permits only for $q \leq 19$. This is not a second route to Lemma IV.11 but the same one in other coordinates: Proposition IV.7 identifies E_L with Y_f, so both arguments apply a Weil-type bound to a single genus-one curve. What the elliptic form adds is not independence but an explanation, since $3\eta_L \leq 12$ is visibly a bounded quantity while $\#E_L(\mathbb{F}_q)$ grows with q. The generic class is settled in characteristic three in [21] and in characteristic two by Ceria and Pavese [22].

What this section contributes is therefore not the pencil-level classification but its redundancy-five syndrome-level source and consequence. Blokhuis–Pellikaan–Szőnyi formulate the codimension-four generalized Reed–Solomon code in syndrome/coset language and determine its extended coset-leader weight enumerator; their line criterion does not supply the map from a redundancy-five syndrome to the pencil classified here. The passage $f \mapsto W_f$ of Lemma III.1 is not present in that literature, and it is not surjective onto the lines: the characteristic-two inseparable unisecants, split-free as lines, are realized by no rank-two syndrome, as the proof of Proposition IV.3 shows. Neither do incidence distributions produce syndromes; Table IV records canonical representatives in divided-power syndrome coordinates with their orbit sizes, stabilizers, and Frobenius fusion, and extracting such a list from [20, Prop. 4.5 and Thm. 5.1] would still require evaluating $\#E_L$ and η_L over all $2q-3+\mu$ generic line orbits. The promotion of this redundancy-five syndrome classification from split-free to deep, through Proposition IV.1, is also not supplied by the cited line-incidence classifications.

Redundancy five is the terminal model for the transverse branch of the recursion. Its proof separates those later tasks cleanly: identify the lower splitting curve, remove its singular, branch, diagonal, and marker incidences, apply a rational-point bound, and read a surviving point as a split witness. Higher redundancy changes how one reaches that curve and which polar lines remain trapped. The contained branch instead requires the component classification and carrier arithmetic developed below.

TABLE IV: Complete sporadic orbit inventory at redundancy five. A representative $[a_0, \ldots, a_4]$ is in the divided-power syndrome coordinates of Section III. The last column records the nontrivial Frobenius fusion; a dash means the orbit is fixed.

q	representative	orbit size	stabilizer	Frobenius
7	`[0,1,0,0,3]`	28	12	–
7	`[0,1,0,0,2]`	112	3	–
7	`[0,0,1,0,3]`	168	2	–
7	`[0,1,0,3,2]`	168	2	–
7	`[1,0,0,2,1]`	168	2	–
8	`[0,1,1,0,2]`	252	2	$\to [0,1,1,0,4]$
8	`[0,1,1,0,4]`	252	2	$\to [0,1,1,0,6]$
8	`[0,1,1,0,6]`	252	2	$\to [0,1,1,0,2]$
9	`[1,0,0,1,2]`	180	4	–
9	`[0,1,0,4,1]`	360	2	$\leftrightarrow [0,1,0,4,4]$
9	`[0,1,0,4,4]`	360	2	$\leftrightarrow [0,1,0,4,1]$
11	`[1,0,0,1,10]`	330	4	–
11	`[1,0,0,1,1]`	660	2	–
13	`[0,1,0,0,4]`	182	12	–
13	`[1,0,0,4,7]`	546	4	–
17	`[1,0,0,1,5]`	1224	4	–
19	`[0,1,0,0,4]`	570	12	–

The certificate additionally records the complete cubic-pencil member histogram of each representative. Repeated orbit sizes in Table IV are therefore visibly distinguished without requiring a working-directory file.

TABLE V
INDEPENDENT R5 COMPLETENESS TOTALS. THE FIRST EQUALITY IS THE DIRECT SYNDROME ENUMERATION VERSUS THE ORBIT–STABILIZER SUM; THE SECOND DECOMPOSES THE SAME TOTAL INTO THE FORMULA IN THEOREM I.2 AND THE SPORADIC ROWS ABOVE.

q	direct = orbit sum	nonsporadic	sporadic
7	$889 = 889$	245	644
8	$1116 = 1116$	360	756
9	$1391 = 1391$	491	900
11	$1848 = 1848$	858	990
13	$2080 = 2080$	1352	728
16	$2432 = 2432$	2432	0
17	$4131 = 4131$	2907	1224
19	$4541 = 4541$	3971	570

Remark IV.15 (No one-column MDS extension at this radius). At $q = 8$ the split-free count of Table V is 1116, splitting as 360 nonsporadic and 756 sporadic directions. These are deep holes of $\mathrm{PRS}_{\mathbb{F}_8}(4)$ by Proposition IV.1 and Corollary III.2; they are not one-column MDS extensions, and no such extension exists.

The two notions differ by one column. A syndrome direction is a deep hole when it lies in the span of no $r-2$ parity-check columns and the radius is $r-1$; adjoining a point to the $(q+1)$-arc of the normal rational curve keeps it an arc, equivalently extends the dual code by one column, only when that point lies in the span of no $r-1$ columns. At redundancy five the first condition gives distance at least four and Proposition IV.1 gives radius exactly four, so no direction achieves the distance five that an extension requires. Dür's equivalence states this as a biconditional: the covering radius is $r-1$ precisely when the arc of every normal rational curve in $\mathrm{PG}(r-1, q)$ is complete, so there is no MDS extension of the dual by one digit [6], as stated by Kaipa [1, Sec. IV]. The same reading applies at every redundancy classified here, since each radius gate proves $\rho = r-1$.

V. Coherent polar induction

A. Spaces, markers, and bad schemes

Fix a field k, a two-dimensional k-space E, and

$$S_n = \mathbb{P}(\Gamma^n E).$$

For $\lambda \in E^\vee$, define divided-power contraction by

$$\langle \iota_\lambda f, g \rangle = \langle f, \lambda g \rangle.$$

Then

$$g \in W_{\iota_\lambda f} \quad \Longleftrightarrow \quad \lambda g \in W_f. \tag{4}$$

The lower witness lifts squarefreely only when $\lambda \nmid g$. Thus the factor selected for contraction must remain as a marked forbidden root.

Definition V.1 (Persistent and modular loci)**.** At redundancy j, put $n = j - 1$. The *persistent scheme* $\mathcal{P}_j \subset S_n$ is the determinantal scheme

$$\mathcal{P}_j = \{[f] : \operatorname{rank} C_f^{(2)} \le 2\},$$

cut out by the maximal minors of the three-row catalecticant $C_f^{(2)} : \operatorname{Sym}^{n-2} E^\vee \to \Gamma^2 E$. For every positive-characteristic normal-rational-curve nucleus $\mathcal{N} \subset \mathbb{P}(\Gamma^{n-1} E)$, that is, an intersection of the relevant osculating subspaces, let $\widetilde{\mathcal{N}}$ be its affine cone. Its coherent lift is the linear scheme

$$\mathcal{M}_n(\mathcal{N}) = \mathbb{P} \ker \left(\Gamma^n E \longrightarrow \operatorname{Hom}(E^\vee, \Gamma^{n-1} E / \widetilde{\mathcal{N}}) \right).$$

The *modular locus* $\mathcal{M}_j \subset S_n$ is the reduced union of these coherent lifts; Lucas' binomial criterion determines the nucleus coordinate spans at each level. A *collision divisor* records marker parameters at which the marked factor becomes a repeated root.

Definition V.2 (Contraction and marker spaces)**.** Let $C_f : E^\vee \to \Gamma^{n-1} E$ be $\lambda \mapsto \iota_\lambda f$, and put $U_f = \mathbb{P}(E^\vee) \setminus \mathbb{P}(\ker C_f)$. The first-polar rational map is

$$\Phi_f : U_f \longrightarrow S_{n-1}, \qquad [\lambda] \longmapsto [\iota_\lambda f].$$

Its image is a point or a line. For $j \ge 1$, let

$$\operatorname{Conf}_j(\mathbb{P}(E^\vee)) = \{(\lambda_1, \ldots, \lambda_j) : \lambda_i \ne \lambda_\ell \text{ for } i \ne \ell\}.$$

The ordered marker space $U_{f,j}$ is the open subset on which every successive contraction is nonzero. The universal j-step polar flag is the map

$$(\lambda_1, \ldots, \lambda_j) \longmapsto \big([\iota_{\lambda_1} f], \ldots, [\iota_{\lambda_j} \cdots \iota_{\lambda_1} f] \big).$$

In an affine coordinate r the first map is

$$\Phi_f(r) = (a_1 - r a_0, \ldots, a_n - r a_{n-1}), \qquad \Phi_f(\infty) = (a_0, \ldots, a_{n-1}).$$

Example V.3 (The complete R5-to-R6 propagation at $q = 29$)**.** Work over $\mathbb{F}_{29}$ at redundancy six and take

$$f = (6 : 1 : 0 : 0 : 5 : 1).$$

The Hankel kernel is the net of quartics

$$W_f = \ker \begin{pmatrix} 6 & 1 & 0 & 0 & 5 \\ 1 & 0 & 0 & 5 & 1 \end{pmatrix} = \langle (0,0,1,0,0), (24,1,0,1,0), (28,1,0,0,1) \rangle.$$

The displayed basis has no common factor, so the persistent test does not decide the syndrome. Choose the first marker $r = 0$. Then

$$\iota_0 f = (1 : 0 : 0 : 5 : 1).$$

This is now a redundancy-five syndrome, and its full cubic pencil is

$$W_{\iota_0 f} = \langle (0,1,0,0), (25,0,1,24) \rangle.$$

The member with pencil parameter $[9 : 28]$ is

$$h(t) = 4 + 9t - t^2 + 5t^3 \in W_{\iota_0 f},$$

because the two Hankel products are $4 + 5 \cdot 5 = 0$ and $5(-1) + 5 = 0$ in $\mathbb{F}_{29}$. Direct factorization gives

$$h(t) = 5(t-2)(t-6)(t-27).$$

None of these roots is the marked root 0. Equation (4) therefore lifts h to

$$t\, h(t) = (0, 4, 9, 28, 5) \in W_f,$$

a squarefree quartic with roots $0, 2, 6, 27$. The Hankel criterion now places f in the span of the corresponding four normal-rational-curve columns, so f is shallow. This one calculation exhibits every interface in the R5-to-R6 propagation: the upper Hankel net, the ordinary common-factor test, contraction to the complete R5 cubic pencil, selection on its splitting cover, marker avoidance, and squarefree lifting. In the general proof the R5 curve argument supplies such a pencil parameter after deleting the marked fibers; the displayed parameter $[9 : 28]$ is one complete instance of that step. Thus Example V.3 exhibits the transverse recursion in miniature; Figure 1 places it beside the contained branch. The next definition records exactly the data needed to repeat the transverse calculation without its chosen coordinates.

Definition V.4 (One-step lower package)**.** Fix $f \in S_n(\mathbb{F}_q)$ with injective contraction map, so that $\Phi_f : \mathbb{P}(E^\vee) \to S_{n-1}$ is defined everywhere. A one-step lower package abstracts the R6 argument: a lower bad locus, the R5 ordered-root curve with its deletions, and the finite set of markers where that construction is unavailable. It consists of:

(i) a closed bad scheme $B_{n-1} \subset S_{n-1}$ and a finite stratification $S_{n-1} \setminus B_{n-1} = \coprod_\alpha X_\alpha$;

(ii) for every $\lambda \in \mathbb{P}(E^\vee)(\mathbb{F}_q)$ with $b = \iota_\lambda f \in X_\alpha(\mathbb{F}_q)$, a specified proper geometrically integral curve $Y_{\alpha,b}$, defined over $\mathbb{F}_q$, with genus bound g_α and point bound
$$\#Y_{\alpha,b}(\mathbb{F}_q) \geq q + \kappa_\alpha - 2g_\alpha\sqrt{q};$$
together with a zero-dimensional closed deletion scheme $D_{\alpha,b,\lambda} \subset Y_{\alpha,b}$, of degree at most δ_α, containing the singular, branch, diagonal, marker, and collision points, such that every rational point outside $D_{\alpha,b,\lambda}$ determines an ordered rational split squarefree $g \in W_b$ with $\lambda \nmid g$;

(iii) a finite parameter scheme $Z_f \subset \mathbb{P}(E^\vee)$, of degree at most d, containing $\Phi_f^{-1}(B_{n-1})$ and every parameter at which the preceding construction is unavailable, including parameters where a fixed factor of the lower system equals a retained marker.

The index α includes any constant-field or Frobenius twist; the assertion in (iii) is the required descent statement. All degrees are geometric degrees on the displayed curve or parameter line.

Figure 1 gave the conceptual recursion before its formal data were introduced. In the present notation, the contained branch is the level-specific assertion that the whole line $\Phi_f(U_f)$ lies in the lower bad scheme. The transverse branch is the package in Definition V.4: it deletes finitely many parameters and points on the lower ordered splitting cover, then lifts a surviving witness by Equation (4).

Theorem V.5 (Polar-flag construction)**.** The maps of Definition V.2 commute with field extension and the natural $\mathrm{GL}(E)$ action. They include the infinity chart displayed above. If $g \in W_{\iota_{\lambda_j} \cdots \iota_{\lambda_1} f}$, then $\lambda_1 \cdots \lambda_j g \in W_f$; the lift is squarefree exactly when g is squarefree and avoids every marker λ_i.

Proof. Contraction is defined by the perfect divided-power/symmetric-power pairing, so it commutes with base change and change of basis. Repeated application of (4) gives the membership assertion. Since the λ_i are distinct on the configuration space, their product is squarefree; the product with g is squarefree precisely when g is squarefree and none of the λ_i divides it. The coordinate formula at infinity follows by evaluating C_f on the second basis covector. □

Lemma V.6 (Three readings of the collision divisor)**.** Assume W_f has trivial gcd, and let $\lambda \in \mathbb{P}(E^\vee)$. The following are equivalent:

(i) every member of W_f divisible by λ is divisible by λ^2;

(ii) λ is a fixed factor of $W_{\iota_\lambda f}$;

(iii) λ lies on the collision divisor of the linear series W_f.

Proof. Equation (4) gives the equality of subspaces
$$\lambda W_{\iota_\lambda f} = W_f \cap \lambda \operatorname{Sym}^{n-2} E^\vee.$$
Thus (i) and (ii) are the same statement after dividing by λ. Since W_f is base-point-free, (i) says exactly that the morphism defined by W_f has zero differential at λ, which is (iii). □

Lemma V.7 (Uniform collision alternatives)**.** Let $j \geq 5$, and let a base-point-free g^{j-4}_{j-2} on $\mathbb{P}^1$ define an inseparable morphism in characteristic p. Then
$$p \mid (j-2), \qquad p(j-4) \leq j-2.$$
The only possible inseparable cases are $(j, p) = (5, 3)$ and $(6, 2)$. In the separable case the collision divisor has degree at most $2j - 6$; in particular this holds unconditionally for $j \geq 7$.

Proof. An inseparable morphism factors through absolute Frobenius. Hence $p \mid (j-2)$, and its $j-3$ sections lie in the $(j-2)/p+1$-dimensional space of p-th powers. This gives the inequality. Since $(j-2)/(j-4) < 2$ for $j \geq 7$, no prime remains; the two smaller cases follow directly. In the separable case some value–derivative minor is nonzero; as a binary Wronskian it has degree $2(j-2) - 2 = 2j - 6$ and contains the collision scheme. □

Lemma V.8 (Linear modular pullbacks)**.** Let $\mathcal{N} \subset S_{j-2}$ be a projective linear modular nucleus. Then $\Phi_f^{-1}(\mathcal{N})$ is empty, one point, or the whole parameter line. In the transverse case its degree is at most one.

Proof. The polar map has linear coordinates in the parameter, so the pullback ideal is generated by linear forms on $\mathbb{P}^1$. $\square$

Lemma V.9 (Old-marker fixed-factor divisor)**.** Assume W_f has trivial gcd, and fix a retained marker $x \in \mathbb{P}(E^\vee)$. The parameters λ for which x divides every member of $W_{\iota_\lambda f}$ form a proper closed subscheme of $\mathbb{P}(E^\vee)$ of degree at most three.

Proof. Append the evaluation row at x to the two consecutive Hankel rows of $\iota_\lambda f$. The maximal minors cut out the stated fixed-factor condition. Their entries are linear in λ, so their common zero scheme has degree at most three.

The minors cannot vanish identically. By Equation (4), identical vanishing would say

$$W_f \cap \lambda \operatorname{Sym}^{n-2} E^\vee \subset x \operatorname{Sym}^{n-2} E^\vee \qquad \text{for every geometric } \lambda.$$

Given a nonzero $g \in W_f$, choose a geometric linear factor $\lambda \mid g$. The displayed containment then gives $x \mid g$. Thus x would divide every member of W_f, contrary to the trivial-gcd hypothesis. $\square$

Lemma V.10 (Exact-linear-gcd transport)**.** Suppose $W_f = LV$ has exact linear gcd L, and let $\lambda \neq L$. Then every member of $W_{\iota_\lambda f}$ is divisible by L. Along a coherent flag chosen outside the rank-at-most-two terminal carrier, the gcd remains exactly L; thus the flag reaches the R5 quadratic-graph package with the same fixed factor.

Proof. The lifting identity gives

$$\lambda W_{\iota_\lambda f} = W_f \cap \lambda \operatorname{Sym}^{n-2} E^\vee \subset L \operatorname{Sym}^{n-2} E^\vee.$$

Since L and λ are coprime, division by λ shows that L divides every lower member. If the lower gcd grows to degree two, the lower syndrome lies in the rank-at-most-two terminal carrier and that parameter is excluded. The dimension bound rules out a larger common factor away from the rank-one boundary. Induction over the marker list proves the assertion. Excluding $\lambda = L$ is exactly the fixed-factor collision needed to keep the final lifted product squarefree. $\square$

Lemma V.11 (Identically colliding stages)**.** If the collision divisor of a base-point-free stage is the whole parameter line, then the stage is inseparable. The only possibilities are the characteristic-three R5 component and the characteristic-two R6 component, and their coherent lifts lie in the declared persistent or modular locus.

Proof. Lemma V.6 identifies whole-line collision with zero differential of the moving linear series. The numerical argument in Lemma V.7 leaves only $(j, p) = (5, 3)$ and $(6, 2)$. Proposition IV.13 identifies the first with the wild rank/fixed-factor boundary, while Propositions A.5 and A.6 identify the second with the binary modular component. Proposition V.15 and Lemma V.8 give their coherent lifts. $\square$

Definition V.12 (Contained pullback and recursive bad scheme)**.** Let $B \subset S_{n-1}$ be a closed scheme with homogeneous ideal I_B. On the open $S_n^\circ \subset S_n$ where C_f is injective, pull each element of I_B back along the universal polar map

$$\mathbb{P}(E^\vee) \times S_n^\circ \longrightarrow S_{n-1}.$$

Expand the resulting bihomogeneous forms in the marker coordinates. The vanishing of all their coefficients defines the *contained-pullback scheme* $\operatorname{Cont}_n(B)$: its geometric points are exactly the f for which the whole polar line lies scheme-theoretically in B. This coefficient ideal is independent of the chosen generators of I_B.

Let $\mathcal{B}_5^{\mathrm{red}} \subset S_4$ be the reduced terminal carrier consisting of the rank-at-most-two Hankel locus, the closure of the cyclic carrier on the rank-two open, its true characteristic-two cyclic plane and characteristic-three wild cone, the rank-one contraction boundary, and the two identically colliding inseparable components. The exact-linear-gcd stratum is not part of this carrier: Proposition IV.3 supplies its separate quadratic-graph lower package. Ordinary collision points are deletion divisors, not components of $\mathcal{B}_5^{\mathrm{red}}$.

Inductively, let $\mathcal{B}_{j+1}^{\mathrm{red}} \subset S_j$ be the reduced union of the closure of $\operatorname{Cont}_j(\mathcal{B}_j^{\mathrm{red}})$, the noninjective-contraction locus, and the locus on which the universal marker-collision divisor is the whole parameter line. We call this the *reduced recursively pointed contained bad locus* at redundancy $j + 1$.

Definition V.13 (One-step contained-component assertion)**.** For a specified bad scheme B_{n-1}, collision divisor, and explicit closed list $\mathcal{C}_n \subset S_n$, the assertion $\mathrm{CC}(n, 1; \mathcal{C}_n)$ says that every f for which the contraction map is noninjective and hence supplies no polar line, $\Phi_f^* B_{n-1}$ is the whole parameter line and hence supplies no transverse marker, or the collision divisor is the whole line and hence every lift repeats its marked root, belongs to $\mathcal{C}_n$. When the list is clear we write $\mathrm{CC}(n, 1)$.

Theorem V.14 (Finite-depth polar escape)**.** Let $s \geq 1$ and $f \in S_n(\mathbb{F}_q)$. Suppose that coherent markers can be chosen in s stages with the following uniform data. After every surviving partial flag of length $i < s$, the unavailable choices for the next marker form a zero-dimensional scheme on $\mathbb{P}(E^\vee)$ of degree at most d_{i+1}. Outside this scheme the next contraction is nonzero, the markers remain distinct, and all earlier markers remain forbidden. After s contractions, every surviving flag lies in one of finitely many strata α carrying a proper geometrically integral curve $Y_{\alpha,\boldsymbol{\lambda}}$ with

$$\#Y_{\alpha,\boldsymbol{\lambda}}(\mathbb{F}_q) \geq q + \kappa_\alpha - 2g_\alpha\sqrt{q}.$$

Assume that a deletion scheme of degree at most δ_α contains the singular, branch, diagonal, collision, and all retained-marker incidences, and that every remaining rational point gives a split squarefree member of the terminal Hankel system avoiding every marker. Put

$$\mathcal{H}_\kappa(g,\delta) = 1 + \left\lfloor \left(g + \sqrt{g^2 + \delta - \kappa}\right)^2 \right\rfloor$$

and assume $\delta_\alpha \geq \kappa_\alpha - g_\alpha^2$ for every α. If

$$q \geq \max_\alpha \mathcal{H}_{\kappa_\alpha}(g_\alpha, \delta_\alpha) \quad \text{and} \quad q + 1 > \max_{1\leq i\leq s} d_i,$$

then W_f contains a split squarefree member.

Proof. Choose the markers successively. At stage i, a zero-dimensional scheme of degree at most $d_i < q+1$ cannot contain all $q+1$ rational points of $\mathbb{P}^1$, so a surviving choice exists. On the resulting terminal stratum the recorded point bound gives

$$q + \kappa_\alpha - 2g_\alpha\sqrt{q} > \delta_\alpha,$$

so there is a rational point outside all deletions. It yields a split squarefree terminal member avoiding every marker. Repeated application of Equation (4) multiplies it by the distinct marker factors; the avoidance hypotheses make the resulting member of W_f squarefree. □

The one-step theorem used below is the case $s = 1$: a lower package in Definition V.4 supplies $d_1 = d$ and the terminal curve data. The theorem isolates the uniform numerical escape and lifting argument at every finite depth. Applying it at a new redundancy still requires a complete contained-component assertion and uniform lower package at every intermediate level. Redundancies six and seven first verify those inputs at depths one and two. Section VI then proves the reduced carrier theorem at arbitrary finite depth and derives the uniform consequence.

B. Contained-component calculations

The ordinary persistent locus admits a uniform contained-line calculation. This does not classify other lower bad schemes, but it closes the persistent part of every level-specific component gate.

Proposition V.15 (Contained rank-two polar lines)**.** Let $n \geq 5$, let $f \in \mathbb{P}(\Gamma^n E)$, and let

$$A = C_f^{(2)} : \operatorname{Sym}^{n-2} E^\vee \longrightarrow \Gamma^2 E$$

be the three-row catalecticant. Assume the contraction map $E^\vee \to \Gamma^{n-1}E$ is injective, so the first-polar map is defined on all of $\mathbb{P}(E^\vee)$. This is exactly the complement of the rank-one boundary: after a change of coordinates, a nonzero kernel vector is ∂_X, and $\partial_X f = 0$ forces f to be a scalar multiple of $Y^{[n]}$. Such a point lies on the normal rational curve and is already excluded from the split-free locus. Then

$$\Phi_f(\mathbb{P}(E^\vee)) \subset \{b : \operatorname{rank} C_b^{(2)} \leq 2\} \quad \Longleftrightarrow \quad \operatorname{rank} A \leq 2$$

scheme-theoretically and in every characteristic. If $\operatorname{rank} A = 3$, the rank-drop parameters form a subscheme of $\mathbb{P}(E^\vee)$ of degree at most three.

Proof. For a nonzero $\lambda \in E^\vee$, multiplication identifies $\operatorname{Sym}^{n-3} E^\vee$ with the hyperplane

$$H_\lambda = \lambda \operatorname{Sym}^{n-3} E^\vee \subset \operatorname{Sym}^{n-2} E^\vee,$$

and functoriality gives

$$C^{(2)}_{\iota_\lambda f} = A|_{H_\lambda}.$$

The forward implication is immediate. Conversely, suppose $\operatorname{rank} A = 3$ and put $K = \ker A$, so $\dim K = n-4$. If the lower rank is at most two, rank–nullity on the $(n-2)$-dimensional space H_λ gives

$$\dim(K \cap H_\lambda) \geq (n-2) - 2 = n - 4 = \dim K.$$

Thus $K \subset H_\lambda$. Containment for every geometric $[\lambda]$ would imply

$$K \subset \bigcap_{[\lambda]\in\mathbb{P}(E_{\bar{k}}^\vee)} \lambda \operatorname{Sym}^{n-3} E_{\bar{k}}^\vee = 0,$$

because a nonzero binary form cannot be divisible by every linear form. This contradicts $\dim K = n-4 > 0$.

The maximal minors of $C^{(2)}_{\iota_\lambda f}$ are homogeneous cubics in λ. Vanishing at every geometric point is therefore identical vanishing, proving the scheme-theoretic assertion. In the noncontained case at least one cubic is nonzero, and the common rank-drop scheme lies in its degree-three zero divisor. ☐

Thus Proposition V.15 covers the entire non-rank-boundary locus on which the first-polar line is defined; the excluded noninjective locus is the rank-one normal-rational-curve locus just identified. After rank one and rational split secants are removed as shallow, the upper rank-two locus consists of the persistent tangent and conjugate-secant families. On a persistent fiber the nonsplit torus acts by $z \mapsto z^n$, giving T/T^n modulo inversion and Frobenius, while the tangent unipotent cocycle is $z \mapsto z + nu$.

Modular loci are equally explicit. If $\mathcal{N}$ is a nucleus in the lower divided-power module, Definition V.1 identifies the complete space whose first-polar line lies in $\mathcal{N}$ with the linear kernel scheme $\mathcal{M}_n(\mathcal{N})$. Lucas' theorem computes the nucleus coordinates, and this kernel computes every coherent lift without an ambient syndrome census. At redundancy r, if $p > r-1$, Lucas' theorem leaves no nucleus coordinate, so $\mathcal{M}_r$ is empty. This removes the modular term only; the R5 cyclic carrier survives in large characteristic.

VI. The recursive carrier theorem

The finite-depth theorem reduces the uniform problem to the geometry of polar lines contained in the terminal bad locus. We now determine that locus after reduction in each geometric fiber and show that all modular descendants merge into one linear carrier.

For $r \geq 6$, put $n = r-1$, and let the ground field have characteristic $p > 0$. In divided-power coordinates define

$$\mathcal{M}^{\max}_{r,p} = \mathbb{P}\left\langle e_j : \binom{r-2}{j} = \binom{r-2}{j-1} = 0 \pmod{p} \right\rangle. \tag{5}$$

An empty span denotes the empty projective scheme. Let $\mathcal{P}_r$ be the rank-at-most-two consecutive catalecticant scheme in $\mathbb{P}(\Gamma^{r-1}E)$.

The next three propositions prove the unconditional geometric layer: they find every reduced terminal component, merge all modular descendants, and select the components that persist up the contraction tower. On a first pass, read their statements and the four-row component table in Proposition VI.3, then resume at Theorem VI.4. The intervening elimination and characteristic-wise substitutions prove exhaustiveness and add no hypothesis to the finite-depth theorem.

Proposition VI.1 (Reduced terminal carrier)**.** In bottom coordinates $c = (c_0, \ldots, c_4)$, put

$$D = \det \begin{pmatrix} c_0 & c_1 & c_2 \\ c_1 & c_2 & c_3 \\ c_2 & c_3 & c_4 \end{pmatrix}.$$

The reduced terminal bad carrier is the irredundant union of the prime cubic $V(D)$ and one residual prime. Away from characteristics two and three the residual prime is the projected Veronese surface

$$[u : v : w] \longmapsto [6u^2 : 3uv : v^2 + 2uw : 3vw : 6w^2], \tag{6}$$

the locus of syndrome quartics that are scalar multiples of squares of binary quadratics. The coordinates in (6) are the bottom c_i; writing $z_i = \binom{4}{i} c_i$ for the plain coefficients of the associated quartic $f_c = \sum_{i=0}^{4} z_i X^{4-i} Y^i$, the parametrization gives $f_c = 6(uX^2 + vXY + wY^2)^2$, so the same surface is the plain-coefficient Veronese of (17), which is written in those rescaled coordinates. In characteristic two it is the plane $V(c_0, c_4)$. In characteristic three it is the cone with c_2 free and prime ideal

$$(c_1c_3 - c_0c_4,\ c_1^3 - c_0^2c_3,\ c_0c_3^2 - c_1^2c_4,\ c_3^3 - c_1c_4^2). \tag{7}$$

There is no flat reduced integral model having exactly these residual fibers.

Proof. The elimination has a concrete geometric source. On $\mathrm{Gr}(2, \Gamma^3 E)$, let $z_0, \ldots, z_5$ be Plücker coordinates for a cubic pencil. Its universal symmetric ordered-root equation is

$$\Phi_z(t,s) = z_0 + z_1(t+s) + z_2(t^2 + ts + s^2) + z_3ts + z_4ts(t+s) + z_5t^2s^2.$$

For the Hankel matrix

$$H_c = \begin{pmatrix} c_0 & c_1 & c_2 & c_3 \\ c_1 & c_2 & c_3 & c_4 \end{pmatrix},$$

the signed maximal minors give the map explicitly:

$$z(c) = (c_2c_4 - c_3^2,\ -c_1c_4 + c_2c_3,\ c_1c_3 - c_2^2, c_0c_4 - c_1c_3,\ -c_0c_3 + c_1c_2,\ c_0c_2 - c_1^2).$$

After the diagonal and collision factors already charged as deletion divisors are saturated away, the terminal bad locus is the pullback, along the Hankel Plücker map $c \mapsto z(c)$, of the factorization locus of Φ_z. The involution $(t,s) \leftrightarrow (s,t)$ leaves three possibilities. Symmetric factors give the residual component below; exchanged factors give $D = 0$; and an anti-invariant factor is proportional to $s - t$. In the last case the only nonzero Plücker coordinates are $z_2 = -\mu$ and $z_3 = 3\mu$, and hence

$$z_0z_5 - z_1z_4 + z_2z_3 = (-\mu)(3\mu) = -3\mu^2.$$

Thus it survives only in characteristic three and is then the diagonal collision already removed. Thus the two displayed primes are the only possible noncollision components before small-characteristic specialization.

The exact saturated bridge can also be read without the elimination file. Let J be the pullback under $z(c)$ of

$$\begin{aligned}&(3z_4^2 - 9z_2z_5 - z_3z_5,\ z_3z_4 - 3z_1z_5,\ z_3^2 - 9z_0z_5,\\ &\qquad z_2z_3 - z_1z_4 + z_0z_5,\ z_1z_3 - 3z_0z_4,\ 3z_1^2 - 9z_0z_2 - z_0z_3).\end{aligned}$$

If V denotes the seven-generator residual ideal printed below, exact reduction gives $J \subseteq V$, $3D\,V \subseteq J$, and $(J : D^\infty) = V$ over $\mathbb{Q}$. The factor 3 is necessary over $\mathbb{Z}$, which is precisely why the characteristic-three fiber is checked separately.

Outside characteristics two and three, elimination of the ordered residual roots gives the prime kernel of (6); one convenient generating set is

$$\begin{aligned}&2c_3^3 - 3c_2c_3c_4 + c_1c_4^2,\\ &6c_2c_3^2 - 9c_2^2c_4 + 2c_1c_3c_4 + c_0c_4^2,\\ &2c_1c_3^2 - 3c_1c_2c_4 + c_0c_3c_4, \quad c_0c_3^2 - c_1^2c_4,\end{aligned}$$

together with their images under $c_i \leftrightarrow c_{4-i}$. Direct reduction gives the plane in characteristic two and (7) in characteristic three. The ordered-Hessian calculation sends its complementary ruling to rank one.

Here exhaustiveness is an exact elimination statement, not a visual factorization heuristic. Certificate SC starts from the terminal ordered-root incidence ideal, saturates the collision factors, eliminates the root variables, and verifies the two generic primes and their characteristic-$2,3$ fibers by Gröbner reduction. Its public Python and Singular inputs are in `supplement/evidence/stable-components/`; every generator and remainder used for this conclusion is hash-pinned there.

Each displayed residual ideal is prime: (6) is a projective parametrization with prime kernel, the characteristic-two ideal is linear, and (7) is the toric prime of the monomial quartic after projection from e_2. None is contained in $V(D)$: use $(3,0,1,0,3)$, which is the image of $(u,v,w) = (1,0,1)$ and has $D = 8$, together with e_2 and $(1,0,1,0,0)$, respectively. Thus the intersection with (D) is an irredundant reduced decomposition and has no embedded prime. The residual degree is four generically and one in characteristic two. Constancy of the Hilbert polynomial in a flat projective family rules out the final assertion. The exact eliminations and the characteristic-$2,3,5$ reductions are replayed by the stable-component certificates described in Section D. ☐

Proposition VI.2 (Maximal Lucas union)**.** Every normal-rational-curve nucleus encountered in a coherent polar flag, and every later lift of it, is contained in $\mathcal{M}^{\max}_{r,p}$. Their reduced union equals that single linear scheme. A polar line not contained in the union meets it in degree at most one.

Proof. For the order-a nucleus of the degree-d curve, its coordinate support is contained in

$$\widehat{S}_{d,p} = \{i : \binom{d}{i} = 0 \pmod{p}\}, \tag{8}$$

because the nucleus conditions include the row d. If a support is $S \subset \{0,\dots,d\}$, one coherent lift has support

$$L_d(S) = \{j : (j = d+1 \text{ or } j \in S),\ (j = 0 \text{ or } j-1 \in S)\}. \tag{9}$$

This operation preserves inclusions. Pascal's identity gives $L_d(\widehat{S}_{d,p}) \subset \widehat{S}_{d+1,p}$. Induction therefore contains every earlier descendant in the last one. Conversely, the top-order nucleus at $d = r-2$ occurs in the union, and its one-step lift is exactly (5). The union is therefore one linear scheme. Pulling its linear ideal back to a projective line gives nonzero linear forms unless the line is contained, proving the degree bound. ☐

The endpoint arithmetic in Proposition VI.2 has a known projective-subline form. Wang–Wu–Hu [31, Prop. 11] prove that the Lucas block with $k = p^a + 1$ is nonempty over $\mathbb{F}_{p^e}$ exactly when $a \mid e$, and then its blocks are projective sublines. We use this criterion at that exact boundary. The adjacent-zero carrier (5), its nesting through coherent lifts, and its split-squarefree Hankel incidence are separate questions addressed here.

Proposition VI.3 (Recursive component selection)**.** The reduced recursively contained carrier generated by the terminal carrier and all intermediate nuclei is exactly

$$\mathcal{P}_r \cup \mathcal{M}^{\max}_{r,p}. \tag{10}$$

Fixed linear gcds retain their quadratic-graph package, and marker collisions remain deletion divisors; neither is included in (10).

Proof. Retain m markers. If h is the coefficient row of their product, the bottom syndrome is

$$h\,\mathrm{Cat}_{m,4}(f), \qquad \mathrm{Cat}_{m,4}(f) = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_m & a_{m+1} & a_{m+2} & a_{m+3} & a_{m+4} \end{pmatrix}. \tag{11}$$

Over an algebraic closure, products of distinct linear forms are dense in the projective space of binary forms. Deleting the projective kernel of the linear map does not change its image closure. Hence the attainable bottom syndromes close to the projectivized row space of (11). This row space is irreducible. If it lies in the finite reduced union of Proposition VI.1, its generic point lies in one component, so the whole row space lies in that component. This is a geometric-marker statement; no density of the finite set of rational marker tuples is used. The density and finite-component-selection implication are also kernel checked by the declarations listed in the formalization ledger.

The remaining component check is summarized by

component	possible linear part	upper lift
$V(D)$	arbitrary	$\mathcal{P}_r$
Veronese $(p \neq 2,3)$	point	rank one
wild cone $(p=3)$	ruling through e_2	rank one
$V(c_0, c_4)$ $(p=2)$	binary-plane subspace	$\mathcal{M}^{\max}_{r,2}$

Here is the calculation behind the four rows. Substitute $c_i = xd_i + yd_{i+1}$. For $V(D)$, the coefficients in x, y are exactly the 3×3 minors of

$$\begin{pmatrix} d_0 & d_1 & d_2 & d_3 \\ d_1 & d_2 & d_3 & d_4 \\ d_2 & d_3 & d_4 & d_5 \end{pmatrix},$$

the consecutive Fano identity. For the generic residual prime their unique minimal prime is the ideal of the 2×2 minors of $\left(\begin{smallmatrix} d_0 & \cdots & d_4 \\ d_1 & \cdots & d_5 \end{smallmatrix}\right)$, so the upper syndrome has rank one; geometrically this is also the fact that the projected Veronese contains no line. The same coefficient ideal in characteristic three is again this rank-one ideal, verifying directly that every positive-dimensional ruling of the wild cone through e_2 lifts into $\mathcal{P}_r$. In characteristic two it has exactly the two minimal primes

$$(d_0, d_1, d_4, d_5) \quad \text{and} \quad (d_3d_4 + d_2d_5, d_1d_4 + d_0d_5, d_3^2 + d_1d_5, d_2d_3 + d_0d_5, d_2^2 + d_0d_4, d_1d_2 + d_0d_3).$$

The second lies in the persistent 3×3-minor ideal; the first is the binary-plane descendant. Successive overlaps cut it to the binary line, then the central point, then the empty projective space, and (9) identifies precisely these spaces as Lucas lifts. On a nucleus, (9) and Proposition VI.2 then place every surviving lift in the maximal modular carrier.

The coherent-Fano part of Certificate SC verifies these consecutive-row substitutions, including the characteristic-three ruling and the characteristic-two successive supports, against the same incidence ideal. This exhausts every component and proves (10) by induction on the flag length. ☐

Theorem VI.4 (Finite-depth recursive-carrier consequence)**.** Let $r \geq 6$, and let q be a prime power with

$$q \geq Q_r := 6r - 15 + \left\lfloor 2\sqrt{6r-17} \right\rfloor. \tag{12}$$

Assume that at every intermediate redundancy the transverse strata admit the one-step lower packages of Definition V.4, with terminal deletion at most $6r-17$, exact-linear-gcd deletion at most $2r-2$, and every other parameter budget below $3r-5$. Every split-free redundancy-r syndrome lies in $\mathcal{P}_r \cup \mathcal{M}^{\max}_{r,p}$. If $p > r-1$, the modular term is empty, and the projective deep-hole syndromes are exactly the persistent tangent and conjugate-secant families. Their cardinality is

$$\frac{q(q+1)^2}{2},$$

and their orbit law is T/T^{r-1} modulo inversion and coefficient Frobenius, with tangent cocycle $z \mapsto z + (r-1)u$.

Proof. Proposition VI.3 supplies the all-level contained-component hypothesis in the finite-depth escape theorem. The lower packages and their bounds are hypotheses here; they must be discharged at each fixed redundancy and are not inferred from component selection. Solving the genus-one inequality gives (12); the modular pullback costs only one parameter by Proposition VI.2. Thus every point outside (10) has a split squarefree kernel member.

If $p > r-1$, no interior entry of Pascal row $r-2$ vanishes, so $\mathcal{M}^{\max}_{r,p} = \varnothing$. Rank one and rational secants are shallow; the remaining rank-two points are precisely the tangent and conjugate-secant families and are split-free. The threshold lies in the

Seroussi–Roth–Dür radius range, so split-free and deep coincide. The count and orbit law are the persistent calculation already used at R6 and R7. ☐

The carrier theorem is fiberwise and reduced, and its finite-depth consequence is explicitly conditional on the displayed lower packages. It makes no assertion about nilpotent structure in a chosen integral incidence model, and in small characteristic it classifies the only possible carrier of additional split-free points, not the arithmetic of every point on that carrier.

Proposition VI.5 (Upper-level radius gate)**.** For the ranges used below,

$$\begin{aligned}\rho(\mathrm{PRS}(q-7)) &= 7 \quad (q \geq 43),\\ \rho(\mathrm{PRS}(q-8)) &= 8 \quad (q \geq 53),\\ \rho(\mathrm{PRS}(q-9)) &= 9 \quad (q \geq 59).\end{aligned}$$

More generally, the radius is $r-1$ in the range of Theorem VI.4.

Proof. The dual GRS code has dimension r. In each displayed range, and also under (12), one has

$$r \leq \lfloor q/2 \rfloor + 2.$$

Seroussi–Roth [5, Thm. 1] therefore gives nonextendability of the corresponding normal rational curve; its stated even-field dimension-three exception is irrelevant because $r \geq 8$. Dür's completeness–covering-radius equivalence [6], in the form recorded by Kaipa [1, Sec. IV and Prop. 4(1)], gives the asserted radius. Thus the promotion from split-free syndromes to deep holes in R8–R10 uses one explicit imported gate. ☐

VII. From the Carrier Theorem to the Fixed Levels

Proposition VI.3 identifies every component on which a polar line can remain trapped; the numerical escape consequence in Theorem VI.4 additionally assumes the intermediate lower packages. At a fixed redundancy, two questions remain. First, away from that carrier, one must construct the successive pointed lower packages required by Theorem V.14. Second, one must decide whether the maximal Lucas carrier itself contains a split-free point. These are finite-depth questions, but their answers use different geometry: the first is controlled by marker, ramification, and deletion divisors; the second by the arithmetic of the adjacent-zero Hankel kernel.

Table II records the resulting field ranges and the separate covering-radius gates. The exact exceptional orbits at redundancies six and seven are stated in Theorems A.10 and A.16; the high-field statements are Theorems B.11, B.22, and Corollary C.18. In each case the geometric argument first classifies split-free syndrome directions and only then invokes Proposition VI.5.

A. Pointed escape through redundancy nine

At redundancies six and seven, the first two pointed packages expose the mechanism in its smallest dimensions. The persistent rank-two carrier, exact-linear-gcd boundary, modular nucleus, and self-collision divisor account for every polar line that can fail to escape. The remaining line meets a proper bad scheme of bounded degree, and the terminal cubic cover supplies a split squarefree witness. Appendix A proves the exhaustive contained-component classification and closes the bounded fields by finite certificates.

The next two levels repeat this argument with three and four retained markers. For redundancy eight, the terminal ordered-root object is the off-diagonal $(2,2)$ curve of a cubic pencil; its geometric S_3 monodromy gives an integral genus-one cover, while the three marker fibers contribute the exact deletion budget. For redundancy nine, a principal-open cover of quartic moduli reduces the last obstruction to a residual quadratic and a one-moving-root slice. The resulting bounds are $q \geq 43$ and $q \geq 53$. Appendix B records the equations, divisor tables, modular lifts, and the proof that none of the named bad factors vanishes identically.

B. The first higher Lucas carrier

The first genuinely new arithmetic carrier occurs at redundancy ten in characteristic two. A six-root divisor leaves a final quadratic pair. After normalization, that pair is split and squarefree exactly when its nonzero linear coefficient s satisfies an Artin–Schreier trace condition $\mathrm{Tr}(p/s^2) = 0$, together with the finitely many marker avoidances. This final-pair criterion covers the quotient directions missed by the projective-additive subspace construction. It proves Theorem C.13: every rational point of the degree-nine Lucas carrier is shallow over every admissible binary field.

For the transverse R10 argument, the unavailable-marker degree after i contractions is

$$3 + 2 + (14 - 2i) + i + 2i = 19 + i \leq 23, \qquad 0 \leq i \leq 4.$$

The five pointed stages therefore reach the same terminal cover without losing a retained marker. In characteristic seven, contraction at infinity lands on the R9 binary-quartic carrier; its seven finite roots lift with the factor at infinity to a squarefree octic. In characteristic two, Theorem C.13 removes the remaining carrier. These two arguments, together with the empty carriers in the remaining odd characteristics, give Corollary C.18. Appendix C contains the invariant-block calculation, the full final-pair proof, the stagewise propriety checks, and the characteristic-seven lift.

VIII. Scope and open problems

The recursive carrier theorem closes the reduced contained-component problem at arbitrary redundancy. The remaining obstruction is arithmetic: small-characteristic points on later maximal Lucas carriers need not share the degree-nine answer.

1) Redundancy five is complete for every $q \geq 7$.
2) Redundancy six is a complete deep-hole classification over every prime power $q \geq 7$. Redundancy seven has a complete split-free classification over the same range; at $q = 8$ its radius is seven and exact extraction gives ten deep directions in the diagonal tangent and central-nucleus orbits; only the $q = 7, 9$ radius gates remain separate.
3) Redundancies eight, nine, and ten have exact persistent-only classifications in the stated high-field ranges.
4) At arbitrary redundancy, the reduced carrier and its degree-one modular union are exact. Split-squarefree arithmetic on later Lucas carriers remains open outside the evaluated levels.

In particular, the paper does not settle the general Reed–Solomon deep-hole conjecture. Its unconditional arbitrary-redundancy geometric conclusion is the exact reduced contained carrier. Conditional on the displayed intermediate lower packages, the numerical split-free containment becomes an exact classification when the characteristic exceeds $r - 1$.

The last sporadic fields at redundancies five, six, and seven are $19, 13, 11$, while the corresponding first prime-power field-order gates are $23, 29, 37$. These opposite trends ask whether sufficiently high redundancy has any sporadic fields outside the persistent and modular loci.

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Appendix A
Redundancies six and seven: complete and gated results

This appendix takes the marked polar-escape theorem as input and discharges its first two pointed lower packages. The bottleneck is exhaustiveness: fixed gcd, modular nucleus, self-collision, and transverse cases must account for every polar line. The resulting structural arguments, together with the bounded certificates, prove Theorems A.10 and A.16.

A. Redundancy six

For

$$f = (a_0 : \cdots : a_5) \in \mathbb{P}(\Gamma^5 E), \qquad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \end{pmatrix},$$

the system W_f is a net of quartics off the quintic normal rational curve. It is split-free exactly when it has no completely split squarefree quartic.

Proposition A.1 (Persistent net and orbit law)**.** If $\gcd(W_f) = Q$ has degree two, then

$$W_f = Q \operatorname{Sym}^2 E^\vee.$$

The split-free persistent locus has size $q(q+1)^2/2$. On the tangent fiber the unipotent coordinate is $x \mapsto x + 5u$; on the irreducible quadratic fiber the norm-one torus acts through $z \mapsto z^5$, with inversion and coefficient Frobenius acting on T/T^5.

Proof. For fixed Q, the syndromes annihilating $Q \operatorname{Sym}^3 E^\vee$ form a projective line. If Q is irreducible, all $q+1$ points remain; if $Q = \lambda^2$, one endpoint has dependent Hankel rows, leaving q points. Hence

$$q(q+1) + \frac{q(q-1)}{2}(q+1) = \frac{q(q+1)^2}{2}.$$

Normalize a square to U^2. The nondegenerate fiber is $[e_4 + xe_5]$, and direct degree-five substitution gives $x \mapsto x + 5u$. For an irreducible Q, its stabilizer identifies the $q+1$-point fiber with $T = \{z \in \mathbb{F}_{q^2}^\times : z^{q+1} = 1\}$ and acts by fifth powers; the normalizer inverts z, while coefficient Frobenius multiplies T/T^5 by the characteristic. This proves the stated orbit law and the characteristic-five tangent splitting. ☐

Lemma A.2 (Exact linear gcd is shallow)**.** Let $q \geq 7$. If $\gcd(W_f)$ has degree one, then W_f contains a split squarefree quartic.

Proof. Write $W_f = LV$, where L is rational and $V \subset \operatorname{Sym}^3 E^\vee$ has vector dimension three. It suffices to find in V a product of three distinct rational factors, none equal to L. Move the root of L to infinity and let ℓ span $V^\perp$. On the cubic with distinct finite roots x, y, z, write

$$\ell\big((T - xU)(T - yU)(T - zU)\big) = c_0 + c_1(x + y + z) + c_2(xy + xz + yz) + c_3xyz =: P(x, y, z).$$

Suppose that this value is nonzero for every distinct triple. For fixed distinct x, y, write $P = A(x, y) + zB(x, y)$, where

$$B(x, y) = c_1 + c_2(x + y) + c_3xy.$$

If $B(x, y) \neq 0$, the unique zero in z must be x or y. Consequently

$$B(x, y)P(x, y, x)P(x, y, y) = 0$$

for every $x \neq y$; it also vanishes when $B(x, y) = 0$. This polynomial has degree at most four in each variable. Since $q-1 > 4$, first varying $y \neq x$ and then x shows that it is the zero polynomial. The polynomial ring is a domain. Neither $P(x, y, x)$ nor $P(x, y, y)$ vanishes identically unless $\ell = 0$, as comparison of the coefficients of $1, x, y, x^2, xy, x^2y$ shows. Hence $B = 0$, so $c_1 = c_2 = c_3 = 0$.

Thus V is the hyperplane of cubics vanishing at infinity, and every member of V is divisible by L. This would make $\gcd(W_f)$ divisible by L^2, contrary to its exact degree. Therefore a required cubic exists, and multiplying it by L gives a split squarefree member of W_f. □

Proposition A.3 (First-polar line and secant degree)**.** For $r \in \mathbb{F}_q$ put

$$b(r) = (a_1 - ra_0, \dots, a_5 - ra_4), \qquad b(\infty) = (a_0, \dots, a_4).$$

Then

$$(T - rU)g \in W_f \iff g \in W_{b(r)}.$$

If W_f has trivial gcd, its polar line meets the lower secant cubic in degree at most three.

Proof. The first equivalence follows by expanding the two Hankel equations of $(T - rU)g$. Let

$$C_f = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \\ a_2 & a_3 & a_4 & a_5 \end{pmatrix}.$$

The lower catalecticant equals

$$C_f \begin{pmatrix} -r & 0 & 0 \\ 1 & -r & 0 \\ 0 & 1 & -r \\ 0 & 0 & 1 \end{pmatrix}.$$

Cauchy–Binet gives the secant determinant

$$D_f(r) = -\Delta_{012}r^3 + \Delta_{013}r^2 - \Delta_{023}r + \Delta_{123}.$$

It vanishes identically exactly when $\operatorname{rank} C_f \leq 2$, equivalently when W_f has quadratic gcd. In the trivial-gcd case it therefore has at most three zeros. □

Proposition A.4 (Complete lower trivial-gcd carrier)**.** Let $c \in \mathbb{P}(\Gamma^4 E)$ and suppose that the cubic pencil W_c has trivial gcd. One of the following applies over an algebraic closure, and the last three alternatives exhaust the cases in which the S_3 package in (i) is unavailable:

(i) the cubic map is separable with geometric monodromy S_3, and its off-diagonal fiber square is geometrically integral;
(ii) the characteristic is neither two nor three and the pencil is cyclic, in which case c lies on the closure of

$$\left\{\operatorname{diag}(1, 4, 6, 4, 1)^{-1}[Q^2] : [Q] \in \mathbb{P}(\operatorname{Sym}^2 E^\vee)\right\};$$

(iii) the characteristic is two and c lies on $\Pi_{\text{cyc}} = \mathbb{P}\langle e_1, e_2, e_3\rangle$;
(iv) the characteristic is three and c lies on $\operatorname{Join}(e_2, C_4)$, the closure of the wild cyclic pencils together with the inseparable nucleus point.

Thus these displayed tame, binary, and ternary loci exhaust the trivial-gcd lower pencils for which the S_3 package is unavailable.

Proof. Proposition IV.4 associates to W_c a degree-three map. If it is separable, its transitive geometric monodromy is C_3 or S_3. In the S_3 case two-transitivity makes the off-diagonal fiber square geometrically integral, giving (i). In the cyclic case, Lemma IV.5 identifies the two ramification points with a binary quadratic Q. Substitution in the two Hankel equations gives the divided-power syndrome $\operatorname{diag}(1, 4, 6, 4, 1)^{-1}[Q^2]$ whenever the displayed diagonal is invertible. This proves (ii) in characteristics other than two and three.

The same coefficient calculation can be made without division. In characteristic two it is

$$g \cdot e_2 = [0 : \delta\gamma : \alpha\delta + \beta\gamma : \alpha\beta : 0], \qquad g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}.$$

Its image is dense in the plane $\mathbb{P}\langle e_1, e_2, e_3\rangle$, so its closure is exactly the binary locus in (iii). In characteristic three every separable cyclic pencil is wild. Lemma IV.5 puts it in the PGL_2-orbit of $e_2 - ae_4$. Here e_2 is fixed and the orbit closure of e_4 is the quartic normal rational curve C_4; hence the complete wild closure is $\mathrm{Join}(e_2, C_4)$. The only inseparable trivial-gcd pencil is the all-cube nucleus pencil, namely its vertex e_2, by Proposition IV.13. This proves (iv) and exhausts the inseparable case. The three alternatives are therefore exhaustive. ☐

Proposition A.5 (Cyclic and collision degrees)**.** For a trivial-gcd net:

(i) in characteristic different from two and three, the lower tame cyclic carrier is a binomially rescaled projected quadratic Veronese surface of degree four and contains no polar line;
(ii) the pointed ramification condition has degree at most six and does not vanish identically;
(iii) an S_3 lower pencil with the marked root deleted has total deletion degree at most nineteen.

Proof. In syndrome coordinates the tame cyclic surface is

$$\left\{\mathrm{diag}(1,4,6,4,1)^{-1}[Q^2] : [Q] \in \mathbb{P}(\mathrm{Sym}^2 E^\vee)\right\}.$$

When 2 and 3 are invertible this is the projected quadratic Veronese surface, of degree four, and contains no line. Characteristic three has a different wild closure, treated in Proposition A.6. For (ii), Lemma V.7 with $j = 6$ gives the degree-six bound except possibly in characteristic two. In that characteristic, inseparability would identify W_f, after a change of coordinates, with the full pure-square space $\langle T^4, T^2U^2, U^4\rangle$. Its annihilator is $\langle e_1^\vee, e_3^\vee\rangle$. For the two consecutive Hankel rows this would require simultaneously

$$a_0 = a_2 = a_4 = 0, \qquad a_1 = a_3 = a_5 = 0,$$

contrary to $f \neq 0$. Thus the bound holds in every characteristic.

Finally the R5 off-diagonal curve has deletion degree thirteen, including its possible rational singular point. The unique cubic through a specified marker contributes at most six ordered pairs, giving $(g, \delta, \kappa) = (1, 19, 1)$ and

$$q + 1 - 2\sqrt{q} > 19.$$

Its first prime-power solution is 29. ☐

Proposition A.6 (Modular contained components)**.** In characteristic three the lower wild carrier is the degree-four cone

$$\mathrm{Join}(e_2, C_4),$$

and no polar line of a trivial-gcd net is contained in it. In characteristic two the cyclic carrier is the plane

$$\Pi_{\mathrm{cyc}} = \mathbb{P}\langle e_1, e_2, e_3\rangle,$$

whose unique contained polar-line component is

$$\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle.$$

Proof. Every line on the wild cone is a ruling. By equivariance it suffices to compare the two consecutive rows on one ruling; in the frame $\langle e_2, e_4\rangle$ the conditions

$$(a_0, \dots, a_4),\ (a_1, \dots, a_5) \in \langle e_2, e_4\rangle$$

force the overlapping coordinates successively to vanish. Undoing the frame gives $f = \mu\nu(t)$, including $f = \mu e_5$ at infinity. These are fixed-factor/rank-boundary syndromes, so no trivial-gcd polar line is contained in the wild cone.

In characteristic two, direct substitution gives

$$g \cdot e_2 = [0 : \delta\gamma : \alpha\delta + \beta\gamma : \alpha\beta : 0],$$

whose closure is Π_{cyc}. The consecutive-row overlap is now literal: the first row lies in the plane exactly when $a_0 = a_4 = 0$, and the second exactly when $a_1 = a_5 = 0$. Hence a polar line lies in the plane exactly when

$$a_0 = a_1 = a_4 = a_5 = 0,$$

namely when $f \in \mathbb{P}\langle e_2, e_3\rangle$. This proves both existence and uniqueness of the contained binary component. ☐

Proposition A.7 (Binary nucleus arithmetic)**.** The scheme $\mathcal{N}_3(\mathbb{F}_q)$ is one projective-semilinear orbit of $q + 1$ points. It is split-free over $q = 2^m$ exactly when m is odd.

Proof. For the representative e_3,

$$W_{e_3} = \langle 1, t, t^4\rangle.$$

If $C = 0$, the member $A + Bt + Ct^4$ has degree at most one and cannot have four distinct roots. Assume $C \neq 0$. The root differences of $A + Bt + Ct^4$ then lie in the kernel of $u \mapsto u^4 + (B/C)u$. Four distinct rational roots exist exactly when all roots of $u^3 = B/C$ are rational, equivalently when $3 \mid q-1$, which for $q = 2^m$ means m is even. The stabilizer of e_3 is the Borel of order $q(q-1)$, so the orbit has $q+1$ points; its lower-unipotent transforms and endpoint are precisely $\mathcal{N}_3(\mathbb{F}_q)$. $\square$

Proposition A.8 (Exhaustive R6 contained-component classification)**.** For a redundancy-six syndrome, the alternatives in Definition V.13 are exhausted as follows:

(i) a noninjective contraction map is the rank-one normal-rational-curve boundary and is shallow;
(ii) a polar line contained in the lower secant cubic comes from the upper rank-two catalecticant locus; after its shallow rational secants are removed, this is the persistent tangent and conjugate-secant locus;
(iii) no trivial-gcd polar line is contained in the tame cyclic surface or the characteristic-three wild cone;
(iv) in characteristic two, containment in the cyclic plane occurs exactly for $f \in \mathcal{N}_3 = \mathbb{P}\langle e_2, e_3 \rangle$;
(v) on the remaining trivial-gcd locus the collision divisor is not identically zero.

The exact-linear-gcd lower stratum is not a contained component: it is the separate quadratic-graph package of Proposition IV.3. Consequently the complete nonshallow contained list is the persistent locus together with $\mathcal{N}_3$ in characteristic two. There is no unlisted contained component.

Proof. The noninjective and secant-cubic alternatives are Proposition V.15 and Proposition A.3. The tame cyclic and collision assertions are Proposition A.5; the wild and binary specializations are Proposition A.6. Proposition A.4 proves that these are the complete trivial-gcd lower failure loci, rather than a selected list of examples. Together they are all the clauses of the terminal carrier in Definition V.13: contraction indeterminacy, containment in each component of the lower bad scheme, and identical collision. Hence the list is exhaustive. $\square$

Proposition A.9 (R6 one-step exhaustion)**.** Assume $q \geq 29$. Then every split-free redundancy-six syndrome lies in the persistent quadratic-gcd locus or, when $q = 2^m$ with odd m, in $\mathcal{N}_3(\mathbb{F}_q)$.

Proof. Contained nets. The complete contained branch is Proposition A.8. The common-factor trichotomy is exhaustive. A quadratic gcd gives the persistent tangent and conjugate-secant loci, together with the shallow rational secant; an exact linear gcd is shallow by Lemma A.2. It remains to treat a trivial-gcd net.

Transverse exclusions. Its first-polar line is not contained in the lower secant cubic, by Proposition A.3. The pullback of that cubic has degree at most three. The tame cyclic surface has degree four and contains no polar line. In characteristic three its wild replacement has the same degree and no contained trivial-gcd polar line; in characteristic two the replacement is a plane, whose only contained component is $\mathcal{N}_3$, by Proposition A.6. Finally, Proposition A.5 gives a nonzero collision divisor of degree at most six. Outside the contained binary component, the excluded parameter scheme therefore has degree

$$d \leq 3 + 4 + 6 = 13.$$

Bottom pencils. For every remaining marker, the lower cubic pencil has exact linear gcd or geometric monodromy S_3. In the first case the graph argument of Proposition IV.3 is pointed as follows: besides its eight original deletions, remove the at most two graph points whose split quadratic contains the new marker. Since $q + 1 > 10$, a split squarefree cubic avoiding that marker remains. Here the fixed gcd λ_0 differs from the marker r. Indeed, $\lambda_0 = r$ says that every cubic in $W_{b(r)}$ vanishes at r, so every lifted quartic has a double root there; by Lemma V.6, this is exactly a point of the degree-six collision divisor already placed in Z_f, so no additional deletion budget is needed. In the S_3 case, Lemma IV.11 supplies the geometrically integral off-diagonal curve. Its singular, diagonal, and branch deletions have degree at most thirteen, and deleting the unique cubic through the marker removes at most six ordered pairs. Thus

$$(g, \delta, \kappa) = (1, 19, 1), \qquad q + 1 - 2\sqrt{q} > 19$$

for every $q \geq 29$. Explicitly,

$$\mathcal{H}_1(1, 19) = 1 + \left\lfloor (1 + \sqrt{19})^2 \right\rfloor = 29, \qquad 30 - 2\sqrt{29} > 19.$$

These data form the one-step package of Definition V.4, with $d = 13$, so Theorem V.14 produces a split squarefree quartic.

The modular residue. This is one field-order threshold, not three characteristic-dependent estimates: the first characteristic-three and characteristic-two prime powers above it happen to be 81 and 32, respectively. On $\mathcal{N}_3$, Proposition A.7 gives the asserted odd-degree split-free orbit and proves shallowness in even extension degree. $\square$

The covering-radius gate is complete. Its exact high-rate endpoint is $q = 8$, since $6 = \lfloor 8/2 \rfloor + 2$. Thus Seroussi–Roth nonextendability and Dür's equivalence give $\rho(\mathrm{PRS}(q-5)) = 5$ for $q \geq 8$, while Certificate R6 checks it directly at $q = 7$ and independently confirms $q = 8$.

Theorem A.10 (Redundancy six)**.** For every prime power $q \geq 7$, the deep syndromes of $\mathrm{PRS}(q-5)$ are the persistent locus of size $q(q+1)^2/2$, together with:

(i) respectively $18, 11, 4, 2, 1$ exceptional $\mathrm{PGL}_2(q)$ orbits at $q = 7, 8, 9, 11, 13$;
(ii) one characteristic-two nucleus orbit of size $q+1$ when $q = 2^m$ and odd $m \geq 5$.

The lower bound $m \geq 5$ avoids double counting: the remaining odd case $m = 3$, namely $q = 8$, is already the $|\mathcal{O}| = 9$ row of the exceptional table. There are no other exceptional orbits. The complete persistent orbit law is T/T^5 modulo inversion and Frobenius, with a tangent fixed/nonzero split exactly in characteristic five. At $q = 7, 8, 9, 11, 13$, the exceptional direction totals are

$$5152, \quad 4713, \quad 1800, \quad 792, \quad 546,$$

and the corresponding total numbers of deep syndrome directions are

$$5376, \quad 5037, \quad 2250, \quad 1584, \quad 1820.$$

Proof. Propositions A.8 and A.9 give the complete conceptual exhaustion for every prime power $q \geq 29$. Certificate R6 exhausts

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27\},$$

finds exceptional projective-orbit counts $18, 11, 4, 2, 1$ only at the five displayed fields, and no further orbit at the remaining fields. Together with the separate radius gate, this covers every prime power $q \geq 7$. ☐

The listed sources are disjoint. The persistent locus has quadratic gcd, every finite exceptional row has trivial gcd, and the separate nucleus term occurs only for binary fields beyond the finite table.

The five small tables are exhaustive computations. For an exceptional net W_f, let $I_f(r)$ be the number of irreducible cubic members of the quotient pencil $W_{b(r)}$, including the infinity chart, and define its shared-root pair count by

$$E_f = \sum_{r \in \mathbb{P}^1(\mathbb{F}_q)} \binom{I_f(r)}{2}.$$

In odd characteristic let

$$F_f(T, U) = \sum_{i=0}^{5} a_i T^{5-i} U^i$$

be the binary quintic attached to $f = (a_0 : \cdots : a_5)$, and let $\tau_5(f)$ be its rational factorization type; for example, $1^3 + 2$ denotes a triple rational factor and an irreducible quadratic factor. Their semilinear orbit counts are $18, 5, 2, 2, 1$; E_f, together in odd characteristic with $\tau_5(f)$, separates the classes up to precisely Frobenius fusion. Table VI prints the canonical representatives, stabilizers, orbit sizes, and fusion lengths.

TABLE VI: Exceptional redundancy-six classes. Each row is one projective-semilinear class. The columns give a canonical syndrome representative, the size and stabilizer order of each constituent $\mathrm{PGL}_2(q)$-orbit, the number ϕ of such orbits fused by coefficient Frobenius, the shared-root pair count E, and the binary-quintic factor type τ_5 in odd characteristic.

q	representative	$\lvert\mathcal{O}\rvert$	$\lvert G_f\rvert$	ϕ	E	τ_5
7	$[0:0:0:1:0:1]$	168	2	1	7	1^3+2
7	$[0:0:1:0:3:1]$	336	1	1	19	1^2+1+2
7	$[0:0:1:0:3:2]$	336	1	1	15	1^2+3
7	$[0:1:0:0:0:1]$	168	2	1	10	$1+2+2$
7	$[0:1:0:0:2:0]$	112	3	1	33	$1+1+3$
7	$[0:1:0:1:1:2]$	336	1	1	9	$1+4$
7	$[0:1:0:1:1:5]$	336	1	1	9	$1+1+3$
7	$[0:1:0:1:3:0]$	336	1	1	21	$1+1+1+2$
7	$[0:1:0:1:3:3]$	336	1	1	7	$1+1+3$
7	$[0:1:0:1:3:6]$	336	1	1	14	$1+1+3$
7	$[0:1:0:3:0:5]$	168	2	1	11	$1+4$
7	$[0:1:0:3:0:6]$	168	2	1	20	$1+4$
7	$[0:1:0:3:2:1]$	336	1	1	13	$1+1+3$
7	$[1:0:0:0:1:2]$	336	1	1	7	$1+4$
7	$[1:0:0:0:1:3]$	336	1	1	23	5
7	$[1:0:0:3:1:3]$	336	1	1	13	$1+4$
7	$[1:0:1:0:5:2]$	336	1	1	8	$1+4$
7	$[1:0:1:0:6:2]$	336	1	1	8	5

q	representative	$\lvert\mathcal{O}\rvert$	$\lvert G_f\rvert$	ϕ	E	τ_5
8	$[0:0:0:1:0:0]$	9	56	1	0	–
8	$[0:1:1:0:2:1]$	504	1	3	20	–
8	$[0:1:1:0:2:5]$	504	1	3	19	–
8	$[1:0:0:1:3:6]$	504	1	3	14	–
8	$[1:0:1:3:5:2]$	168	3	1	54	–
9	$[0:1:0:0:0:4]$	180	4	2	28	$1+4$
9	$[0:1:0:1:4:2]$	720	1	2	29	$1+4$
11	$[0:0:0:1:0:0]$	132	10	1	60	1^3+1^2
11	$[0:1:0:2:0:10]$	660	2	1	30	$1+4$
13	$[0:1:0:0:0:2]$	546	4	1	60	$1+4$

For $q=8$, integers are binary polynomial-basis codes with $\alpha^3=\alpha+1$; for $q=9$, they are ternary polynomial-basis codes with $\alpha^2=-1$. Thus $\sum\phi=(18,11,4,2,1)$ and $\sum\phi|\mathcal{O}|=(5152,4713,1800,792,546)$, in the field order $7,8,9,11,13$.

B. Redundancy seven

For

$$f=(a_0:\ldots:a_6)\in\mathbb{P}(\Gamma^6E),\qquad W_f=\ker\begin{pmatrix}a_0&a_1&a_2&a_3&a_4&a_5\\a_1&a_2&a_3&a_4&a_5&a_6\end{pmatrix},$$

the system W_f is a four-dimensional web of quintics. For an affine marker r the contractions are

$$b(r)=(a_1-ra_0,\ldots,a_6-ra_5).$$

Coefficient collection gives

$$(t-r)g\in W_f\quad\iff\quad g\in W_{b(r)};$$

the lift is squarefree exactly when $g(r)\neq 0$.

The contracted net $W_{b(r)}$ has quadratic gcd exactly when its three-row catalecticant has rank at most two. Unless the first-polar line is contained in that rank-two locus, these parameters form the degree-at-most-three intersection of Proposition A.3; the contained case is Corollary A.15.

Proposition A.11 (Complete two-marker lower package)**.** Let W_h be a trivial-gcd quartic net and let x be a prescribed forbidden root. Assume in characteristic two that $h\notin\mathcal{N}_3$. For every prime power $q\geq 37$, the net contains a split squarefree quartic avoiding x.

Proof. Choosing the second marker. By equivariance choose coordinates in which x is finite; all parameter equations below are then homogenized in s, so the infinity chart is included. Choose the second contraction marker s. The unavailable values of s comprise: $s=x$, of degree one; at most three intersections with the lower secant cubic, by Proposition A.3; at most four intersections with the cyclic surface or its characteristic-three wild replacement; and at most six points of the collision divisor, by Propositions A.5 and A.6. In characteristic two the cyclic replacement is a plane, so its pullback has degree at most one unless the polar line is contained in it, the excluded case $h\in\mathcal{N}_3$, by Proposition A.6. We must also exclude the parameters for which the lower pencil has fixed factor x. They are cut out by the 3-minors obtained by appending the fixed evaluation row $(1,x,x^2,x^3)$ to the two Hankel rows of $b(s)$, and hence have degree at most two in s. These minors are not all zero. If they vanished identically, then

$$W_h\cap\{g:g(s)=0\}\subseteq\{g:g(x)=0\}$$

for every geometric s. Given $g\in W_h$, choose a geometric root s of g; then $g(x)=0$. Thus x would divide every member of W_h, contradicting the trivial-gcd hypothesis. Hence the complete second-step parameter scheme has degree at most

$$1+3+4+6+2=16<q+1.$$

Choose s outside it.

The linear-gcd branch. The resulting cubic pencil is neither persistent nor cyclic. If it has exact linear gcd λ_0, then $\lambda_0\notin\{x,s\}$: equality with x is excluded by the fixed-factor minors above, while equality with s is the pointed collision excluded by Proposition A.5. The graph proof of Proposition IV.3 has eight original deletions; avoiding the two additional markers costs at most two graph points each. Since $q+1>12$, a split squarefree cubic avoiding x and s remains.

The S_3 branch. Otherwise the bottom cover has geometric monodromy S_3. Its off-diagonal curve is geometrically integral of arithmetic genus one. The singular, diagonal, and branch deletions cost at most thirteen, and the fibers through x and s cost at most six ordered pairs each. Hence

$$(g,\delta,\kappa)=(1,25,1),\qquad q+1-2\sqrt{q}>25,$$

whose first prime-power solution is 37. In either case the cubic lifts through s to a split squarefree quartic in W_h avoiding x. $\square$

Proposition A.12 (Exact-gcd-one avoidance)**.** Suppose the lower net is $\ell_x U$, where $U = \ker \lambda \subset \operatorname{Sym}^3 E^\vee$ has dimension three and $\gcd(U) = 1$, so that ℓ_x is the exact gcd of the lower net. If $x = r$, the parameter belongs to the collision divisor. If $x \neq r$ and $q \geq 16$, then U contains a split cubic avoiding both x and r.

Proof. If $x = r$, every lifted quintic has the repeated factor $(t-r)^2$. Lemma V.6 identifies this with the degree-eight collision divisor of Proposition A.13. This is the unavailable branch, so assume $x \neq r$. Put $S = \mathbb{P}^1(\mathbb{F}_q) \setminus \{x, r\}$. For ordered distinct $u, v \in S$, the map $w \mapsto \lambda(\ell_u \ell_v \ell_w)$ is linear. If it vanishes identically, choose any remaining w; otherwise its homogenization is a nonzero linear form on $\mathbb{P}^1$, so it has exactly one rational projective zero, including the point at infinity when appropriate. The equations $w = x, r, u, v$ have respective bidegrees $(1,1), (1,1), (2,1), (1,2)$ in (u, v). None is identically zero: the first two would give a common factor of U, contrary to $\gcd(U) = 1$, and the last two are excluded because double-root cubics span $\operatorname{Sym}^3 E^\vee$. A nonzero bidegree-$(a,b)$ divisor on $\mathbb{P}^1 \times \mathbb{P}^1$ has at most $(a+b)(q+1)$ rational points. Thus the bad pairs number at most $10(q+1)$, whereas $(q-1)(q-2) > 10(q+1)$ for $q \geq 16$. $\square$

Proposition A.13 (Collision divisor)**.** After fixed factors are removed, the collision divisor of the moving g^3_5 has degree at most eight.

Proof. This is Lemma V.7 with $j = 7$: the series is separable and $2j - 6 = 8$. $\square$

The characteristic-two nucleus line at redundancy six has exactly one coherent lift, the central point e_3, with

$$W_{e_3} = \langle 1, t, t^4, t^5 \rangle.$$

It is split-free for odd m, and hence deep whenever the separate radius-six gate holds. For even m, the homogenization of $t^4 + t$ has the five roots $\mathbb{P}^1(\mathbb{F}_4)$.

Proposition A.14 (Central binary lift)**.** The complete inverse image of the R6 nucleus line under consecutive sextic contraction is $[e_3]$. It is split-free over $\mathbb{F}_{2^m}$ exactly when m is odd.

Proof. Putting both consecutive rows in $\mathbb{P}\langle e_2, e_3 \rangle$ gives $a_0 = a_1 = a_2 = a_4 = a_5 = a_6 = 0$. Thus only e_3 remains, with $W_{e_3} = \langle 1, t, t^4, t^5 \rangle$. For odd m, contraction at a root of a hypothetical split quintic would give a split quartic on the R6 nucleus line, contradicting Proposition A.7. For even m, $\mathbb{F}_4 \subset \mathbb{F}_q$ and the homogenization of $t^4 + t$ has its four affine $\mathbb{F}_4$ roots and the simple root at infinity. $\square$

Corollary A.15 (Exhaustive R7 contained-component classification)**.** The assertion $\mathrm{CC}(6,1)$ is exhaustive in every characteristic. A noninjective contraction is the shallow rank-one boundary. Containment in the lower persistent locus gives upper catalecticant rank at most two; after the shallow rational secants are removed, these are exactly the persistent tangent and conjugate-secant families. The complete lift of the only additional lower component, the characteristic-two nucleus line, is the central point e_3. The remaining collision divisor is nonzero. Thus the nonshallow contained list is precisely the persistent locus and, in characteristic two, e_3; there is no further contained component.

Proof. Apply Proposition V.15 with $n = 6$. The three rational types of the quadratic recurrence are split squarefree, repeated rational, and irreducible. They give respectively the shallow rational secant, tangent, and conjugate-secant loci. The remaining lower binary nucleus line has the separate complete lift computed in Proposition A.14. Proposition A.13 shows that the collision divisor is nonzero. These are respectively the contraction-indeterminacy, lower-component-containment, and identical-collision alternatives of Definition V.13, so the list is exhaustive. $\square$

Theorem A.16 (Redundancy-seven classification)**.** Certificate R7 unconditionally classifies

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32\}.$$

For every prime power $q \geq 13$, the split-free syndromes at redundancy seven are the persistent locus of size $q(q+1)^2/2$, together with the central nucleus point when $q = 2^m$ and m is odd. At $q = 7, 8, 9, 11$ the additional split-free orbit-size profiles, after removing the persistent locus and also the central nucleus singleton at $q = 8$, are:

$$\begin{array}{r|l} 7 & 56,\ 84^5,\ 112^2,\ 168^{45},\ 336^{141} \\ 8 & 63,\ 72,\ 84^3,\ 168^4,\ 252^{24},\ 504^{86} \\ 9 & 180^3,\ 240^6,\ 360^{18},\ 720^{27} \\ 11 & 264^2,\ 440,\ 660^2. \end{array}$$

This is the complete split-free classification for all prime powers $q \geq 7$. Since Seroussi–Roth nonextendability and Dür's equivalence give $\rho(\mathrm{PRS}(q-6)) = 6$ for every prime power $q \geq 11$, it is also the complete deep-hole classification in that range. At $q = 7, 9$ the table is not promoted without a separate covering-radius calculation. At $q = 8$, where [3, Thm. 17(1)] gives radius seven, the diagonal tangent orbit (nine directions) and the fixed central nucleus are exactly the distance-seven rows; exhaustive replay puts all 45 other frozen representatives and the remaining three persistent representatives at distance six. Thus the $q = 8$ classification is complete. The persistent orbit law is T/T^6 modulo inversion and Frobenius, with tangent splitting in characteristics two and three.

Proof. For fixed irreducible quadratic Q, the persistent fiber has $q+1$ points, while for $Q = \lambda^2$ it has q non-rank-one points. There are $q(q-1)/2$ irreducible quadratics and $q+1$ rational squares. Hence the persistent count is

$$\frac{q(q-1)}{2}(q+1) + (q+1)q = \frac{q(q+1)^2}{2}.$$

For $q \geq 37$, choose a first marker outside the at most three catalecticant intersections, eight collision parameters, and, in characteristic two, one intersection with the nucleus line. If the contracted quartic net has trivial gcd, Proposition A.11 supplies the complete second one-step package: it excludes the cyclic/wild carrier and fixed-gcd/marker coincidences before applying either the pointed linear-gcd graph or the S_3 cover. If the contracted net has exact linear gcd, write it as $\ell_x U$. Exactness gives $\gcd(U) = 1$, so Proposition A.12 applies; its excluded $x = r$ case is already among the eight collision parameters. These first-marker bounds and Proposition A.11 are exactly the depth-two data of Theorem V.14; the terminal curve has $(g, \delta, \kappa) = (1, 25, 1)$. Corollary A.15 leaves exactly the persistent locus and Proposition A.14. Certificate R7 covers

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32\}$$

by the exact primary quotient enumeration. A separate replay checks the recorded representatives and their orbits. An independent direct-locus replay imports no R7 generator or quotient partition: below 16 it constructs the literal four-secant complement, from 16 onward it constructs the classified R6 pointed locus directly, including the transient marked orbit at $q = 19$, and in every field it intersects all contraction conditions and rebuilds the complete $\mathrm{PGL}_2(q)$-orbit and Frobenius partition. It agrees exactly with the public record in all fourteen fields. The radius theorem promotes the split-free list only from $q \geq 11$. $\square$

For the four finite rows Certificate R7 contains the canonical representative, stabilizer, persistent/central flags, and Frobenius link of every orbit. The displayed size profile is only a compact summary; repeated sizes are distinguished by those canonical records. There are respectively $194, 119, 54, 5$ exceptional representatives at $q = 7, 8, 9, 11$; the complete lists are printed in `supplement/CLASSIFICATION-RECORDS.md`, while the matching JSON is the machine-readable record. The following gives one printed canonical representative for every orbit-size block in the displayed profile; repeated orbits within a block are distinguished in the full list:

q	$\lvert\mathcal{O}\rvert$	representative
7	56	$[0:0:0:0:1:0:0]$
	84	$[0:1:0:0:0:1:0]$
	112	$[0:0:1:0:0:2:0]$
	168	$[0:0:0:0:1:0:1]$
	336	$[0:0:0:1:0:1:1]$
8	63	$[0:0:0:1:0:0:1]$
	72	$[0:0:0:0:1:0:0]$
	84	$[0:1:1:1:1:3:6]$
	168	$[1:0:1:1:3:7:7]$
	252	$[0:0:1:0:1:0:0]$
	504	$[0:0:1:0:1:0:1]$
9	180	$[0:1:0:0:0:4:0]$
	240	$[0:0:1:0:1:1:1]$
	360	$[0:1:0:1:3:3:2]$
	720	$[0:0:1:1:0:2:2]$
11	264	$[0:1:0:0:0:0:2]$
	440	$[1:0:1:1:4:3:3]$
	660	$[1:0:1:0:1:4:8]$.

For $q = 8, 9$, the coordinates use the same polynomial-basis encoding specified below Table VI.

The $q = 19$ lower net

$$W = \langle 1, t^3, t^4 \rangle$$

shows why the parameter must remain marked. Its split squarefree members are exactly

$$U(T^3 - uU^3), \qquad u \in (\mathbb{F}_{19}^\times)^3 = \{1, 7, 8, 11, 12, 18\}.$$

All six contain the marked point at infinity. The obstruction is pointed rather than intrinsic to W: at the marker 0, for example, $U(T^3 - U^3)$ is an admissible member. No coherent sextic polar line lies in the infinity-pointed orbit. Thus the R7 proof escapes over the marker parameter; classifying unpointed bad lower fibers would retain a false exception.

TABLE VII
FIELD-RANGE CLOSURE LEDGER. "CERTIFICATE" MEANS A COMPLETE FINITE CLASSIFICATION WITH CANONICAL REPRESENTATIVES, NOT MERELY AN ORBIT COUNT.

Result	prime powers	closure mechanism
R5	$7, 8, 9, 11, 13, 17, 19$	direct census and independent replay
	16	characteristic-two branch budget, Corollary IV.10; certificate row retained as a regression check
	$q \geq 23$	cubic-cover geometry, Lemma IV.11
R6	$7, 8, 9, 11, 13, 16$	direct syndrome/radius census
	$17, 19, 23, 25, 27$	finite structural bridge certificate
	$q \geq 29$	marked-cover point bound and contained components
R7	$7, 8, 9, 11$	direct split-free census
	$13, 16, 17, 19, 23, 25, 27, 29, 31, 32$	coherent-polar finite bridge certificate
	$q \geq 37$	marked-cover point bound and contained components

The R7 rows classify split-free syndromes. Conversion to code deep-hole directions uses the separate covering-radius gate and is made only for $q \geq 11$. The public certificate schema and exact replay boundaries are given in the supplement; the table does not promote orbit-size profiles to identifiers.

There is no prime power in any of the open numerical intervals $(19, 23)$, $(27, 29)$, or $(32, 37)$. Thus adjacent finite and geometric rows in Table VII leave no unlisted field order.

APPENDIX B
FIXED-LEVEL PROOFS AT REDUNDANCIES EIGHT AND NINE

This appendix discharges the three- and four-marker packages needed after the recursive carrier theorem. At R8 the bottleneck is the integral off-diagonal $(2, 2)$ cover of a cubic pencil; at R9 it is rational selection on the residual-quadratic slices of quartic moduli. The component and deletion calculations below close Theorems B.11 and B.22.

A. Redundancy eight

Proposition B.1 (Generic three-marker bottom cover)**.** After three contractions, the generic bottom object is the R5 off-diagonal $(2, 2)$ curve. The diagonal and branch deletion costs twelve and three marked roots cost eighteen, so $\delta = 30$. The inequality

$$q + 1 - 2\sqrt{q} > 30$$

holds at every prime power $q \geq 43$. The first-polar line has transverse/collision budget

$$3 + 1 + 10 = 14,$$

where ten is the ramification degree of a moving g^4_6.

Proof. Markers do not change the bottom curve or its geometric monodromy; each removes at most the six ordered pairs in its unique cubic fiber. The point bound is the $\kappa = 1$ Aubry–Perret form used throughout. A noncontained line meets the lower rank-two catalecticant carrier in degree at most three, and the lower binary central point in at most one parameter. After fixed factors are removed, self-collision is ramification of a separable g^4_6, of degree $(4+1)(6-4) = 10$. Separability follows because the p-th-power subspace has dimension smaller than the five-dimensional series. □

Definition B.2 (Three-marker lower-package gate)**.** The assertion $\mathrm{LP}(6, 1)$ says that the pointed R7 splitting incidence, after the persistent rank-two carrier, the binary central point, and the marked-collision divisor are removed, has the geometrically integral identity twists and deletion bounds used in Proposition B.1. It includes equations and monodromy checks for the recursive cyclic, wild, inseparable, gcd, and branch strata; a finite-field regression is not a substitute for those geometric data.

The ordered cover retains the marker labels needed for monodromy, but the contraction itself descends to the unordered marker cubic. Indeed, for affine markers r, s, t,

$$a_i^{\langle r,s,t\rangle} = a_{i+3} - (r + s + t)a_{i+2} + (rs + rt + st)a_{i+1} - rst\, a_i. \tag{13}$$

Thus it depends only on the elementary symmetric functions of the markers, equivalently on the monic cubic $T^3 - (r + s + t)T^2 + (rs + rt + st)T - rst$. This separates the symmetric contraction parameter from the ordered-root cover used below.

Lemma B.3 (Bottom strata)**.** For the two-step contraction of a pointed sextic, the lower rank, gcd-one, cyclic, wild, inseparable, and branch strata are exactly the closed schemes displayed below. The reversed homogeneous chart introduces no additional stratum.

Proof. For a pointed sextic $f=(a_0,\dots,a_6)$ with old marker x, contract successively:

$$b_i(r)=a_{i+1}-ra_i,\qquad c_i(r,s)=b_{i+1}(r)-sb_i(r).$$

For $c=(c_0,\dots,c_4)$ put

$$H_c=\begin{pmatrix}c_0&c_1&c_2&c_3\\c_1&c_2&c_3&c_4\end{pmatrix}.$$

On the rank-two-kernel chart, if p_0,p_1 span $\ker H_c$, the bottom ordered-root curve is

$$G_c(u,v)=\frac{p_0(u)p_1(v)-p_1(u)p_0(v)}{[u,v]}=0. \tag{14}$$

The rank-at-most-two secant carrier, including the gcd-two and rank-drop boundary strata, is

$$D(c)=\det\begin{pmatrix}c_0&c_1&c_2\\c_1&c_2&c_3\\c_2&c_3&c_4\end{pmatrix}=0. \tag{15}$$

For an exact common root z, the gcd-one incidence is cut out by

$$\operatorname{rank}\begin{pmatrix}H_c\\1\quad z\quad z^2\quad z^3\end{pmatrix}\le 2. \tag{16}$$

with the evident homogeneous row at infinity. After substitution, (15) has degree at most three in s, while each 3-minor in (16) has degree at most two for fixed z.

The remaining reducible-monodromy carriers are explicit. Put $z_i=\binom{4}{i}c_i$. Outside characteristics two and three the cyclic closure is

$$[u:v:w]\longmapsto[u^2:2uv:v^2+2uw:2vw:w^2], \tag{17}$$

a degree-four projected Veronese surface containing no line. In characteristic three it is the degree-four cone $\bowtie(e_2,C_4)$; its only lines are rulings, and the consecutive-row overlap excludes every ruling as a polar line. In characteristic two it is

$$\Pi_{\rm cyc}=\mathbb{P}\langle e_1,e_2,e_3\rangle,\qquad c_0=c_4=0. \tag{18}$$

whose unique contained polar-line component is $\mathcal{N}_3=\mathbb{P}\langle e_2,e_3\rangle$. The inseparable degeneration is the derivative-minor locus of p_0,p_1 and introduces no further stratum. Outside characteristics two and three a nonconstant degree-three map cannot factor through Frobenius. In characteristic three an inseparable cubic pencil is projectively $\langle X^3,Y^3\rangle$; the two Hankel rows then force $c_0=c_1=c_3=c_4=0$, the vertex e_2 of the wild cone. In characteristic two, after cancelling a nonzero pencil gcd, an inseparable ratio has degree at most two and is a rational function of t^2. Homogenizing its numerator and denominator to cubics supplies the same linear factor to both, so it belongs to the exact-gcd-one boundary. Finally, the branch equation on this chart is $p_0'p_1-p_0p_1'=0$. These carrier and branch loci are the scheme-theoretic images of the displayed incidences (equivalently their Fitting-minor ideals); the reversed homogeneous chart supplies infinity. Thus these are closed strata over the coefficient ring, not a classification only of their rational points. □

Lemma B.4 (Bottom monodromy and deletion)**.** Off the strata of Lemma B.3, the bottom ordered-root curve is geometrically integral of arithmetic genus one and has deletion degree at most 30. On the exact-gcd-one stratum, the residual deck graph supplies a squarefree pair avoiding the three markers for $q\ge 43$.

Proof. Off (15)–(18), the separable cubic map is noncyclic, so its geometric monodromy is S_3. Transitivity on ordered pairs of distinct roots makes (14) geometrically integral. It has arithmetic genus one. The diagonal costs four, the degree-four different costs at most eight off-diagonal ordered pairs, and each of the three marker fibers x,r,s costs at most six. Thus

$$\delta\le 4+8+3\cdot 6=30. \tag{19}$$

The Aubry–Perret bound for a reduced geometrically integral projective curve of arithmetic genus one supplies a surviving rational point for every $q\ge 43$ [30, p. 468].

It remains to show that the strata removed above do not hide another component. On an exact-gcd-one bottom stratum write the cubic pencil as ℓ_zQ, with Q a base-point-free pencil of quadratics. In the separable case its deck involution ι is rational, so its graph has $q+1$ rational points. At most two are fixed and at most two ordered graph points involve each of z,x,r,s. A point remains for $q\ge 43$, giving three distinct roots outside the markers. The inseparable degree-two case is on the rank/boundary carrier. □

Lemma B.5 (Two-marker selection)**.** Fix two old markers x,r. Outside the declared contained strata, the bad divisor on the internal s-line has degree at most 19. For $q\ge 43$ one can choose s so that the lower package supplies a split squarefree quartic avoiding x and r.

Proof. The complete divisor table on the s-line is

failure	defining mechanism	degree
rank at most two	$D(c(s))$	3
cyclic/wild monodromy	pullback of (17)–(18)	4
new gcd root s	$\mathrm{Ram}(g_d^2),\ d \le 4$	6
gcd root x or r	one nonzero minor in (16)	$2+2$
marker collision	$s = x,\ s = r$	2

In characteristic two the cyclic entry has degree one, so degree four is a uniform bound. The total is

$$3 + 4 + 6 + 2 + 2 + 2 = 19 < q + 1. \tag{20}$$

The rank pullback cannot be the whole line: Proposition A.3 identifies that containment exactly with quadratic gcd of the original net. The only possible contained cyclic line is the declared binary nucleus. For a fixed marker, not all minors in (16) can vanish identically without giving the original quartic net that fixed factor: otherwise the hyperplanes $W \cap \ker(\mathrm{ev}_s)$ and $W \cap \ker(\mathrm{ev}_x)$ agree for dense s, making the evaluation map constant. Choose one nonzero quadratic minor. For self-collision, take a generic pencil projection of the moving g_d^2, $d \le 4$. It is separable: in characteristic two the only dimension-three equality case is the pure-square space $\langle 1, t^2, t^4 \rangle$, which the consecutive Hankel overlap equations do not realize; all other p-power spaces have smaller dimension. Riemann–Hurwitz gives degree at most $2d - 2 \le 6$, and every self-collision maps into this ramification divisor. Choosing s outside this divisor and applying the preceding gcd-zero or gcd-one bottom analysis gives a split squarefree quartic avoiding the two old markers. This proves the two-old-marker lemma. □

Lemma B.6 (Outer marker selection)**.** For a pointed sextic outside the persistent and modular contained strata, the bad divisor on the outer first-marker line has degree at most 15. A surviving outer marker lifts the witness of Lemma B.5 to a split squarefree quintic avoiding the original marker.

Proof. The lower rank-two pullback costs three parameters and the binary nucleus costs one. If a contracted net has exact gcd ℓ_z, the condition $z = r$ maps into the ramification of a generic separable pencil projection of the moving g_d^3, $d \le 5$. Riemann–Hurwitz bounds it by $2d - 2 \le 8$. The condition $z = x$ is again (16), now with the quartic evaluation row, and costs at most two because not all of its quadratic minors vanish off the fixed-marker boundary, by the same evaluation-hyperplane argument. Together with $r = x$, the internal divisor has degree

$$3 + 1 + 8 + 2 + 1 = 15. \tag{21}$$

If the contracted net has trivial gcd, the two-old-marker lemma applies. If it has exact gcd with $z \notin \{x, r\}$, the ordered-pair argument has three $(1,1)$ and two $(2,1)/(1,2)$ bad equations, at most $12(q+1) < (q-2)(q-3)$. The cases $z = x, r$ are precisely the excluded quadratic and ramification divisors. Lifting first by ℓ_s and then by ℓ_r gives a split squarefree quintic avoiding x. □

Proposition B.7 (Pointed lower package)**.** The assertion $\mathrm{LP}(6,1)$ holds. On the outer first-marker line its exceptional divisor has degree at most fifteen; on the recursive second-marker line the corresponding divisor has degree at most nineteen. Its only positive-genus identity twist is the geometrically integral off-diagonal $(2,2)$ curve of genus at most one with deletion degree 30.

Proof. Lemmas B.3 and B.4 identify the bottom strata and the unique positive-genus identity twist. Lemmas B.5 and B.6 give the two parameter-line budgets and the coherent squarefree lift. For auditability, the complete exhaustion is summarized here.

Stratum	Defining equation	Degree	Contained case	Transverse conclusion
rank at most two	(15)	3 in the marker	quadratic-gcd persistent carrier	finite intersection of degree at most 3
cyclic or wild	(17) or (18)	at most 4	only the binary nucleus line	finite intersection of degree at most 4
exact gcd one	minors in (16)	2 per fixed marker	fixed-factor boundary	the residual deck graph supplies a squarefree pair
inseparable	derivative minors of p_0, p_1	absorbed above	wild vertex, rank drop, or binary nucleus	no additional stratum
geometric S_3	(14)	bidegree $(2,2)$	only the preceding carriers	integral genus-one cover with deletion degree 30
marker collision	diagonals and ramification	1 per diagonal; at most 6 for self-collision	declared collision boundary	finite divisor on the marker line

The gcd has degree zero, one, or at least two, and a separable gcd-zero cubic cover has monodromy S_3 or C_3; hence the displayed list is exhaustive. No certificate or finite-field regression is used for this integrality assertion. Notice also that the outer and recursive parameter budgets 15 and 19 are already strictly below $q + 1$ for $q \ge 19$; the threshold 43 is imposed solely by the genus-one deletion inequality (19), not by a hidden parameter-choice bound. □

Proposition B.8 (Modular lifts at degree seven)**.** The consecutive-row Lucas calculation leaves no nonzero lift in characteristic two, the line $\mathbb{P}\langle e_2, e_5\rangle$ in characteristic three, and the line $\mathbb{P}\langle e_3, e_4\rangle$ in characteristic five. The latter two lines are shallow in the range $q \geq 43$.

Proof. Lucas' criterion applied to the consecutive binomial rows gives the three stated supports; the exact row reductions are replayed by Certificate R8. In characteristic five, the homogenization of $t^5 - t$ supplies six distinct projective roots and satisfies the two required middle-coefficient equations throughout the lift.

In characteristic three, $\mathrm{PGL}_2(q)$ is transitive on the rational points of $\mathbb{P}\langle e_2, e_5\rangle$, so it suffices to treat e_2. Choose nonzero a, b, c with no two equal up to sign and $a^2+b^2+c^2 = 0$. The nonsingular conic has $q+1$ points; deleting the three coordinate and six equality/antiequality sections costs at most eighteen, so a choice exists for $q \geq 27$. The reciprocal pairs $\pm a^{-1}, \pm b^{-1}, \pm c^{-1}$ give six distinct roots and the two required elementary-symmetric coefficients vanish. □

Proposition B.9 (Degree-seven contained components)**.** The assertion $\mathrm{CC}(7, 1)$ for the structural components of Definition B.2 holds in every characteristic. The rank-two component lifts to the upper rank-two carrier; the lower binary central point has no nonzero coherent lift; and the marked-collision scheme is never the whole polar line. The only additional nucleus lifts are

$$\mathbb{P}\langle e_2, e_5\rangle \quad \text{in characteristic three,} \qquad \mathbb{P}\langle e_3, e_4\rangle \quad \text{in characteristic five,}$$

and both are shallow in the range $q \geq 43$.

Proof. The rank-two assertion is Proposition V.15 with $n = 7$. For the binary central point e_3, requiring both consecutive septic rows to be supported at its single coordinate permits a_3 in the first row and a_4 in the second; their overlap is zero. Thus its projective coherent lift is empty. The full nucleus-support overlap is exactly Proposition B.8, which also supplies the shallow witnesses in characteristics three and five.

It remains to exclude an identically colliding series. Remove the fixed divisor from the five-dimensional W_f and let the moving degree be $d \leq 6$. If its projective map were inseparable in characteristic p, it would factor through Frobenius, so its section space would have dimension at most

$$\lfloor d/p \rfloor + 1 \leq 4,$$

contrary to $\dim W_f = 5$. A generic pencil projection is therefore a separable map of degree at most six. Every marked self-collision is a ramification point of that projection, so Riemann–Hurwitz bounds the collision divisor by $2d - 2 \leq 10$. It is finite and cannot be an additional contained component. □

Remark B.10 (Exact remaining R8 gate). Proposition B.9 closes the contained-component problem, while Proposition B.7 closes the pointed lower package. The certificate checks the nucleus overlaps, witnesses, and numerical budgets; the geometric integrality and recursive carrier equations are proved separately above.

Theorem B.11 (Redundancy eight)**.** For every prime power $q \geq 43$, the deep syndromes of $\mathrm{PRS}(q-7)$ are exactly the persistent tangent and conjugate-secant families, of total size $q(q+1)^2/2$. Their orbit law is T/T^7 modulo inversion and Frobenius. The tangent family splits exactly in characteristic seven.

Proof. Outside the contained locus, Proposition B.1 chooses a good first contraction, and Proposition B.7 supplies a three-marker witness. Propositions B.8 and B.9 eliminate every nonpersistent contained lift. Proposition VI.5 promotes the resulting split-free classification to deep holes. □

This theorem makes no assertion about additional deep orbits for $q < 43$.

B. Redundancy nine

Definition B.12 (R9 pointed domain)**.** The assertion $\mathrm{LP}(7, 1)$ is made on the pointed splitting incidence after removing the persistent rank-two carrier, the complete characteristic-specific Lucas-nucleus support, and the fixed-marker and self-collision components. At every recursive level, a polar line contained in one of these supports belongs to that declared closed boundary; the parameter-divisor argument applies only to the noncontained line. Thus a section that vanishes identically is never counted as a finite divisor of its nominal degree.

Proposition B.13 (Four-marker pointed lower package)**.** The assertion $\mathrm{LP}(7, 1)$ holds for every $q \geq 53$. Its bottom identity twist is the geometrically integral off-diagonal $(2, 2)$ curve of Proposition B.7, now with four forbidden markers and deletion degree

$$4 + 8 + 4 \cdot 6 = 36. \tag{22}$$

The three recursive parameter-line budgets, from the bottom upward, are at most 22, 18, and 18. At the top degree-eight first-polar line the transverse/component budget is at most

$$3 + 2 + 12 = 17. \tag{23}$$

No geometric-integrality, gcd, or contained-component assertion is included merely as a number in these budgets.

Proof. We retain the notation and the scheme-theoretic strata (14)–(18). Their homogeneous incidence ideals commute with further contraction, so adding a marker changes only the corresponding evaluation and diagonal pullbacks; it does not change the bottom cover. We prove the package in three steps.

Three-pointed quartic net. Let W be a trivial-gcd Hankel net of quartics, let x_1, x_2, x_3 be distinct prescribed markers, and exclude the characteristic-two nucleus already listed in (18). Contract at a fourth parameter s. On the resulting cubic pencil the rank pullback (15), the cyclic/wild pullback (17)–(18), and self-collision have degrees at most 3, 4, and 6, respectively. For each x_i, choose a nonzero quadratic minor in (16); the three fixed-root pullbacks cost at most 6, and the three equations $s = x_i$ cost 3. Thus the parameter divisor has degree at most

$$3 + 4 + 6 + 3 \cdot 2 + 3 = 22. \tag{24}$$

None of these equations vanishes identically. For the rank and cyclic carriers this is the no-contained-line calculation following (20). For a fixed marker, identical vanishing of all the minors in (16) would make the kernels of $\mathrm{ev}_s|_W$ and $\mathrm{ev}_{x_i}|_W$ agree for dense s. The two nonzero functionals would then be proportional, making the projective evaluation map of the three-dimensional trivial-gcd net W constant, a contradiction. For self-collision, a generic pencil projection of the moving g^2_d, $d \leq 4$, is separable by the pure-power dimension argument following (20), and Riemann–Hurwitz gives degree at most 6.

Away from this divisor, a gcd-zero cubic pencil is on the geometric S_3 stratum. Its ordered-pair cover (14) is geometrically integral of genus at most one. The diagonal and different delete at most $4+8$ points, and each of x_1, x_2, x_3, s deletes at most the six ordered pairs in its cubic fiber. This proves (22). If instead the cubic pencil has exact gcd ℓ_z, cancel it and use the graph of the rational deck involution of the residual quadratic pencil. The graph has $q+1$ rational points; at most two are fixed, and at most two involve each of z, x_1, x_2, x_3, s. Since $q + 1 > 12$, a graph point remains. In the inseparable quadratic case the characteristic is two and, after a root-coordinate change, the residual pencil is $\langle T^2, U^2 \rangle$. On the affine chart $\ell_z = t - z$, its two kernel cubics have coefficient vectors

$$(-z, 1, 0, 0), \qquad (0, 0, -z, 1).$$

The two overlapping Hankel rows give successively

$$c_1 = zc_0, \quad c_2 = z^2 c_0, \quad c_3 = z^3 c_0, \quad c_4 = z^4 c_0.$$

Hence the determinant (15) vanishes; the homogeneous $z = \infty$ chart gives the endpoint. Thus the inseparable case really lies on the rank-drop carrier. Multiplication by ℓ_s gives a split squarefree quartic avoiding x_1, x_2, x_3.

Two-pointed quintic series. Let $f \in S_6$ carry two distinct markers x_1, x_2 and lie outside the lower persistent, binary-nucleus, fixed-marker, and collision boundary of Definition B.12; contract at r. The lower rank-two and binary-nucleus pullbacks then cost at most $3 + 1$. If the contracted quartic net has exact gcd ℓ_z, the condition $z = r$ lies in the ramification divisor of a generic separable projection of the moving g^3_d, $d \leq 5$, and costs at most 8. The conditions $z = x_i$ are the two fixed-evaluation incidences (16), each of degree at most two; $r = x_i$ contributes two diagonals. Hence

$$3 + 1 + 8 + 2 \cdot 2 + 2 = 18. \tag{25}$$

The same rank, evaluation-hyperplane, and separable-projection arguments show that no listed pullback is the whole r-line off the declared boundary.

For a trivial-gcd contracted net, the preceding three-pointed lemma applies to x_1, x_2, r. For exact gcd ℓ_z away from these markers, cancel ℓ_z. In the ordered-pair parametrization of the residual cubic hyperplane, avoiding z, x_1, x_2, r gives four bidegree-$(1, 1)$ equations; the two root self-collisions have bidegrees $(2, 1)$ and $(1, 2)$. Their rational zero sets contain at most $14(q+1)$ ordered pairs, whereas

$$(q - 3)(q - 4) > 14(q + 1) \qquad (q \geq 53).$$

Thus a split squarefree quartic remains. The cases $z \in \{x_1, x_2, r\}$ are exactly the excluded fixed-evaluation and ramification divisors. Lifting by ℓ_r gives a split squarefree quintic avoiding x_1, x_2.

One-pointed sextic series. Now let $f \in S_7$ have marker x and lie in the open pointed domain of Definition B.12; contract at r. The pullback of the lower rank-two carrier has degree at most three. The complete consecutive-row Lucas calculation gives at most two linear nucleus-support pullbacks. A whole binary-central line belongs to the binary Lucas-nucleus component, and the other whole-line possibilities are exactly the remaining declared modular supports, so these pullbacks are finite on the pointed domain. After removing fixed factors, a generic projection of the moving g^4_d, $d \leq 6$, is separable: an inseparable series of degree at most six has dimension at most $\lfloor 6/p \rfloor + 1 < 5$. Its ramification degree is at most $2d - 2 \leq 10$, and it contains the collision in which the new gcd root equals r. The fixed-root incidence at x costs at most two, and $r = x$ costs one. The total is

$$3 + 2 + 10 + 2 + 1 = 18. \tag{26}$$

Choose r off this divisor and apply the two-pointed quintic result with markers x, r. Its quintic avoids both, so multiplication by ℓ_r gives a split squarefree sextic in W_f avoiding x. This is $\mathrm{LP}(7, 1)$.

Finally, let $f \in S_8$ lie outside the persistent, modular, and self-collision components of $\mathrm{CC}(8,1)$. Its first-polar line meets the lower rank-two carrier in degree at most three and the complete linear nucleus support in degree at most two. The new-marker collision is ramification of a separable g^5_d, $d \le 7$: an inseparable series would have dimension at most $\lfloor 7/p \rfloor + 1 < 6$, and Riemann–Hurwitz gives at most $2d - 2 \le 12$. This proves (23) and excludes an additional contained collision component.

The gcd at every recursive level has degree zero, one, or at least two. The last case lies in the degree-appropriate rank-two catalecticant carrier, whose bottom equation is (15) and whose contained-line pullback is Proposition V.15; the degree-one case is treated after cancellation. A separable gcd-zero cubic map has geometric monodromy S_3 or C_3, with the latter and every inseparable case in (17)–(18) or the derivative-minor boundary. Hence the strata are exhaustive. Since

$$q + 1 - 2\sqrt{q} > 36$$

at $q = 53$ and the left side is increasing for $q > 1$, (22) leaves a rational point for every $q \ge 53$. The parameter budgets (24)–(26) and (23) are all smaller than $q + 1$. The first prime power satisfying the genus-one inequality is 53. $\square$

The only substantial modular carrier occurs in characteristic seven and is a projective binary-quartic space. Fixing a split quintic $P(t) = \sum_{i=0}^{5} p_i t^i$ and a carrier point $h = (a_0, \dots, a_4)$, put

$$H_j = \sum_{i=0}^{4} a_i p_{i+j}$$

with $p_k = 0$ outside $0 \le k \le 5$, and

$$D = H_0 H_2 - H_1^2, \quad N_s = H_{-1} H_2 - H_0 H_1, \quad N_u = H_{-1} H_1 - H_0^2.$$

Off $D = 0$, the unique residual quadratic is

$$Dt^2 - N_s t + N_u,$$

with branch polynomial $K = N_s^2 - 4 N_u D$. The exact residual identities are proved next; the geometric exhaustion of transported slices is kept as a separate printed gate.

Proposition B.14 (Residual quadratic identities)**.** With $p_k = 0$ outside $0 \le k \le 5$, the two consecutive Hankel equations for

$$P(t)(t^2 - st + u)$$

are

$$H_0 - sH_1 + uH_2 = 0, \qquad H_{-1} - sH_0 + uH_1 = 0.$$

On $D \ne 0$, Cramer's rule gives

$$s = N_s/D, \qquad u = N_u/D.$$

The branch divisor is $K = 0$, fixed/residual collision is $\operatorname{Res}(P, Dt^2 - N_s t + N_u) = 0$, and fixed-root collision is $\operatorname{Disc}(P) = 0$.

Proof. Expand the two consecutive coefficients annihilated by the carrier quartic. The displayed linear system has determinant D, so Cramer's rule gives the residual factor. Its discriminant is K. The two resultants have their usual intrinsic collision meanings; consequently the construction commutes with scalar, PGL_2, and Frobenius transport. $\square$

Proposition B.15 (One-moving-root slice)**.** Let $P = R(t)(t - x)$ with four fixed distinct roots. Then D, N_s, N_u have degree at most two in x, and K has degree at most four. If the squarefree reduction of K is nonconstant and not a square, the normalization of $y^2 = K(x)$ is geometrically integral of genus at most one.

Proof. Each H_j is affine in x, so its 2×2 minors are quadratic and K is quartic. In odd characteristic the squarefree nonsquare equation defines a geometrically integral double cover; the hyperelliptic genus formula gives $\lfloor (\deg K_{\mathrm{red}} - 1)/2 \rfloor \le 1$. $\square$

Proposition B.16 (Binary-quartic slices and rational base selection)**.** Every nonzero geometric binary quartic has a four-root slice whose reduced residual cover is geometrically integral of genus at most one. If the quartic is rational over a characteristic-seven field $\mathbb{F}_q$ with $q > 102$, the four roots may be chosen distinct and rational.

Proof. Let

$$\mathcal{H} = \mathbb{P}(\operatorname{Sym}^4 E), \qquad \mathcal{B} = \operatorname{Conf}_4(\mathbb{P}^1).$$

Over $\mathcal{H} \times \mathcal{B}$, the residual formulas define a universal branch quartic $K_{h,\mathbf{r}}(x)$. Its subdiscriminants cut out the locus where the squarefree part drops degree. Hence the good-slice incidence

$$\mathcal{U} = \{(h, \mathbf{r}) : K_{h,\mathbf{r},\mathrm{red}} \text{ is nonconstant and separable}\}$$

is a PGL_2-stable open set. The geometric component problem is the surjectivity of $\mathcal{U} \to \mathcal{H}$. One successful specialization would show only generic nonemptiness; fiberwise surjectivity requires an open cover of quartic moduli.

$$\begin{array}{ccccc} h_L & \xrightarrow{\text{six sections}} & (\Delta_1,\dots,\Delta_6) & \xrightarrow{\sum b_i\Delta_i=1} & \bigcup_i D(\Delta_i)=\mathbb{A}^1_L \\ \{4,\ 3+1,\ 2+2,\ 2+1+1\} & \xrightarrow{\text{disc}=3,3,1,5} & & \text{good slices on every boundary stratum.} & \end{array}$$

Fig. 2. The redundancy-nine good-slice argument. The Bézout identity covers the entire squarefree normal-form atlas; four root-partition calculations cover its boundary. The conclusion is fiberwise surjectivity, not generic success of one specialization.

The quotient has one one-dimensional squarefree stratum and four multiple-root boundary strata. On the standard even normal-form atlas for the squarefree stratum, write $h_L = [1, 0, L, 0, 1]$. The six four-point sets and the coefficient vectors of $\Delta_i(L)$ and $b_i(L)$ are supplied in the electronic supplement. They are six test sections of the universal base family over this atlas. Pullback of the branch discriminant gives $\Delta_i(L)$, and direct multiplication gives

$$\sum_{i=1}^{6} b_i(L)\Delta_i(L) = 1 \quad \text{in } \mathbb{F}_7[L]. \tag{27}$$

Thus the principal opens $D(\Delta_i)$ cover the normal-form atlas; these are six sections, not six isolated successful tests. Transport moves the quartic and its four-point base together, so the existence statement descends despite the many-to-one normal form. On the boundary, root multiplicity gives exactly the partitions $4, 3+1, 2+2, 2+1+1$: there are at most three support points, and triple transitivity gives one orbit per partition. Their reduced branch discriminants are $3, 3, 1, 5$, respectively. Thus every boundary orbit also meets $\mathcal{U}$, proving surjectivity.

Now fix $h \in \mathcal{H}(\mathbb{F}_q)$. Over $\mathbb{F}_q(\mathbf{r})$ let $d(h) \geq 1$ be the number of distinct roots of the generic $K_{h,\mathbf{r}}$. Define $B_h(\mathbf{r})$ to be its $d(h)$-th subdiscriminant, equivalently the corresponding nonzero Hermite principal minor. This is a polynomial, not a discriminant after an unspecified gcd division. Its nonvanishing after specialization guarantees a nonconstant separable squarefree part. Surjectivity over the algebraic closure proves $B_h \neq 0$. The affine chart of four finite roots is dense in $\mathcal{B}$, so this nonempty good open meets the chart used below. The bad affine base locus is contained in

$$B_h(\mathbf{r}) \prod_{i<j} (r_i - r_j) = 0, \tag{28}$$

so rational selection is avoidance of a proper divisor in configuration space, not a second monodromy assertion.

Every coefficient of $R_{\mathbf{r}}(t)(t-x)$ is affine in x, of degree at most one in each r_i and total $\mathbf{r}$-degree at most four. Hence D, N_s, N_u have individual degree at most two and total degree at most eight; K has individual degree at most four and total degree at most sixteen. Every quartic subdiscriminant is a Hermite minor of coefficient-degree at most six. Therefore

$$\deg_{r_i} B_h \leq 24, \qquad \deg B_h \leq 96. \tag{29}$$

Multiplying by the four-root Vandermonde gives a nonzero polynomial of total degree at most 102. Schwartz–Zippel leaves a rational ordered base with distinct roots whenever $q > 102$. $\square$

Proposition B.17 (Good-slice deletion)**.** Once a rational base with an integral slice is fixed, the moving/fixed diagonal, determinant, branch, four fixed-root collisions, and moving-root collision delete at most, respectively,

$$8,\ 4,\ 4,\ 8,\ 8$$

points on the normalized double cover. Their total is 32, so $q + 1 - 2\sqrt{q} > 32$ supplies a split witness.

Proof. The projection of the normalized residual cover to the moving-root line has degree two. Therefore the degree-four fixed-root diagonal and the degree-two determinant pull back to at most eight and four points. The degree-four branch divisor is ramified and contributes at most four points. For each of the four fixed roots the residual-root collision is linear in the moving parameter; its pullback contributes at most two points, for a total of eight. The moving-root resultant has degree at most four and contributes at most eight points after pullback. Removing their union from the normalized genus-at-most-one slice proves the claim. $\square$

Proposition B.18 (Characteristic-seven finite bridges)**.** At $q = 7$, the formal carrier has exactly 819 projective rootless quartics. At $q = 49$, Certificate R9-49 exhausts all

$$\frac{49^5 - 1}{48} = 5{,}884{,}901$$

projective carrier points and finds a split witness for each.

Proof. At $q = 7$, a split squarefree degree-seven projective divisor is the complement of one of the eight rational points. Inclusion–exclusion gives

$$\frac{7^5 - 1}{6} - 8\frac{7^4 - 1}{6} + 28\frac{7^3 - 1}{6} - 56\frac{7^2 - 1}{6} + 70\frac{7 - 1}{6} = 819.$$

The $q = 49$ clause is deliberately certificate-only: one lexicographic five-root search stops after every carrier point has a witness. The residual algebra is independently replayed, but there is no second exhaustive carrier implementation. ☐

Proposition B.19 (R9 persistent orbit law)**.** Put $d = \gcd(8, q + 1)$. The projective orbits in the conjugate-secant fiber are the inversion orbits in $C_d = T/T^8$, and coefficient Frobenius acts by multiplication by the characteristic. The tangent fiber is transitive unless the characteristic is two, when its zero and nonzero parts are separate.

Proof. The nonsplit torus acts by eighth powers and its normalizer inverts the torus coordinate. Thus inversion-fixed and paired classes have stabilizers $2d$ and d, respectively; coefficient Frobenius is the p-power map. For $d = 8$, it fuses the two odd pairs exactly when $p \equiv \pm 3 \pmod 8$. On the tangent fiber the unipotent action is $z \mapsto z+8u$, which vanishes exactly in characteristic two. ☐

Proposition B.20 (Other modular lifts)**.** The consecutive Lucas supports leave the point $\mathbb{P}\langle e_4\rangle$ in characteristic five and the quartic carrier above in characteristic seven; no other characteristic has a nonzero coherent lift. The characteristic-five point is shallow for every $q > 5$.

Proof. Lucas' theorem and the consecutive-row overlap give the two supports. Here $r = 9$, so (5) reads off Pascal row $r-2 = 7$: modulo five that row is $(1, 2, 1, 0, 0, 1, 2, 1)$, and $\binom{7}{j} \equiv \binom{7}{j-1} \equiv 0$ holds only at $j = 4$, leaving the single point $\mathbb{P}\langle e_4\rangle$. In characteristic five, homogenize $t^5 - t$ to include its simple root at infinity and multiply by a linear factor $t - a$ with $a \notin \mathbb{F}_5$. The seven roots are distinct and the two required Hankel coefficients vanish. Such an a exists for every $q > 5$. ☐

Proposition B.21 (Degree-eight contained components)**.** The assertion $\mathrm{CC}(8, 1)$ holds. The ordinary contained component is the upper rank-two carrier. The only modular lifts are the characteristic-five point and the characteristic-seven binary-quartic carrier of Proposition B.16; the former is shallow for every field in the theorem range. The marked self-collision scheme is never the whole polar line.

Proof. Proposition V.15 gives the ordinary component, and Proposition B.20 gives the complete Lucas overlap and the characteristic-five witness.

For collision, remove the fixed divisor from the six-dimensional Hankel series and let its moving degree be $d \le 7$. If the associated map were inseparable in characteristic p, its sections would lie in the pullback of a series of degree $\lfloor d/p\rfloor$, of dimension at most

$$\lfloor d/p\rfloor + 1 \le 4 < 6.$$

Thus the moving series is separable. Its ramification divisor has degree at most $(5 + 1)(7 - 5) = 12$, so collision is finite and supplies no additional contained component. ☐

Theorem B.22 (Redundancy nine)**.** For every prime power $q \ge 53$, the deep syndromes of $\mathrm{PRS}(q-8)$ are exactly the persistent tangent and conjugate-secant families, of total size $q(q + 1)^2/2$. Their orbit law is T/T^8 modulo inversion and Frobenius; the tangent family splits exactly in characteristic two.

Proof. Proposition B.13 handles every transverse nonmodular point. Proposition B.21 leaves the persistent and modular supports. Proposition B.20 eliminates characteristic five. For characteristic seven, the only powers below 343 are 7 and 49, both below the headline range; at every $q = 7^m \ge 343$, Propositions B.16 and B.17 produce a witness. Proposition VI.5 completes the promotion to deep holes. ☐

Proposition B.18 is a carrier boundary, not part of the headline classification: the $q = 7$ count is not a deep-hole statement for the inadmissible parameter $\mathrm{PRS}(-1)$, and the $q = 49$ pass is not an ambient $\mathbb{P}^8$ census.

Appendix C
Lucas-carrier arithmetic and redundancy ten

This appendix evaluates the modular carrier left by the recursive theorem. It first separates the additive endpoint orbit from the full degree-nine carrier, then proves that the full carrier is shallow, and finally combines that arithmetic with five pointed contractions to obtain redundancy ten.

Let $d = 2^s$ with $s \ge 2$ in characteristic two. Lucas' theorem gives the top nucleus of the degree-d normal rational curve as

$$\mathbb{P}\langle e_1, \dots, e_{d-1}\rangle,$$

whose coherent degree-$(d + 1)$ lift is

$$\mathcal{M}_{d+1} = \mathbb{P}\langle e_2, \dots, e_{d-1}\rangle.$$

At the distinguished Borel endpoint e_{d-1} the Hankel kernel contains

$$\mathcal{U}_s = \langle 1, t, t^d\rangle.$$

We use the standard linearized/subspace-polynomial terminology of [32], [33]; the carrier and root-cover calculations below are specific to the present Hankel system.

Proposition C.1 (Lucas overlap kernel)**.** For $d = 2^s$, the lower nucleus and its coherent lift are exactly

$$\mathbb{P}\langle e_1, \dots, e_{d-1}\rangle, \qquad \mathbb{P}\langle e_2, \dots, e_{d-1}\rangle.$$

At e_{d-1}, $\mathcal{U}_s$ is endpoint-specific; the intersection common to the full carrier is only $\langle 1, t^d\rangle$.

Proof. Lucas' theorem gives $\binom{d}{i} \equiv 0 \pmod 2$ for $0 < i < d$. Applying this support condition to the two consecutive contraction rows removes the two endpoint coordinates and gives the displayed lift. For a basis point e_i, the Hankel equations kill the adjacent coefficients b_{i-1}, b_i. At $i = d-1$, the monomials $1, t, t^d$ survive. Varying i over the carrier leaves only $1, t^d$ in the intersection. $\square$

Proposition C.2 (Linearized root cover)**.** On the generic locus $BC \neq 0$, the normalized ordered-root cover of $A + Bt + Ct^d$ has geometric monodromy $\mathrm{AGL}_1(\mathbb{F}_d)$, minimal constant field $\mathbb{F}_d$, and based deck group C_{d-1}. Coefficientwise Frobenius acts on the based group by multiplication by two, with exact order s. The net contains a split squarefree member over $\mathbb{F}_{2^m}$ if and only if $s \mid m$.

$$K = \overline{\mathbb{F}}_2(a,b) \quad \subset \quad K_1 = K(\beta), \quad \underbrace{\beta^{d-1} = b}_{\substack{\mathbb{F}_d^\times \\ \text{scalings},\ d-1}} \quad \subset \quad K_1(y), \quad \underbrace{y^d + y = a/\beta^d}_{\substack{(\mathbb{F}_d,+) \\ \text{translations},\ d}}$$

$$(\mathbb{F}_d, +) \rtimes \mathbb{F}_d^\times = \mathrm{AGL}_1(\mathbb{F}_d)$$

Fig. 3. The linearized ordered-root cover. The Kummer scaling layer and connected additive layer have coprime roles; their explicit semidirect action accounts for the full geometric monodromy.

Proof. Work first over the geometric coefficient field

$$k = \overline{\mathbb{F}}_2, \qquad K = k(A/C, B/C),$$

on the open set $BC \neq 0$. Put $a = A/C$ and $b = B/C$. The Kummer extension

$$K_1 = K(\beta), \qquad \beta^{d-1} = b,$$

has degree $d-1$: the valuation of b at its simple zero is one. Since k contains $\mathbb{F}_d$, this is the scaling part of the geometric root cover, not an extension of constants. Over K_1 choose

$$y^d + y = a/\beta^d.$$

The roots are $\{\beta(y+c) : c \in \mathbb{F}_d\}$. The right side has a simple pole at the pole of a. If the $\mathbb{F}_d$-torsor were disconnected, its geometric translation group would lie in a proper additive subgroup of $\mathbb{F}_d$. A nonzero $\mathbb{F}_2$-linear functional annihilating that subgroup would then give an Artin–Schreier equation $w^2 + w = \gamma a/\beta^d$ with $\gamma \in \mathbb{F}_d^\times$. This is impossible: the right side has a simple pole, whereas every pole of $w^2 + w$ has even order. Thus $K_1(y)/K_1$ is connected of degree d. Its translations $y \mapsto y + c$ give the normal subgroup $(\mathbb{F}_d, +)$. The $d-1$ Kummer conjugates act by

$$(\beta, y) \longmapsto (\zeta\beta, \zeta^{-1}y), \qquad \zeta \in \mathbb{F}_d^\times,$$

and conjugate translations by multiplication by ζ. We have thus exhibited $d(d-1)$ automorphisms of an extension of degree $d(d-1)$. Consequently the full geometric group, rather than a proper subgroup, is

$$(\mathbb{F}_d, +) \rtimes \mathbb{F}_d^\times = \mathrm{AGL}_1(\mathbb{F}_d).$$

Return now to the arithmetic coefficient field $\mathbb{F}_2(A/C, B/C)$. Ratios of nonzero root differences recover every element of $\mathbb{F}_d$, so any splitting field contains $\mathbb{F}_d$ in its constant field. The preceding regular construction over $\mathbb{F}_d$ introduces no larger constants, proving minimality. Coefficient Frobenius acts on $\mathbb{F}_d^\times$ by $c \mapsto c^2$, and hence has exact order s on the based component group C_{d-1}. Boundary members with $C = 0$ have degree at most one, while those with $B = 0$ have zero derivative; neither is split squarefree of degree d. Thus a split member forces $\mathbb{F}_d \subset \mathbb{F}_{2^m}$, equivalently $s \mid m$. Conversely, $t^d + t$ splits simply under that inclusion. The equality $\mathrm{ord}_{2^s-1}(2) = s$ follows from

$$\gcd(2^r - 1, 2^s - 1) = 2^{\gcd(r,s)} - 1.$$

$\square$

At $s = 2$ this recovers the odd/even extension law at redundancies six and seven. The first fresh carrier is $s = 3$, in syndrome degree nine. The $\mathbb{F}_8$-linear section has order-three component Frobenius, but its four quotient directions change the full answer.

Proposition C.3 (The e_7 orbit)**.** The stabilizer of $e_7 \in \mathbb{P}(\Gamma^9 E)$ is the upper Borel, so its PGL_2-orbit is $\mathbb{P}^1$. At the distinguished point,

$$W_{e_7} = \langle 1, t, t^2, t^3, t^4, t^5, t^8 \rangle.$$

Proof. Lucas parity and extraction of the x^2y^7 coefficient give

$$ge_7 = (ad + bc)^2 \big(b^5 e_2 + b^4 d\, e_3 + bd^4 e_6 + d^5 e_7\big).$$

This is proportional to e_7 exactly when $b = 0$. The apolar action is contragredient, so it transports both W_f and split root divisors. The displayed kernel follows from the two Hankel equations. □

Proposition C.4 (Framed Artin–Schreier quotient)**.** Order roots $0, 1, a, b, c, d, u, v$ and put

$$S = a + b + c + d, \quad E = ab + ac + ad + bc + bd + cd, \quad h = 1 + S, \quad p = 1 + E + S + S^2.$$

On the open set where the six fixed roots are distinct, $h, p, 1 + h + p$ are nonzero, and $x^2 + hx + p \neq 0$ for $x = a, b, c, d$, the remaining pair is the geometrically integral Artin–Schreier cover

$$y^2 + y = p/h^2.$$

Its rational lifting law is $\mathrm{Tr}_{\mathbb{F}_{2^m}/\mathbb{F}_2}(p/h^2) = 0$.

Proof. The first elementary-symmetric equation gives $u + v = 1 + S = h$. In the second equation the root 1 contributes $S + u + v = 1$, so

$$0 = E + S(u + v) + uv + S + u + v = E + Sh + uv + 1,$$

and hence $uv = p$. Thus u, v are the roots of $z^2 + hz + p$. The divisor $h = 0$ is exactly their collision divisor. On $h \neq 0$, putting $y = u/h$ gives the displayed Artin–Schreier equation.

It remains to prove that its class is geometric, rather than a constant-field artifact. Write $B = b + c + d$, so $h = a + 1 + B$. Along the prime divisor $h = 0$, the leading coefficient of the order-two pole of p/h^2 is

$$D = 1 + bc + bd + cd + B + B^2,$$

in the residue field $\overline{\mathbb{F}}_2(b, c, d)$. Its derivative $\partial D/\partial b = 1 + c + d$ is a nonzero rational function, so D is not a square there. If $p/h^2 = g^2 + g$, the order-two pole would force g to have a simple pole whose principal coefficient squares to D, a contradiction. The class therefore stays nonzero after algebraic extension of the constants, proving geometric integrality and geometric monodromy C_2.

The remaining open conditions have exact collision meanings: $p \neq 0$ excludes $u, v = 0$, $1 + h + p \neq 0$ excludes $u, v = 1$, and $x^2 + hx + p \neq 0$ excludes $u, v = x$ for $x \in \{a, b, c, d\}$. Over $\mathbb{F}_{2^m}$ the standard Artin–Schreier criterion gives a rational pair precisely when $\mathrm{Tr}_{\mathbb{F}_{2^m}/\mathbb{F}_2}(p/h^2) = 0$. □

Proposition C.5 (Additive frame cover)**.** The family

$$t^8 + A_4 t^4 + A_2 t^2 + A_1 t + A_0, \qquad A_1 \neq 0,$$

has a connected affine-frame cover with geometric monodromy $\mathrm{AGL}_3(\mathbb{F}_2)$.

Proof. For an $\mathbb{F}_2$-space U and $w \notin U$,

$$L_{U + \langle w \rangle} = L_U^2 + L_U(w) L_U.$$

Induction gives exponents $8, 4, 2, 1$ and $A_1 = \prod_{v \neq 0} v \neq 0$. An affine frame is an origin plus three independent differences.

Let $\mathcal{F}$ be the open affine frame space of tuples $(a; v_1, v_2, v_3)$ with independent v_i. It is geometrically integral. Sending a frame to the monic polynomial with root set

$$a + \langle v_1, v_2, v_3 \rangle_{\mathbb{F}_2}$$

lands in the displayed coefficient family. Conversely, the eight geometric roots of a generic member form an affine three-space: the difference set is the kernel of its additive part, and any root is an origin. Its ordered affine frames are therefore exactly one free orbit of

$$\mathbb{F}_2^3 \rtimes \mathrm{GL}_3(\mathbb{F}_2).$$

Thus the generic fiber has exactly $8|\mathrm{GL}_3(\mathbb{F}_2)| = 1344$ points, every two of them differ by one affine transformation, and the invariant function field is the coefficient field. The normalization is connected and has exact geometric deck group $\mathrm{AGL}_3(\mathbb{F}_2)$. □

Remark C.6 (Logical boundary). The generic Artin–Schreier cover describes the full framed quotient, and Proposition C.5 identifies a proper additive subcover. The shallowness theorem below does not require a rational point on the generic cover: its witnesses are constructed directly from rational three-dimensional subspaces.

Theorem C.7 (Shallowness of the degree-nine e_7 orbit)**.** For every $m \geq 3$, the entire PGL_2-orbit of e_7 is shallow over $\mathbb{F}_{2^m}$. The number of monic additive-affine witnesses is

$$\frac{q(q-1)(q-2)(q-4)}{1344}.$$

Proof. Every three-dimensional $\mathbb{F}_2$-subspace $V \subset \mathbb{F}_{2^m}$ gives

$$L_V(t) = \prod_{v \in V}(t - v) = t^8 + A_4 t^4 + A_2 t^2 + A_1 t.$$

The t^7, t^6 coefficients vanish, so $L_V \in W_{e_7}$, and $A_1 \neq 0$ makes its roots simple. Every admissible binary field in the statement contains such a three-space. Proposition C.3 transports the witness around the orbit.

There are

$$\binom{m}{3}_2 = \frac{(q-1)(q-2)(q-4)}{168}$$

three-spaces. Each has $q/8$ affine cosets, and $L_{a+V}(t) = L_V(t) + L_V(a)$. A split polynomial recovers its root coset, whose difference set recovers V; hence distinct cosets or three-spaces cannot be counted twice. Multiplying the two factors gives the displayed formula. $\square$

Thus the order-three law of Proposition C.2 belongs to a proper $\mathbb{F}_8$-linear section, not to deepness of the full e_7 kernel. We now treat the complete carrier.

A. The complete degree-nine carrier

Write

$$\mathcal{M}_9 = \mathbb{P}\langle e_2, e_3, e_4, e_5, e_6, e_7 \rangle, \qquad U = \langle e_2, e_3, e_6, e_7 \rangle.$$

The PGL_2-module filtration gives

$$\mathcal{M}_9/U \simeq \overset{4}{\det} \otimes E. \tag{30}$$

a) Proof route: The proof of $\mathcal{M}_9(\mathbb{F}_{2^m})$ shallowness has four distinct steps.

(1) On the invariant block $\mathbb{P}(U)$, reciprocal subspace polynomials settle the rank-one strata. The remaining rank-two twists reduce to the Artin–Schreier final-pair cover in Proposition C.8.
(2) Off $\mathbb{P}(U)$, projective transport fixes the nonzero quotient coordinate. The coefficient map in (36) has rank three or four; in either case the nonsquare double-pole calculation makes the Artin–Schreier cover geometrically integral.
(3) Fixing five roots leaves one moving base root. The good-base conditions have bounded degree, and the normalized cover has genus at most one with deletion degree 48. This gives every complement point for $q \geq 64$.
(4) The finite boundary is explicit. Full-carrier quotient certificates close $q = 16, 32$. At $q = 64$, only the invariant block is certificate-backed; the complement is covered by the preceding mathematical point-count argument. For $q \geq 128$, both parts are mathematical.

Thus the rank and selector calculations carry the uniform theorem, while the certificates close exactly the stated finite residue.

Proposition C.8 (Invariant-block arithmetic)**.** Every rational point of $\mathbb{P}(U)$ is shallow over every $\mathbb{F}_{2^m}$ with $m \geq 4$.

Proof. Identify U with 2×2 matrices by $e_2, e_3, e_6, e_7 \leftrightarrow E_{00}, E_{01}, E_{10}, E_{11}$. A rank-one matrix is $u^{(4)} v^{\mathsf{T}}$. The diagonal $[u] = [v]$ is the Frobenius-graph orbit and is handled by the subspace polynomial of a three-dimensional $\mathbb{F}_2$-space. The off-graph orbit has representative e_3, whose kernel equations are $f_2 = f_3 = 0$. If

$$L_V(X) = X^8 + A_4 X^4 + A_2 X^2 + A_1 X, \qquad A_1 \neq 0,$$

is the subspace polynomial of a three-space $V \subset \mathbb{F}_{2^m}$, then

$$A_1^{-1}(t + A_4 t^5 + A_2 t^7 + A_1 t^8)$$

has the eight distinct roots $\{0\} \cup \{v^{-1} : 0 \neq v \in V\}$ and satisfies those equations. Projective transport closes both rank-one strata.

For rank two write the rational syndrome as $z = (a, b, c, d)$, with $ad + bc \neq 0$. This is the rational A_5-twist family of $e_3 + e_6$, but the construction below is defined over z itself and requires no chosen cocycle. Choose six roots with polynomial $h = \sum_{i=0}^{6} h_i t^i$, and put

$$\begin{aligned} A_0 &= bh_0 + ch_3 + dh_4, & A_1 &= ah_0 + bh_1 + ch_4 + dh_5, \\ A_2 &= ah_1 + bh_2 + ch_5 + d, & A_3 &= ah_2 + bh_3 + c. \end{aligned}$$

and set

$$\Delta = A_1A_3 + A_2^2, \quad Q = A_0A_3 + A_1A_2, \quad N = A_1^2 + A_0A_2.$$

The final pair has sum Q/Δ, product N/Δ, and rational distinct roots precisely when

$$y^2 + y = N\Delta/Q^2 \tag{31}$$

has a rational point away from the root-collision divisors. These equations descend the cover uniformly across all twists. Fix five roots and let x be the sixth. Then Q has degree at most two in x, $N\Delta$ has degree at most four, and Q' is a nonzero polynomial in the five-root parameters; the good-base open below imposes $Q' \neq 0$. At a root of Q, Artin–Schreier reduction leaves a simple pole exactly when

$$R = ((N\Delta)')^2 + (Q')^2 N\Delta \neq 0.$$

At the alternating representative $e_3 + e_6$, the four A_i are independent; on $Q = 0$ and $A_3 \neq 0$ one has

$$A_0 = A_1A_2/A_3, \qquad N\Delta = \Delta^2 A_1/A_3.$$

Thus the double-pole leading coefficient is A_1/A_3, which is not a square in the residue function field. Every geometric rank-two point is projectively conjugate to that representative, so the induced root-coordinate change is birational. Ordering the roots is finite separable; it cannot make the coefficient a square, because adjoining its square root in characteristic two would be purely inseparable. Thus the same nonsquare test holds on every rational twist and says $Q \nmid R$. The cover is geometrically integral and has at most two reduced simple poles, so genus at most one. A nonzero good-base selector has total degree at most 102 in the five ordered roots. For a good base the forbidden x-values have degrees $5, 2, 2, 2, 10, 4$, totaling 23, and therefore delete at most $2 \cdot 23 + 2 = 48$ points of the normalized double cover. The zero bound gives a good base for $q \geq 128$, while the Aubry–Perret bound for a geometrically integral projective curve of arithmetic genus at most one then supplies a surviving point [30, p. 468]. The complete rank-two invariant-block quotient certificate at $q = 64$ and the $q = 16, 32$ quotient certificates cover the remaining admissible fields. Their checkers rebuild the orbit partitions, twists, root polynomials, and both Hankel equations; the generator and separately written replay are the `rank-two-artin-schreier-avoidance` and `higher-lucas-modular-carriers` artifacts under `supplement/evidence/lucas-m9/`. Section D records their exact trust boundary. □

Proposition C.9 (Exact final-pair criterion)**.** Let $z = (z_2, \ldots, z_7) \in \mathcal{M}_9(\mathbb{F}_{2^m})$. Choose six distinct finite roots with monic polynomial $h = \sum_{i=0}^{6} h_i t^i$, and define

$$A_j = \sum_{i=2}^{7} z_i h_{i+j-3} \quad (0 \leq j \leq 3),$$

with coefficients outside $0, \ldots, 6$ equal to zero. If the final pair has sum s and product p, the two Hankel equations are exactly

$$A_0 + sA_1 + pA_2 = 0, \qquad A_1 + sA_2 + pA_3 = 0. \tag{32}$$

On

$$\Delta = A_1A_3 + A_2^2 \neq 0, \qquad Q = A_0A_3 + A_1A_2 \neq 0,$$

their unique solution is

$$s = Q/\Delta, \qquad p = (A_1^2 + A_0A_2)/\Delta, \tag{33}$$

and the pair is rational and distinct exactly when

$$\mathrm{Tr}_{\mathbb{F}_{2^m}/\mathbb{F}_2}\left(\frac{(A_1^2 + A_0A_2)\Delta}{Q^2}\right) = 0. \tag{34}$$

Collision with an old root r is

$$r^2\Delta + rQ + (A_1^2 + A_0A_2) = 0. \tag{35}$$

The rank-deficient cases are exactly the corresponding rank-one or rank-zero alternatives in (32).

Proof. Substitute $f = h(t)(t^2 + st + p)$ in the two consecutive Hankel equations. This gives (32); Cramer's rule gives (33). In characteristic two the quadratic is split with distinct roots precisely when $s \neq 0$ and $\mathrm{Tr}(p/s^2) = 0$, which is (34). Evaluation of the quadratic at r gives (35). Conversely, every split divisor of nonfull support can be moved off infinity and partitioned into six roots and a final pair. The full-support divisor is checked directly in homogeneous coordinates. Thus the criterion loses neither infinity nor the degenerate linear-system cases. □

The next three lemmas carry the all-field part of the theorem below: the first fixes the rank of the coefficient map off the invariant block, the second makes the final-pair cover geometrically integral, and the third selects a rational base of five roots with an effective degree bound.

Lemma C.10 (Rank of the coefficient map off the invariant block)**.** Let $z = (z_2, \dots, z_7)$ and let $h = (h_0, \dots, h_5)^\mathsf{T}$ collect the free coefficients of the monic sextic of Proposition C.9. Then

$$(A_0, A_1, A_2, A_3)^\mathsf{T} = M_z h + (0, 0, z_7, z_6)^\mathsf{T}, \quad M_z = \begin{pmatrix} z_3 & z_4 & z_5 & z_6 & z_7 & 0 \\ z_2 & z_3 & z_4 & z_5 & z_6 & z_7 \\ 0 & z_2 & z_3 & z_4 & z_5 & z_6 \\ 0 & 0 & z_2 & z_3 & z_4 & z_5 \end{pmatrix}, \tag{36}$$

and $\operatorname{rank} M_z \geq 3$ whenever $z_4 \neq 0$ or $z_5 \neq 0$. In that case exactly one of the following holds.

(i) $\operatorname{rank} M_z = 4$, and $h \mapsto (A_0, A_1, A_2, A_3)$ is onto $\mathbb{A}^4$.

(ii) $\operatorname{rank} M_z = 3$, the left kernel of M_z is a line $\langle \ell \rangle$, and the image of $h \mapsto (A_0, A_1, A_2, A_3)$ is the affine hyperplane

$$\ell_0 A_0 + \ell_1 A_1 + \ell_2 A_2 + \ell_3 A_3 = \kappa, \qquad \kappa = \ell_2 z_7 + \ell_3 z_6. \tag{37}$$

Proof. Reading $A_j = \sum_{i=2}^{7} z_i h_{i+j-3}$ row by row, with $h_6 = 1$ and $h_k = 0$ for $k \notin \{0, \dots, 6\}$, gives the displayed affine form; the constant vector collects the two h_6 terms, which occur in A_2 and A_3 only. A row vector ℓ satisfies $\ell M_z = 0$ exactly when

$$\begin{aligned} \ell_0 z_3 + \ell_1 z_2 &= 0, & \ell_0 z_4 + \ell_1 z_3 + \ell_2 z_2 &= 0, \\ \ell_0 z_5 + \ell_1 z_4 + \ell_2 z_3 + \ell_3 z_2 &= 0, & \ell_0 z_6 + \ell_1 z_5 + \ell_2 z_4 + \ell_3 z_3 &= 0, \\ \ell_0 z_7 + \ell_1 z_6 + \ell_2 z_5 + \ell_3 z_4 &= 0, & \ell_1 z_7 + \ell_2 z_6 + \ell_3 z_5 &= 0, \end{aligned} \tag{38}$$

one equation per column, read left to right.

Suppose $\operatorname{rank} M_z \leq 2$, so the left kernel K has dimension at least two. Either K contains a nonzero vector with $\ell_0 = \ell_1 = 0$, or $\ell \mapsto (\ell_0, \ell_1)$ is injective on K and hence, $\dim K \geq 2$, also onto.

In the first case take $\ell = (0, 0, \ell_2, \ell_3) \neq 0$. If $\ell_2 \neq 0$, normalize $\ell_2 = 1$: the second, third, fourth and fifth equations of (38) read $z_2 = 0$, $z_3 = \ell_3 z_2$, $z_4 = \ell_3 z_3$ and $z_5 = \ell_3 z_4$, so $z_2 = z_3 = z_4 = z_5 = 0$ in turn. If $\ell_2 = 0$ then $\ell_3 \neq 0$, and the third through sixth equations read $\ell_3 z_2 = \ell_3 z_3 = \ell_3 z_4 = \ell_3 z_5 = 0$, with the same conclusion.

In the second case K contains $(1, 0, \ell_2, \ell_3)$ and $(0, 1, \ell_2', \ell_3')$. The first equation applied to the second vector gives $z_2 = 0$, and applied to the first gives $z_3 = 0$; the second and third equations applied to the first vector then give $z_4 = \ell_2 z_2 = 0$ and $z_5 = \ell_2 z_3 + \ell_3 z_2 = 0$.

Either case contradicts $z_4 \neq 0$ or $z_5 \neq 0$, so $\operatorname{rank} M_z \in \{3, 4\}$. At rank four the linear map $h \mapsto M_z h$ is onto, hence so is its translate. At rank three the left kernel is a line $\langle \ell \rangle$, the image of $h \mapsto M_z h$ is the linear hyperplane $\sum_i \ell_i A_i = 0$, and translating by $(0, 0, z_7, z_6)$ moves the right-hand side to $\kappa = \ell_2 z_7 + \ell_3 z_6$. ☐

Lemma C.11 (Geometric nontriviality of the final-pair Artin–Schreier cover)**.** Let k be algebraically closed of characteristic two, let $z \in \mathcal{M}_9(k)$ satisfy $z_4 \neq 0$ or $z_5 \neq 0$, and let $L \subseteq \mathbb{A}^4$ be the image of the coefficient map of Lemma C.10. Then the reduced divisor $Q = 0$ in L has an irreducible component $\mathfrak{q}$ on which A_3 and Δ do not vanish identically and on whose function field A_1/A_3 is not a square. Ordering the six roots leaves that conclusion intact, so the double pole of $N\Delta/Q^2$ along $\mathfrak{q}$ does not reduce and (31) is geometrically integral.

Proof. On $Q = 0$ with $A_3 \neq 0$ one has $A_0 = A_1 A_2 / A_3$, hence

$$N = A_1^2 + \frac{A_1 A_2^2}{A_3} = \frac{A_1(A_1 A_3 + A_2^2)}{A_3} = \frac{A_1 \Delta}{A_3}, \qquad N\Delta = \frac{A_1}{A_3} \Delta^2. \tag{39}$$

Since $u^2 \equiv u$ modulo $\wp(u) = u^2 + u$, the double pole of $N\Delta/Q^2$ along a component of $Q = 0$ reduces to a simple pole exactly when A_1/A_3 is a square in the function field of that component. That square class is what the two rank cases test.

Rank four. Here $L = \mathbb{A}^4$ and the coordinates A_i are independent. The partial derivatives of $Q = A_0 A_3 + A_1 A_2$ are A_3, A_2, A_1, A_0, whose only common zero is the origin, so the projective quadric is smooth and $X = V(Q)$ is irreducible; in particular A_3 and Δ do not vanish on it identically. Inside X lie the planes $\Pi_1 = V(A_0, A_1)$, $\Pi_{13} = V(A_1, A_3)$ and $\Pi_2 = V(A_2, A_3)$, and $V(A_1) \cap X = \Pi_1 \cup \Pi_{13}$, $V(A_3) \cap X = \Pi_{13} \cup \Pi_2$. At the generic point of Π_1 the functions A_2, A_3 are units and $A_0 = A_1 A_2 / A_3$, so the local ring of X there is a discrete valuation ring with uniformizer A_1; symmetrically A_3 is a uniformizer at Π_2, where A_0, A_1 are units and $A_2 = A_0 A_3 / A_1$; and at Π_{13} the two relations $A_3 = A_1 A_2 / A_0$ and $A_1 = A_0 A_3 / A_2$ show that A_1 and A_3 generate the same maximal ideal. Hence

$$\operatorname{div}\left(\frac{A_1}{A_3}\right) = [\Pi_1] - [\Pi_2],$$

with both multiplicities equal to one. A square has even multiplicity along every prime divisor, so A_1/A_3 is not a square in $k(X)$, and $\mathfrak{q} = X$ serves.

Rank three, principal case. Let (37) be the relation and assume first $(\ell_0, \ell_1) \neq (0, 0)$. We claim some component $\mathfrak{q}$ of $Q = 0$ in L carries A_2 and A_3 as algebraically independent functions. Otherwise every component lies in an affine plane $\alpha A_2 + \beta A_3 = \gamma$ with $(\alpha, \beta) \neq (0, 0)$. Such a plane is not all of L, because $(0, 0, \alpha, \beta)$ is not proportional to ℓ; and a component of the quadric surface $Q|_L$ is either that whole surface, which has degree two and so lies in no plane, or a plane. Hence every component is one of the planes $\alpha A_2 + \beta A_3 = \gamma$, and $Q|_L$, being of degree two, is a scalar multiple of a product of two of those linear forms, hence a polynomial in A_2, A_3 alone. That is impossible: if $\ell_0 \neq 0$, eliminating A_0 leaves $Q|_L = A_3(\kappa + \ell_1 A_1 + \ell_2 A_2 + \ell_3 A_3)/\ell_0 + A_1 A_2$, whose coefficient of A_1 is $\ell_1 A_3/\ell_0 + A_2 \neq 0$; and if $\ell_0 = 0$, $\ell_1 \neq 0$, eliminating A_1 leaves $Q|_L = A_0 A_3 + A_2(\kappa + \ell_2 A_2 + \ell_3 A_3)/\ell_1$, which contains $A_0 A_3$. Fix such a component $\mathfrak{q}$. On it $A_3 \not\equiv 0$ and $\ell_0 A_2 + \ell_1 A_3 \not\equiv 0$, since either vanishing is a linear dependence between A_2 and A_3.

Substituting $A_0 = A_1 A_2/A_3$ into (37) and clearing A_3 gives $A_1(\ell_0 A_2 + \ell_1 A_3) = A_3(\kappa + \ell_2 A_2 + \ell_3 A_3)$, that is

$$\frac{A_1}{A_3} = \frac{\kappa + \ell_2 A_2 + \ell_3 A_3}{\ell_0 A_2 + \ell_1 A_3} \tag{40}$$

on $\mathfrak{q}$. Together with $A_0 = A_1 A_2/A_3$ and (37) this writes every coordinate as a rational function of A_2, A_3, so $k(\mathfrak{q}) = k(A_2, A_3)$. In characteristic two a rational function of two independent variables is a square exactly when both of its partial derivatives vanish. Writing P and D for the numerator and denominator of (40),

$$\frac{\partial}{\partial A_2}\frac{P}{D} = \frac{\ell_2 D + \ell_0 P}{D^2} = \frac{(\ell_1\ell_2 + \ell_0\ell_3)A_3 + \ell_0\kappa}{D^2}, \qquad \frac{\partial}{\partial A_3}\frac{P}{D} = \frac{(\ell_1\ell_2 + \ell_0\ell_3)A_2 + \ell_1\kappa}{D^2},$$

so A_1/A_3 is a square on $\mathfrak{q}$ only if

$$\ell_0\kappa = \ell_1\kappa = 0, \qquad \ell_1\ell_2 = \ell_0\ell_3. \tag{41}$$

Those equalities force $z_4 = z_5 = 0$. If $\kappa \neq 0$ then $\ell_0 = \ell_1 = 0$, and the first case of the proof of Lemma C.10 applies verbatim to ℓ. If $\kappa = 0$ and $\ell_0 = 0$ then $\ell_1\ell_2 = 0$; for $\ell_1 = 0$ the same first case applies, while for $\ell_2 = 0$ and $\ell_1 \neq 0$ the first four equations of (38) read $\ell_1 z_2 = 0$, $\ell_1 z_3 = 0$, $\ell_1 z_4 + \ell_3 z_2 = 0$ and $\ell_1 z_5 + \ell_3 z_3 = 0$, giving $z_2 = z_3 = z_4 = z_5 = 0$. Finally let $\ell_0 \neq 0$; then $\kappa = 0$, and scaling $\ell_0 = 1$ and writing $\lambda = \ell_1$, $\mu = \ell_2$, $\ell_3 = \lambda\mu$, the first three equations of (38) give

$$z_3 = \lambda z_2, \qquad z_4 = (\lambda^2 + \mu)z_2, \qquad z_5 = \lambda(\lambda^2 + \mu)z_2, \tag{42}$$

the fourth gives $z_6 = (\lambda^4 + \lambda^2\mu + \mu^2)z_2$, and the fifth substituted into the sixth gives $(\lambda^2 + \mu)z_6 + \lambda^2\mu z_4 = (\lambda^2 + \mu)^3 z_2 = 0$. If $z_2 = 0$ then (42) and the remaining equations force $z = 0$; otherwise $\lambda^2 + \mu = 0$ and (42) gives $z_4 = z_5 = 0$. Both outcomes are excluded, so A_1/A_3 is not a square on $\mathfrak{q}$.

Rank three, relation in A_2, A_3 only. Suppose $\ell_0 = \ell_1 = 0$, so the relation is $\ell_2 A_2 + \ell_3 A_3 = \kappa$ and A_0, A_1 are unconstrained on L. If $\ell_2 \neq 0$, use (A_0, A_1, A_3) as coordinates on L and substitute $A_2 = (\kappa + \ell_3 A_3)/\ell_2$:

$$Q|_L = A_3 A_0 + A_1(\kappa + \ell_3 A_3)/\ell_2.$$

This is of degree one in A_0, so it is irreducible unless A_3 divides its constant term, which happens only for $\kappa = 0$, where $Q|_L = A_3(A_0 + \ell_3 A_1/\ell_2)$. In the irreducible case the unique component carries A_1 and A_3 as independent coordinates; in the split case the component $A_0 = \ell_3 A_1/\ell_2$ does. On such a component $\partial(A_1/A_3)/\partial A_1 = 1/A_3 \neq 0$, so A_1/A_3 is not a square. If instead $\ell_2 = 0$, then $\ell_3 \neq 0$ and $A_3 = \kappa/\ell_3$ is constant on L. For $\kappa = 0$ the function A_3 vanishes identically, so the last row of M_z vanishes and $z_2 = z_3 = z_4 = z_5 = 0$, which is excluded; for $\kappa \neq 0$ the surface $Q|_L = \kappa A_0/\ell_3 + A_1 A_2$ is irreducible with A_1, A_2 independent on it, and A_1/A_3 is a nonzero constant multiple of A_1, again not a square.

The component avoids $\Delta = 0$. On $A_3 \neq 0$ the locus $Q = \Delta = 0$ is $\{(u^3c, u^2c, uc, c)\}$, the two-dimensional irreducible cone over the twisted cubic, and on $A_3 = 0$ it is the plane $A_2 = A_3 = 0$. The chosen $\mathfrak{q}$ has $A_3 \not\equiv 0$, so if $\mathfrak{q}$ were contained in $\Delta = 0$ it would equal that cone, forcing $\ell_0 u^3 c + \ell_1 u^2 c + \ell_2 uc + \ell_3 c = \kappa$ for all u and c, hence $\ell = 0$ and $\kappa = 0$. In the rank-four case $\mathfrak{q}$ has dimension three and the cone has dimension two. Either way $\Delta \not\equiv 0$ on $\mathfrak{q}$, so (39) really computes the double-pole coefficient there.

Passage to ordered roots. The coefficient map $h \mapsto A$ is onto L with linear fibers, so the preimage of $\mathfrak{q}$ is irreducible and its function field is a purely transcendental extension of $k(\mathfrak{q})$, which acquires no new square roots of elements of $k(\mathfrak{q})$. Ordering the six roots is a finite separable base change, and adjoining a square root in characteristic two is purely inseparable, so it cannot make A_1/A_3 a square either. Hence the double pole survives and (31) is geometrically integral. □

Lemma C.12 (Good-base selector and degree bound)**.** Let z satisfy $z_4 \neq 0$ or $z_5 \neq 0$. Fix five roots $r_1, \ldots, r_5$ with $g(t) = \prod_{i=1}^{5}(t + r_i) = \sum_{i=0}^{5} g_i t^i$, let x be the sixth root, and put

$$B_j = \sum_{i=2}^{7} z_i g_{i+j-4} \quad (0 \le j \le 4), \qquad \text{so that} \qquad A_j = B_j + x B_{j+1}. \tag{43}$$

Then $Q = q_2 x^2 + q_1 x + q_0$ with

$$q_2 = [x^2]Q = B_1 B_4 + B_2 B_3, \qquad q_1 = Q' = B_0 B_4 + B_2^2, \qquad q_0 = B_0 B_3 + B_1 B_2, \tag{44}$$

neither q_2 nor q_1 vanishes identically in $g_0,\dots,g_4$, and $N\Delta$ and $R=((N\Delta)')^2+(Q')^2N\Delta$ have degree at most four in x. Write $R=\sum_{i=0}^4 R_ix^i$ and let $\rho=\rho_1x+\rho_0$ be the pseudo-remainder of R by Q, defined by $q_2^3R=SQ+\rho$; explicitly

$$\begin{aligned}\rho_1&=q_2^3R_1+q_2^2q_1R_2+(q_2^2q_0+q_2q_1^2)R_3+q_1^3R_4,\\ \rho_0&=q_2^3R_0+q_2^2q_0R_2+q_2q_1q_0R_3+(q_2q_0^2+q_1^2q_0)R_4.\end{aligned}\tag{45}$$

Some ρ_i is a nonzero polynomial, and the product of that ρ_i with q_2, q_1 and the Vandermonde $\prod_{i<j}(r_i+r_j)$ is a nonzero polynomial of degree at most

$$14+2+2+4=22\tag{46}$$

in each root coordinate. Consequently every binary $q\geq 32$ admits five distinct rational roots for which Q is a genuine quadratic in x with $Q'\neq 0$ and $Q\nmid R$.

Proof. Expanding $A_0A_3+A_1A_2$ with $A_j=B_j+xB_{j+1}$ gives (44); the two middle terms B_1B_3 cancel, and in characteristic two $Q'=\mathrm{d}Q/\mathrm{d}x$ is the coefficient q_1. Each A_j is linear in x, so Δ, N and Q have degree at most two, $N\Delta$ degree at most four, $(N\Delta)'$ degree at most two, and R degree at most four.

Neither q_2 nor q_1 is the zero polynomial. Written out,

$$\begin{aligned}&B_0=z_4g_0+z_5g_1+z_6g_2+z_7g_3, && B_1=z_3g_0+z_4g_1+z_5g_2+z_6g_3+z_7g_4,\\ &B_2=z_2g_0+z_3g_1+z_4g_2+z_5g_3+z_6g_4+z_7, && B_3=z_2g_1+z_3g_2+z_4g_3+z_5g_4+z_6,\\ &B_4=z_2g_2+z_3g_3+z_4g_4+z_5.\end{aligned}$$

In $q_1=B_0B_4+B_2^2$ the square B_2^2 contributes only square monomials. The coefficients of g_0^2, g_4^2, g_2^2, g_1^2, g_3^2 and 1 are z_2^2, z_6^2, $z_4^2+z_2z_6$, z_3^2, $z_5^2+z_3z_7$ and z_7^2; read in that order they force $z_2=z_6=z_4=z_3=z_5=z_7=0$. In $q_2=B_1B_4+B_2B_3$ the coefficients of g_0g_1, g_1g_2, g_1g_4 and g_3g_4 are z_2^2, z_3^2, $z_4^2+z_3z_5+z_2z_6$ and $z_5^2+z_3z_7$; read in that order they force $z_2=z_3=z_4=z_5=0$. Both conclusions contradict the hypothesis.

Pseudo-division by the degree-two polynomial Q with leading coefficient q_2 takes three steps, each replacing R by q_2R plus a multiple of Q; carrying them out gives (45). Over the fraction field of $k[g_0,\dots,g_4]$ the element q_2 is invertible, so $\rho=0$ would say $Q\mid R$. Lemma C.11 rules that out: the double pole along the component $\mathfrak{q}$ does not reduce, which is exactly the nonvanishing of R modulo Q. Hence some $\rho_i\neq 0$.

For the degrees, each g_j is elementary symmetric in $r_1,\dots,r_5$ and therefore multi-affine, so each B_j is multi-affine, each of q_2,q_1,q_0 has degree at most two in every root coordinate, each coefficient of $N\Delta$ at most four, and each R_i at most eight. By (45) each ρ_i has degree at most $6+8=14$. The Vandermonde $\prod_{i<j}(r_i+r_j)$ has degree four in each root, which gives (46).

Finally, a nonzero polynomial of degree at most d in each of n variables separately does not vanish identically on S^n for any set S with $|S|>d$: fix all but one variable at values where the polynomial is nonzero, and use that a nonzero univariate polynomial of degree at most d has at most d roots. With $d=22$ and $S=\mathbb{F}_q$, every $q>22$, hence every binary $q\geq 32$, carries a point where the product above is nonzero. There the Vandermonde factor makes the five roots distinct, $q_2\neq 0$ makes Q quadratic in x, $q_1=Q'\neq 0$, and $\rho\neq 0$ gives $Q\nmid R$. $\square$

Theorem C.13 (Empty first higher Lucas carrier)**.** For every $m\geq 4$,

$$\mathcal{M}_9(\mathbb{F}_{2^m})\cap\{\text{split-free syndromes}\}=\varnothing.$$

Equivalently, every one of

$$1+q+q^2+q^3+q^4+q^5$$

projective carrier points has a split squarefree degree-eight member in its Hankel kernel.

Proof. Proposition C.8 handles $\mathbb{P}(U)$. Outside it, projective transport normalizes the quotient in (30) to $[e_4]$, leaving the affine slice

$$(z_2,z_3,z_4,z_5,z_6,z_7)=(a,b,1,0,c,d)$$

modulo its upper-Borel stabilizer.

In particular $z_4\neq 0$, so Lemma C.10 applies: (36) has rank three or four, and in either case Lemma C.11 makes the final-pair cover (31) geometrically integral, with a component of $Q=0$ along which the double pole does not reduce. The rank split, the derivative identity (41) and its elimination are manuscript calculations. The public rank-two replay reconstructs the complete $q=64$ rational-twist mass and its five witnesses; it is not a symbolic all-field proof.

Fix five roots and let x be the sixth, as in (43). Lemma C.12 then supplies, for every binary $q\geq 32$, five distinct rational roots at which Q is quadratic in x with $Q'\neq 0$ and $Q\nmid R$; its selector product has individual root degree at most 22, which is what the grid bound needs. The reduced cover has at most two simple poles, hence genus at most one. The exact deletion table on the normalized cover is

condition	$x = r_i$	$Q = 0$	$\Delta = 0$	final/old collision	final/x collision
$\deg_x$	5	2	2	10	4,

so at most 48 rational points are removed. Since $64 + 1 - 2\sqrt{64} = 49$, every complement point is shallow for $q \geq 64$.

For $q = 16, 32$, the intrinsic Borel quotients contain respectively 292 and 1090 orbits. The compact certificates give one exact eight-root witness for every orbit, and an independently written replay reconstructs the quotient and both Hankel equations. This closes the two fields below the point-count threshold and proves the theorem. □

Projective-additive three-space divisors are sufficient but not necessary. At $q = 16$ their complete PGL_2-closure has 510 divisors, while 101 of the 292 complement orbits contain none of them; ordinary split witnesses still exist on all 101. Thus the endpoint criterion of Wang–Wu–Hu and Proposition C.2 describe a proper section. The exact uniform replacement is (32)–(35).

Proposition C.14 (Pointed R10 stage budgets)**.** Let $f \notin \mathcal{P}_{10} \cup \mathcal{M}_{10,p}^{\max}$, and retain i distinct markers after i contractions, $0 \leq i \leq 4$. The unavailable choices for the next marker form a zero-dimensional scheme of degree at most $19 + i$, hence at most 23. None of its defining factors vanishes identically off the displayed carrier union.

Proof. At syndrome degree $9 - i$, the pullback of the lower persistent carrier has degree at most three and the complete lower Lucas union has degree at most two. After its fixed factors are removed, the moving separable linear series has ramification degree at most

$$2(9 - i) - 4 = 14 - 2i.$$

Each retained marker contributes one linear marker diagonal and one fixed-marker gcd incidence cut out by quadratic minors, of degree at most two. Thus the full pointed degree is

$$3 + 2 + (14 - 2i) + i + 2i = 19 + i \leq 23. \tag{47}$$

The persistent and Lucas pullbacks can vanish identically only on the contained schemes classified by Proposition VI.3. Identically zero ramification is the inseparable case routed to the Lucas carrier. A marker diagonal is a proper point. Finally, the evaluation-hyperplane argument in the proof of Proposition B.13 shows that an identically vanishing fixed-marker minor forces the exact-linear-gcd package after cancellation, whose contained case is persistent. Hence every factor in the bound (47) is proper on the stated complement. □

The binary top step of the next proposition needs its own slice, because in characteristic two the divided cubic discriminant is the square $(AD_3 + BC)^2$ and cannot separate a split cubic from a nonsplit one. The replacement keeps an ordered residual root.

Lemma C.15 (Characteristic-two ordered-Hessian slice and selector)**.** Let q be a power of two, let $f \in \mathbb{P}(\Gamma^9 E)(\mathbb{F}_q)$ with $f \notin \mathcal{P}_{10} \cup \mathcal{M}_{10,2}^{\max}$, and put $k = \overline{\mathbb{F}_q}$; the geometric assertions below are taken after base change to k, while the selector polynomials of (iii) have coefficients in the field generated by those of f, hence are defined over $\mathbb{F}_q$. For $R \in \mathrm{Sym}^5 E^\vee$ squarefree let

$$\ell_{f,R} = \mathbb{P}\{\, \iota_{R\lambda} f : \lambda \in E^\vee \,\} \subset \mathbb{P}(\Gamma^3 E) \tag{48}$$

be the pencil obtained by iterating the contraction of (4) along the six linear factors of $R\lambda$, spanned by the divided cubics $U = (A, B, C, D_3)$ and $V = (a, b, c, d)$, and write $c_{u,v} = uU + vV$. Put $h_R = \iota_R f \in \Gamma^4 E$, so that U and V are its two coordinate shifts,

$$U = (h_0, h_1, h_2, h_3), \qquad V = (h_1, h_2, h_3, h_4), \tag{49}$$

and let

$$\mathcal{D}(R) = \det \begin{pmatrix} h_0 & h_1 & h_2 \\ h_1 & h_2 & h_3 \\ h_2 & h_3 & h_4 \end{pmatrix} \tag{50}$$

be the terminal persistent determinant of h_R, that is the cubic D of Proposition VI.1 evaluated at h_R. Write C_3 for the twisted cubic of divided cubes in $\mathbb{P}(\Gamma^3 E)$, which is the base locus of the divided Hessian. Then the following hold.

(i) The Hankel kernel of $c_{u,v}$ is spanned by the divided Hessian $P_0 X^2 + P_1 XY + P_2 Y^2$, where

$$\begin{aligned} P_0 &= (BD_3 + C^2)u^2 + (Bd + bD_3)uv + (bd + c^2)v^2, \\ P_1 &= (AD_3 + BC)u^2 + (Ad + aD_3 + Bc + bC)uv + (ad + bc)v^2, \\ P_2 &= (AC + B^2)u^2 + (Ac + aC)uv + (ac + b^2)v^2. \end{aligned} \tag{51}$$

The ordered-Hessian incidence

$$\mathcal{C}_{f,R} : \quad P_0 X^2 + P_1 XY + P_2 Y^2 = 0 \subset \mathbb{P}^1_{u:v} \times \mathbb{P}^1_{X:Y} \tag{52}$$

has bidegree $(2,2)$ and arithmetic genus one, and off $P_1=0$ it is the Artin–Schreier cover of class P_0P_2/P_1^2.

(ii) After the vertical factors $\ell_{f,R}\cap C_3$ are removed, (52) is geometrically integral unless one of three things happens: the pencil has a common kernel quadratic, which for the shift pencil (49) is exactly $\mathcal{D}(R)=0$; or $\ell_{f,R}$ lies in the Veronese surface Σ of lines with a common quadratic factor, whose member attached to $[Q]=[a:b:c]\in\mathbb{P}(\mathrm{Sym}^2 E^\vee)$ is the line spanned by the two shifts $(a,b,c,0)$ and $(0,a,b,c)$ of Q; or $\ell_{f,R}$ lies in one of the two Fano rulings of the quadric $\mathcal{Q}_0=\{AD_3+BC=0\}$. The first case is the horizontal-factor degeneracy and the last is the inseparable one. In Plücker coordinates $(z_0,\dots,z_5)=(p_{01},p_{02},p_{03},p_{12},p_{13},p_{23})$ the surface Σ has the parametrization $[a^2:ab:ac:b^2{+}ac:bc:c^2]$ and is cut out by the 2×2 minors of

$$\begin{pmatrix} z_0 & z_1 & z_2\\ z_1 & z_3+z_2 & z_4\\ z_2 & z_4 & z_5\end{pmatrix}, \tag{53}$$

by $z_1=z_4=0$, $z_2=z_3$, $z_0z_5=z_2^2$ for the tangent ruling, and by

$$\mathcal{N}:\quad z_0=z_5=0,\quad z_2=z_3,\quad z_1z_4=z_2^2 \tag{54}$$

for the complementary ruling. Both rulings lie in $\mathcal{D}=0$: the three coefficients of P_1 are $e_1=h_0h_3{+}h_1h_2$, $e_2=h_0h_4{+}h_2^2$ and $e_3=h_1h_4+h_2h_3$, and

$$\mathcal{D}=h_1e_3+h_2e_2+h_3e_1 \tag{55}$$

identically, so $P_1\equiv 0$ forces $\mathcal{D}=0$. Write $\mathcal{C}^{\mathrm{nv}}_{f,R}$ for the residual curve obtained from (52) by dividing P_0,P_1,P_2 by their common factor in (u,v). It has bidegree $(a,2)$ with $a\le 2$, hence arithmetic genus at most one.

(iii) Neither $\mathcal{D}$ nor a suitable pullback $\mathcal{A}$ of a minor of (53) vanishes identically in R. Their degrees in every individual root of R are at most 3 and 4, so the product $\mathcal{A}\mathcal{D}$ vanishes on the whole degeneracy locus of (ii) and has degree at most 7 in every root. Consequently a squarefree R with $\ell_{f,R}$ outside that locus exists over $\mathbb{F}_q$ as soon as

$$q\ge \min\{7\cdot 5+\tbinom{5}{2}+1,\ 8\cdot 5\}=\min\{46,40\}=40, \tag{56}$$

where $40=8\cdot 5$ counts field elements used as candidate root values, split into five pairwise disjoint eight-element blocks.

(iv) For such an R the residual curve $\mathcal{C}^{\mathrm{nv}}_{f,R}$ is geometrically integral and its normalization has genus at most one. The affine parameters that fail to produce a squarefree witness are cut by equations of total degree at most $5+2+2+10+4=23$.

(v) Every rational point of that normalization outside those parameters gives a completely split squarefree octic in W_f whose roots avoid the five roots of R.

Proof. (i) For a divided cubic $c=(A,B,C,D_3)$ the Hankel matrix is $H_c=\left(\begin{smallmatrix} A & B & C\\ B & C & D_3\end{smallmatrix}\right)$, so its kernel in $\mathrm{Sym}^2 E^\vee$ is spanned by the signed maximal minors $(BD_3+C^2,\ AD_3+BC,\ AC+B^2)$, which are the coefficients of the divided Hessian of c. Substituting $c=c_{u,v}$ makes each minor a binary quadratic in (u,v): the u^2 and v^2 terms are the minors of U and of V, and the uv term is their polarization, which is (51). A point of (52) is a member of the pencil together with one ordered root of its residual quadratic, the equation is of degree two in each factor, and a $(2,2)$ curve on $\mathbb{P}^1\times\mathbb{P}^1$ has arithmetic genus one. Dividing $P_0X^2+P_1XY+P_2Y^2=0$ by P_1^2/P_0 and setting $y=P_0X/(P_1Y)$ turns it into $y^2+y=P_0P_2/P_1^2$.

(ii) A vertical factor divides P_0,P_1,P_2 simultaneously, that is, the divided Hessian vanishes at the corresponding member of the pencil; its base locus is the twisted cubic C_3, so the vertical factors are exactly $\ell_{f,R}\cap C_3$. A horizontal factor $m(X,Y)$ says that m lies in the Hankel kernel of every member of the pencil, so it is a common kernel quadratic. For the shift pencil (49) the two members U and V contribute the rows (h_0,h_1,h_2), (h_1,h_2,h_3) and (h_1,h_2,h_3), (h_2,h_3,h_4), so a common kernel quadratic is exactly a kernel vector of the matrix in (50): the horizontal-factor locus is $\mathcal{D}(R)=0$. This family is a genuine third case, not part of Σ. For instance $h_R=(1,0,0,0,1)$ gives the chord $\ell_{f,R}=\mathbb{P}\langle e_0,e_3\rangle$, whose incidence is $uv\,XY=0$ and stays reducible after its two vertical factors are removed, while its Plücker vector $(0,0,1,0,0,0)$ satisfies neither (53) nor (54).

Away from that locus no horizontal factor occurs, and after the vertical factors are removed bidegree additivity makes any separable factorization $(1,1)+(1,1)$, so both factors are graphs of projectivities. Normalize the first to the identity by independent coordinate changes on the parameter and root lines. Over an algebraically closed field of characteristic two the second projectivity is the identity, semisimple with two fixed points and conjugate to $g(t)=ct$ with $c\neq 1$, or unipotent and conjugate to $g(t)=t+1$. In the semisimple case normalize the two cubics at the fixed parameters to $U=(a,1,0,0)$ and $V=(0,0,\gamma,d)$, for which (51) gives

$$P_0=d\,uv+\gamma^2v^2,\qquad P_1=(ad+\gamma)uv,\qquad P_2=u^2+a\gamma\,uv.$$

The graph of the identity is $uY+vX$ and the graph of the second projectivity is $uY+cvX$, so the comparison to make is with

$$(uY+vX)(uY+cvX)=cv^2X^2+(1+c)uvXY+u^2Y^2,$$

the roles of X and Y being fixed by the convention of (i), in which P_0 is the coefficient of X^2. Its Y^2 coefficient leaves no scalar freedom, and matching the three coefficients forces $d=0$, $\gamma^2=c$, $\gamma=1+c$ and $a\gamma=0$; since $c\neq 0$ also $\gamma\neq 0$, so

$a = 0$ and $c^2 + c + 1 = 0$. The normalized line is spanned by e_1 and e_2, which is the member of Σ attached to $Q = XY$. The identity is the case $c = 1$ of the same comparison and fails it. In the unipotent case the second graph is $(u + v)Y + vX$ and

$$(uY + vX)\big((u + v)Y + vX\big) = v^2X^2 + v^2XY + (u^2 + uv)Y^2,$$

whose XY coefficient is a multiple of v^2 while P_1 is a scalar multiple of uv, so the two cannot match and no such separable factorization exists. The case $P_1 \equiv 0$ is the inseparable locus, treated next. Since the coordinate changes used are basis changes on $\ell_{f,R}$ and on E, and both (52) and Σ are equivariant for them, undoing the normalization introduces no further component. Finally P_1 vanishes identically exactly when $\ell_{f,R}$ lies on the quadric $\mathcal{Q}_0$, whose Fano scheme is the two stated ruling conics; that is the inseparable case. The Plücker equations are those of the parametrization $[z_0 : \cdots : z_5] = [a^2 : ab : ac : b^2 + ac : bc : c^2]$ of Σ attached to $Q = aX^2 + bXY + cY^2$, and of the two rulings of $\mathcal{Q}_0$. For the shift pencil the inseparable case needs no separate equation: expanding the determinant (50) gives

$$\mathcal{D} = h_0h_2h_4 + h_0h_3^2 + h_1^2h_4 + h_2^3 = h_1e_3 + h_2e_2 + h_3e_1$$

with e_1, e_2, e_3 the coefficients of P_1 named in (ii), so $P_1 \equiv 0$ implies $\mathcal{D} = 0$ and both Fano rulings lie in $\mathcal{D} = 0$. The degeneracy locus of (ii) is therefore contained in the union of $\mathcal{D} = 0$ and Σ. Containment in that union is all the selector argument uses; no equality is claimed, and indeed $\mathcal{D} = 0$ also meets lines that are not degenerate.

Finally, dividing out the common factor of P_0, P_1, P_2 in (u, v) removes exactly the vertical components, so $\mathcal{C}^{\mathrm{nv}}_{f,R}$ is what the integrality statement applies to, and its bidegree $(a, 2)$ with $a \le 2$ gives arithmetic genus at most one.

(iii) The pencil (48) is spanned by $\iota_{Rx}f$ and $\iota_{Ry}f$, each linear in the coefficients of R, so its Plücker coordinates are quadratic in those coefficients, and the quadratic equations of (53) pull back to degree at most four in them. Suppose every pullback of a minor of (53) vanished identically in R. Then the generic root-line lies in Σ, so every iterated contraction pencil has a common quadratic; Proposition V.15 applied to the overlapping contraction flags returns the persistent rank-two carrier, and when a consecutive contraction coefficient vanishes in the ground characteristic the same overlap equations and Lucas' theorem return the contraction-kernel nucleus flags, so $f \in \mathcal{P}_{10} \cup \mathcal{M}^{\max}_{10,2}$. Suppose instead that $\mathcal{D}$ vanished identically on squarefree R. Split forms are dense, so $\mathcal{D}$ vanishes for every R, that is every bottom syndrome $\iota_R f$ lies in the prime cubic $V(D)$ of Proposition VI.1. The attainable bottom syndromes close to the projectivized row space of the marker catalecticant (11), so that row space lies in the reduced terminal carrier, and Proposition VI.3 puts f in $\mathcal{P}_{10} \cup \mathcal{M}^{\max}_{10,2}$. Both alternatives are excluded, so nonzero $\mathcal{A}$ and $\mathcal{D}$ exist. Each coefficient of h_R is linear in the coefficients of R, so $\mathcal{D}$, a 3×3 determinant in those coefficients, has degree at most three in every root, while $\mathcal{A}$ has degree at most four; by the containment established in (ii) their product vanishes on the whole degeneracy locus and has degree at most seven in each root.

Write $R = \prod_{i=1}^{5}(t + r_i)$, so each coefficient of R is at most linear in each r_i and $\mathcal{AD}$ has degree at most seven in each r_i separately, hence total degree at most $7 \cdot 5 = 35$ in the five roots together. The Vandermonde $\prod_{i<j}(r_i + r_j)$ has total degree $\binom{5}{2} = 10$, so Schwartz–Zippel, which reads the total degree, gives a squarefree base once $q > 35 + 10 = 45$. Alternatively partition $8 \cdot 5 = 40$ field elements into five pairwise disjoint eight-element blocks $S_1, \ldots, S_5$, which reads the individual degrees instead: a nonzero polynomial of degree at most seven in each variable cannot vanish on $S_1 \times \cdots \times S_5$, since successive univariate interpolation would make it zero, and disjointness makes every tuple have distinct coordinates. This gives (56); the quantity being counted is available field elements serving as candidate roots, not deleted points and not projective points.

(iv) For such an R the residual curve $\mathcal{C}^{\mathrm{nv}}_{f,R}$ is geometrically integral by (ii), so its normalization has genus at most one, its Artin–Schreier class P_0P_2/P_1^2 is nonconstant, and the cover is neither reducible nor a constant-field extension. Write $\lambda = (u : v)$ for the base parameter and $h_\lambda = P_0X^2 + P_1XY + P_2Y^2$. The parameters to delete are

failure	equation	$\deg_\lambda$
moving root meets a marker	$R(\lambda) = 0$	5
residual quadratic degenerates	$P_0(\lambda) = 0$	2
residual quadratic is a square	$P_1(\lambda) = 0$	2
a residual root meets a marker	$\mathrm{Res}_X(h_\lambda, R) = 0$	10
a residual root meets the moving root	$h_\lambda(\lambda) = 0$	4

of total degree 23. The degrees are read off from (51): P_0 and P_1 are binary quadratics in λ; the resultant of the quadratic h_λ with the quintic R is of degree two in the coefficients of R and of degree five in those of h_λ, hence of degree 10 in λ; and evaluating h_λ at λ itself adds two to the degree two of its coefficients. None of the five vanishes identically, and integrality is what rules each out: R is squarefree of degree five and nonzero; $P_0 \equiv 0$ would leave $Y(P_1X + P_2Y) = 0$, so $\mathcal{C}^{\mathrm{nv}}_{f,R}$ would contain the horizontal line $Y = 0$; $P_1 \equiv 0$ is the inseparable degeneracy already excluded by $\mathcal{AD} \neq 0$; $\mathrm{Res}_X(h_\lambda, R) \equiv 0$ would put one of the five horizontal lines $X = r_iY$ inside $\mathcal{C}^{\mathrm{nv}}_{f,R}$; and $h_\lambda(\lambda) \equiv 0$ would put the diagonal inside it. An integral curve of bidegree $(a, 2)$ contains no such component. A parameter divided out as a vertical factor has $P_0 = P_1 = P_2 = 0$, so it is already counted among the $P_0 = 0$ and $P_1 = 0$ parameters and needs no separate budget. The displayed degrees are those of the affine equations; each such parameter carries at most two points of the slice, and at most two more lie over the parameter at infinity, so at most $2 \cdot 23 + 2 = 48$ rational points are lost.

(v) A surviving rational point is a parameter λ with $R(\lambda) \neq 0$ together with a rational root of h_λ; since $P_0(\lambda) \neq 0$ and the coefficients are rational, the second root is rational too, and $P_1(\lambda) \neq 0$ makes the two distinct. By (i) the quadratic h_λ spans the Hankel kernel of $\iota_{R\lambda} f$, so six applications of (4) give $R\lambda h_\lambda \in W_f$. Its degree is $5 + 1 + 2 = 8$, it splits completely, and the deleted divisors make its eight roots distinct: the markers are distinct because R is squarefree, λ avoids them, and both residual roots avoid the markers and λ. ☐

Proposition C.16 (R10 fixed-depth escape)**.** For odd $q \geq 59$, and for even $q \geq 64$, every split-free redundancy-ten syndrome belongs to $\mathcal{P}_{10} \cup \mathcal{M}^{\max}_{10,p}$.

Proof. Apply the finite-depth escape theorem through five markers. Proposition C.14 supplies all five successive unavailable-marker schemes, including the retained-marker incidences omitted by an unpointed iteration. The terminal lower packages are the R8 package of Propositions B.1 and B.7, followed by the R9 package of Propositions B.13 and B.21; the R5–R7 terminal packages are those used in their respective classification theorems. At the top step, the generic ordered-pair cover has genus at most one. Its base diagonal and branch delete 12 points, and each of the five forbidden markers deletes at most 6, for the exact total

$$12 + 5 \cdot 6 = 42.$$

These are all terminal transverse, modular, and self-collision strata. Since

$$q + 1 - 2\sqrt{q} > 42$$

first holds at a prime power $q = 59$, the odd assertion follows. In characteristic two the argument above is not used. Lemma C.15 applies directly to f: it chooses all five roots of its base at once, and its witness lies in W_f by repeated application of (4), so it needs neither the stagewise marker selection nor the intermediate lower packages. Its selector is nonzero as soon as $q \geq \min\{46, 40\} = 40$ field elements are available as candidate roots, and for such a base it gives a geometrically integral ordered-Hessian slice whose normalization has genus at most one and whose deletion divisor has degree $5+2+2+10+4 = 23$ on the base line, hence at most $2 \cdot 23 + 2 = 48$ deleted rational points. Since $64 + 1 - 2\sqrt{64} = 49 > 48$, and since 32 has fewer than 40 elements while 64 has more, both conditions first hold at $q = 64$; part (v) of the lemma turns a surviving point into a split squarefree octic in W_f avoiding all five markers. In both characteristics Proposition VI.3 identifies the sole contained locus as the displayed carrier union. ☐

Proposition C.17 (Characteristic-seven coherent lift at R10)**.** Let $q = 7^m \geq 343$. Every point of

$$\mathcal{M}^{\max}_{10,7} \subseteq \mathbb{P}\langle e_3, e_4, e_5, e_6 \rangle \subset \mathbb{P}(\Gamma^9 E)$$

has a split squarefree octic in its Hankel kernel.

Proof. Choose the marker at infinity. In the affine-coordinate convention of Section V, contraction deletes the last coordinate, so a nonzero $f \in \langle e_3, e_4, e_5, e_6 \rangle$ contracts to the same coefficient row

$$b = \iota_\infty f \in \mathbb{P}\langle e_3, e_4, e_5, e_6 \rangle \subset \mathbb{P}\langle e_2, e_3, e_4, e_5, e_6 \rangle,$$

the characteristic-seven R9 binary-quartic carrier. Since $q > 102$, Proposition B.16 selects four distinct finite base roots. The construction in Proposition B.15 then fixes a fifth finite root as its moving parameter; because $q+1-2\sqrt{q} > 32$, Proposition B.17 selects a point whose residual quadratic is separable and avoids all five fixed roots. Its nonzero leading coefficient is part of the deleted determinant locus, so both residual roots are finite. The resulting split squarefree septic $g \in W_b$ therefore avoids the marked root at infinity. Equation (4) gives $\lambda_\infty g \in W_f$, and marker avoidance makes this octic squarefree. ☐

Corollary C.18 (Redundancy ten)**.** For every prime power $q \geq 59$, the projective deep-hole syndromes of $\mathrm{PRS}(q - 9)$ are exactly the persistent tangent and conjugate-secant families. Their cardinality is $q(q + 1)^2/2$, and their orbit law is T/T^9 modulo inversion and Frobenius, with tangent cocycle $z \mapsto z + 9u$.

Proof. For odd $q \geq 59$, Proposition C.16 confines a nonpersistent split-free point to the maximal Lucas carrier. Its support is empty except in characteristic seven; the first field in this range is 343, where Proposition C.17 applies. For even prime powers in the range one has $q \geq 64$; Proposition C.16 confines every split-free point to $\mathcal{P}_{10} \cup \mathcal{M}_9$, and Theorem C.13 eliminates the second term. Proposition VI.5 holds throughout the stated range, so the split-free equality promotes to deep holes. The persistent count and orbit law are the specialization $r = 10$ of Theorem VI.4. ☐

Appendix D
Verification and formal boundary

No geometric classification theorem in this paper is claimed to be formalized in Lean. The formal layer checks coordinate identities and conditional synthesis implications with the component geometry, rational-point existence, group actions, and certificate semantics kept as explicit inputs.

The conceptual theorems above are accompanied by deterministic evidence bundles. Table VIII records which claims use exhaustive computation. Each public label resolves, through the supplement schema, to its generator, certificate, replay where stated, and SHA-256 manifest. Table I summarizes the corresponding mathematical mechanisms and closing gates.

TABLE VIII: Public verification map. Exact filenames, commands, hashes, domains, and stop conditions belong to the electronic supplement.

Public label	Verified domain	Replay and trust boundary
Certificate R5	nineteen audited field records; seven-field finite bridge, with $q = 16$ covered as a regression check rather than a dependency	independent pointwise and orbitwise replay; canonical representatives and exhaustion records
Certificate R6	direct scan through $q = 16$; finite bridge below 29	independent syndrome replay and separate radius gate
Certificate R6-NF	small exceptional normal forms	deterministic normal-form reconstruction; not a second implementation
Certificate R7	$q = 7, 8, 9, 11$ and finite bridge below 37	primary quotient enumeration and representative/orbit replay; split-free and radius flags remain separate
Certificate R7 direct locus	all fourteen R7 certificate fields	candidate-domain and marker-intersection reconstruction, orbit and Frobenius replay; shares the direct-locus engine, R5 field layer, and R6 pointed theorem from $q \geq 16$, and is not a second field implementation
Certificate SC	terminal factorization and coherent-Fano identities	Python and Singular elimination replays; density and finite-component selection are separately kernel checked
Certificate R8	threshold, nucleus, witness, and budget data	generator plus separately written replay; no ambient syndrome census
Certificate R9	residual-quadratic and characteristic-seven data	independent algebra replay and exact $q = 49$ carrier bridge
Certificate R10	threshold and persistent orbit arithmetic	separate prime-power and cyclic-orbit replay
Certificate Lucas M9	rank-two twists; full $q = 16, 32$ carrier; $q = 64$ invariant block	independent action and Hankel replay; the $q = 64$ complement and the uniform genus-one argument are mathematical

Table VI is a deterministic projection of the hash-pinned public R6 and R6-NF records. The top-level verifier regenerates it and checks byte-for-byte equality with the included source.

The supplement's `CERTIFICATE-SCHEMA.md` defines the public labels and replay classes; its reproduction guide and two manifests record commands, locks, hashes, and byte counts. The entry point `supplement/verify.py` checks those records, the classification projections, and the local PDF; `--replay` runs the paper-local Python replays.

The exact Lean declarations, hypotheses, axiom audit, and replay command are listed in `LEAN-STATEMENTS.md`. They check the coordinate, budget, density, closure, component-selection, and conditional-synthesis interfaces. Concrete component geometry, fixed-level carrier arithmetic, classical radius inputs, and certificate semantics remain explicit inputs. The formal layer also checks the field-eight record but draws no quantum consequence from it, because the required one-column MDS extension does not exist at radius $r - 1$ (Remark IV.15).

A. Data, code, and disclosure

The electronic supplement accompanying this manuscript contains the source, canonical classification representatives and stabilizers, certificate schemas, generators, independent replays where available, expected hashes, toolchain locks, and a top-level verification command. This appendix assigns each computational or formal claim its exact trust route.

The repository includes Projective Reed–Solomon Toolkit under `software/`. This proof-carrying executable lets a reader move from a versioned syndrome request to an independently replayable result. For every $r \geq 5$ and $q \geq r$, `canonicalize` computes an exact semilinear canonical form without a coding verdict, while `distance` and `decode` return exact nearest-error data with a locator-and-magnitude certificate, subject to an explicit candidate budget. The `classify` command consults the hash-pinned R5–R10 theorem registry and returns `DEEP`, `NOT_DEEP`, `UNRESOLVED`, or `UNSUPPORTED`. Only `DEEP` carries a positive certificate; `verify` recomputes its transporter, family route, and radius premise.

This interface makes the software's broader reach visible without enlarging the theorem. Locator candidates are streamed, so the budget bounds search work rather than a stored candidate list. Canonicalization and decoding above R10 do not promote a syndrome to a deep hole without a covering-radius theorem and family exhaustion. The sole enabled higher-redundancy family has even q and $r = q - 1$; its verifier replays the e_{r-2} terminal-locator obstruction and uses the dimension-two radius theorem of [3, Thm. 17(1)]. It is a certified family, not a complete R11-or-higher classification.

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